

Scanning for Significance: False Discovery Control for Impulse Responses*

Giorgi Nikolaishvili
Wake Forest University
nikolag@wfu.edu

Noah D. Gade
Wake Forest University
noah.gade@gmail.com

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Abstract

Applied impulse response analysis often follows a workflow: researchers identify a shock, estimate a large set of responses, select statistically significant estimates, and use their features (e.g. sign, magnitude, and timing) to construct an economic narrative. Two inferential approaches dominate current practice. Pointwise inference treats each response in isolation, so false rejections accumulate among many pointwise intervals. Simultaneous inference bounds the probability of even one false rejection, so its bands widen as the response family expands. Researchers therefore face a stark tradeoff between reliability and power. We propose false discovery rate (FDR) and false coverage rate (FCR) control as an alternative approach that balances this tradeoff by matching inference to the selection and interpretation steps of the workflow. FDR control governs which responses are declared significant, and FCR control ensures the confidence intervals used to interpret them remain valid post-selection. As a byproduct, calibrating coverage to the selected set can deliver greater power than simultaneous inference. The procedure integrates into standard vector autoregression and local projection bootstraps. Applications show the strategy recovers effects lost under simultaneous bands while discarding fragile pointwise findings, at times materially altering the economic narrative.

JEL Codes: C12; C22; C32; E00

Keywords: impulse response; multiple testing; false discovery; VAR; local projection

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1 Introduction

Impulse response functions (IRFs) are among the most widely reported empirical objects in macroeconomics and finance, used to build narratives about how structural shocks propagate through the economy. A typical application estimates responses across dozens or hundreds of horizon–variable–shock–regime combinations. In a common workflow, researchers scan the resulting grid for significant effects, identifying which variables respond, when they peak, and how long they persist. The resulting economic narrative often rests on pointwise confidence bands, which offer no multiplicity control — each band is valid, but only in isolation. However, as the number of reported responses grows, so does the expected number of false rejections at an unbounded rate.¹ Even a modestly-sized exercise produces a large hypothesis family. For example, a five-variable vector autoregression (VAR) with a single identified shock examined over thirty-seven horizons yields 185 coefficients. State-dependent or multi-shock extensions easily involve several hundred. A growing body of work addresses this problem with simultaneous inference methods that control the family-wise error rate (FWER), the probability of at least one false rejection in the response family. This guarantee is well suited to claims about an entire response path, but it is stricter than the scanning workflow requires. A researcher who selects and interprets a subset of responses can tolerate a small, controlled proportion of false discoveries among them. FWER-controlling bands widen as the response family grows, often to the point where they fail to detect any true dynamic effects, discouraging researchers from investigating transmission mechanisms thoroughly.² For this workflow, the result is an uncomfortable choice between overstating evidence and understating it, one that most researchers resolve in favor of pointwise bands, leaving the resulting economic narratives exposed to an unknown share of false rejections.

This paper argues that false discovery rate (FDR) and false coverage rate (FCR) control provide an inferential target matched to the previously-described impulse response scanning workflow: they govern the reliability of the subset of responses the researcher selects and interprets, rather than the full grid of estimates searched by the researcher. FDR control bounds the expected proportion of false rejections among those responses

¹To name a few familiar applications: [Christiano et al. \(2005\)](#) and [Romer and Romer \(2004\)](#) build their monetary policy transmission narratives by reading significance patterns across variables and horizons; [Gertler and Karadi \(2015\)](#) identify a persistent credit-spread response by scanning both dimensions of a proxy SVAR grid; [Ramey \(2011\)](#) and [Ramey and Zubairy \(2018\)](#) adjudicate the fiscal multiplier debate by inspecting significance across horizons and variables, with [Auerbach and Gorodnichenko \(2012, 2013\)](#) and [Ramey and Zubairy \(2018\)](#) further expanding the hypothesis family through state dependence.

²The adoption pattern in applied work is consistent with this cost. Although simultaneous inference methods have been available in the impulse response literature for some time ([Jordà, 2009](#); [Inoue and Kilian, 2016](#); [Montiel Olea and Plagborg-Møller, 2019](#)), pointwise bands remain the norm in empirical practice.

declared significant. Heuristically, if the procedure selects twenty significant responses at a 10% FDR level, at most two of those twenty are expected to be false discoveries. FCR control complements this guarantee by delivering post-selection confidence intervals for the selected responses. If twenty intervals are reported, at least eighteen are expected to cover their true parameter values. Together, FDR and FCR control discipline the economic narrative against false discoveries while preserving the power to detect true dynamic effects, balancing the tradeoff that constrains the narratives that researchers can credibly report from a scanned response grid. Unlike FWER-controlled inference, neither guarantee requires a loss in power as the dimension of the response family grows. A researcher who examines six variables over fifty horizons (e.g. [Känzig \(2021\)](#)) faces the same reliability standard for their reported findings as one who examines only four variables over thirty horizons (e.g. [Bu et al. \(2021\)](#)). Adopting this framework leaves the empirical workflow intact and formalizes only its inferential steps. The researcher still estimates the same high-dimensional IRF grid, but no longer overlays pointwise bands and informally selects salient features. Instead, they first apply an FDR-controlling selection rule to the full collection of test statistics to obtain a formal rejection set, satisfying a specified tolerance. Then, they construct FCR-controlled confidence intervals only for the selected responses.

The existing literature on impulse response inference has approached multiplicity through simultaneous bands that target FWER control, leaving the selective-reporting problem unaddressed. Analytic approaches based on Scheffé-type and Bonferroni adjustments have been proposed for VARs ([Jordà, 2009](#); [Lütkepohl et al., 2015, 2020](#)), and resampling-based methods have been developed for both VAR and LP estimators, ranging from bootstrap joint Wald confidence sets ([Inoue and Kilian, 2016](#)) and balanced bootstrap bands ([Bruder and Wolf, 2018](#)) to sup- t bands ([Montiel Olea and Plagborg-Møller, 2019](#); [Jordà, 2023](#); [Jordà and Taylor, 2025](#); [Inoue et al., 2026](#)). This body of work has sharpened our understanding of the pointwise-versus-simultaneous tradeoff considerably. But the menu of inferential targets under consideration has remained binary: marginal error control for each response, or joint error control over the entire family. The possibility of targeting an intermediate error rate that accounts for multiplicity without requiring joint coverage of the entire response family, and that directly governs the reliability of the selected findings rather than the full grid, has not been explored in the IRF setting. Yet precisely such a framework exists in the statistics literature on multiple testing, where FDR control ([Benjamini and Hochberg, 1995](#)) and FCR control ([Benjamini and Yekutieli, 2005](#)) have become standard tools for high-dimensional inference problems with similar structure.

Both existing inferential targets are defined relative to the fixed grid of tests. Pointwise intervals control the marginal error of each response in isolation, while simultaneous bands control the probability of any error across the entire family. The scanning workflow builds its economic narrative from neither object. Instead, it builds from the responses flagged as significant and the confidence intervals used to interpret them, a data-selected subset of the estimated impulse response grid. Because that subset is assembled precisely on the basis of significance, guarantees that hold over the full grid do not carry over to it. Scanning an IRF grid is therefore a selective reporting problem, and FDR and FCR control address it directly, governing the reliability of the reported findings rather than that of the full test family. This reframing does not require the true response surface to be sparse. Coefficients outside the rejection set are not asserted to be zero, and the FDR guarantee holds for any number of true nulls, including the limiting case in which every response is nonzero.

Building on this reframing, the paper makes three specific contributions. First, we adapt an FDR- and FCR-controlling procedure to impulse response inference. We implement FDR control using the resampling-based stepwise methodology of [Romano et al. \(2008\)](#) (henceforth RSW selection) which accommodates general dependence characteristics of impulse responses by directly estimating the joint distribution of test statistics. We then implement post-selection FCR adjustment using the approach of [Benjamini and Yekutieli \(2005\)](#). We establish asymptotic validity of this procedure for both VAR and LP estimators under mild regularity conditions, showing that FDR and FCR are controlled at the nominal level in large samples. The only requirement on the bootstrap is that it consistently estimates the joint distribution of the studentized IRF vector. We verify this for the workhorse resampling schemes in applied macroeconometrics, including the bias-corrected i.i.d. residual bootstrap for VARs ([Kilian, 1998](#)), the residual-based moving block bootstrap for VARs and proxy SVARs with serially dependent or conditionally heteroskedastic innovations ([Brüggemann et al., 2016](#); [Jentsch and Lunsford, 2022](#)), and the VAR-based wild recursive bootstrap for local projections ([Montiel Olea and Plagborg-Møller, 2021](#)).

Second, we provide Monte Carlo evidence on finite-sample performance in designs calibrated to empirically relevant macroeconomic settings. The simulated data-generating process is a six-variable VAR(4) with block-diagonal autoregressive and impact matrices, so cross-block responses are exactly zero by construction and false discoveries can be counted rather than approximated. We vary the persistence of the process, the estimator and its associated bootstrap (VAR and local projection), the sample size, the horizon range, and the share of true nulls in the family. Across

these configurations the procedure delivers systematic power gains over simultaneous inference while holding FDR at or below the target level. Interval width is the mechanism behind the gain. FCR-adjusted intervals are between one and 1.3 times as wide as pointwise intervals, while sup-t bands are 1.3 to 2.0 times as wide, and the sup-t penalty grows with the size of the response family while the FDR/FCR penalty does not. This is why the gains are largest when the response family is high-dimensional. Pointwise inference, by contrast, exhibits elevated FDR and FCR throughout, confirming that marginal guarantees do not survive selection.

Finally, we apply our methods to two prominent empirical settings: monetary policy shocks (Bu et al., 2021) and oil supply news shocks (Känzig, 2021). In both applications, FWER-controlling confidence bands recover only part of the transmission narrative captured by FDR/FCR-controlled intervals. In the monetary policy application, the simultaneous band detects little beyond the policy stance itself and a short window of the price response, and it detects no output response at any horizon. In the oil application it is more informative, recovering the oil price, the later portion of the inventory buildup, and the U.S. output contraction, but it detects essentially no response of U.S. consumer prices, so the stagflationary narrative is unavailable to the researcher. FDR/FCR-controlled inference recovers the missing effects in both settings without reverting to pointwise reporting. In the monetary policy application, it preserves the qualitative transmission narrative implied by pointwise inference while placing it on formally reliable footing: both the delayed output contraction and the full disinflationary path are selected. In the oil shock application, FDR/FCR-controlled inference retains the core oil price, inventory, and U.S. stagflationary findings but discards the pointwise evidence that oil supply shocks persistently reduce world oil production and world industrial production, altering the economic narrative along two dimensions.

Related literature. Our paper connects the statistics and econometrics literatures on multiple testing to the macroeconometrics literature on joint IRF inference. The foundational references on FDR control are Benjamini and Hochberg (1995) and Benjamini and Yekutieli (2001), with FCR control for post-selection confidence intervals introduced by Benjamini and Yekutieli (2005). Our baseline implementation uses the bootstrap step-down procedure of Romano et al. (2008), which controls FDR under general dependence by estimating the joint distribution of the test statistic vector directly; this procedure builds on the broader resampling-based multiple testing framework of Westfall and Young (1993), White (2000), Romano and Wolf (2005a,b), and Romano and Wolf (2007), surveyed for econometric practice by Romano et al. (2010).

In economics and finance, FDR-style reasoning has been applied to mutual fund performance evaluation ([Barras et al., 2010](#)) and asset pricing anomaly discovery ([Harvey and Liu, 2020](#)), where strong cross-test dependence and selective reporting create concerns analogous to those arising in IRF families. More broadly, the applied microeconomics and program evaluation literatures have increasingly recognized that scanning across many outcomes, subgroups, or time periods requires multiplicity adjustment ([Anderson, 2008](#); [Lee and Shaikh, 2014](#); [List et al., 2019](#)), and [Freyaldenhoven et al. \(2019\)](#) address a closely related problem in panel event-study designs. The macroeconometrics literature has not had the same reckoning, even though the structure of the problem is directly analogous.

In a related relaxation of familywise criteria, [Armstrong et al. \(2022\)](#) develop robust empirical Bayes confidence intervals satisfying an average coverage guarantee: across a large set of parameters, at least a fraction $1 - \alpha$ of the reported intervals cover their true values on average. Their guarantee attaches unconditionally to the full parameter vector and derives its precision gains from shrinkage. FCR control instead attaches to a data-selected subset of findings and derives its gains from calibrating the error budget to the selection, which is the object at issue in the scanning workflow. The broader econometric model-selection literature has long emphasized that significance thresholds encode a trade-off between false inclusion and the power to retain relevant effects, rather than constituting immutable conventions ([Hendry, 1995](#), Sections 13.10.4–13.10.5); see also [Hendry and Krolzig \(2005\)](#) and [Hendry \(2009, Section 1.5.2\)](#).

The IRF inference literature has approached multiplicity exclusively through FWER-controlling simultaneous bands, with the methodological arc moving from analytic to resampling-based approaches. Early work constructed Scheffé-type regions ([Jordà, 2009](#)) and Bonferroni or projection-based bands ([Lütkepohl et al., 2015, 2020](#)), establishing that pointwise bands can substantially undercover IRF paths. Resampling-based methods then emerged as a more powerful alternative. [Inoue and Kilian \(2016\)](#) construct bootstrap joint Wald confidence sets for structurally identified VARs, [Bruder and Wolf \(2018\)](#) propose balanced bootstrap joint bands, and [Montiel Olea and Plagborg-Møller \(2019\)](#) develop sup- t bands, subsequently extended to local projections ([Jordà, 2023](#); [Jordà and Taylor, 2025](#); [Inoue et al., 2026](#)). The literature has largely converged on the sup- t implementation. From a Bayesian perspective, [Sims and Zha \(1999\)](#) and [Inoue and Kilian \(2022\)](#) develop posterior-based joint credible regions that provide a posterior analog of FWER-style coverage, arriving at the same inferential target from different foundations. Frequentist and Bayesian approaches alike thus share a single inferential target: zero tolerance for false rejections anywhere in the response family. A different target, one that

bounds the share of false rejections among the selected findings rather than prohibiting them entirely, is better matched to the scanning workflow that motivates this paper: applications in which researchers select and interpret a subset of responses from a large estimated grid.

Outline. Section 2 uses an empirical monetary policy application and a controlled simulation design to show that the pointwise-versus-simultaneous tradeoff is severe in practice: pointwise bands produce high rates of false rejection across the response grid, while simultaneous sup- t bands lose the ability to detect effects that are plausibly present in the data. Section 3 develops our FDR- and FCR-adjusted procedure, establishes its asymptotic properties for VAR and LP estimators, and shows through Monte Carlo experiments that it recovers a substantial share of pointwise power while maintaining false discovery control at the target level. Section 4 applies the methods to oil supply news shocks identified through OPEC announcement surprises, illustrating how FDR/FCR-controlled inference can alter which transmission patterns the data support. Section 5 provides practical guidance for applied researchers. Section 6 concludes.

2 The Pointwise-Simultaneous Tradeoff in IRF Inference

This section documents the severity of the tradeoff that researchers face when scanning an estimated impulse response grid for significant effects. Pointwise confidence bands are valid for each individual coefficient, but provide no joint guarantee once the full grid is scanned. Simultaneous bands offer joint validity, but in impulse response families of the size routinely reported in empirical applications, they widen enough to leave the researcher with only a fragment of the estimated response grid for narrative formation. An empirical monetary policy application shows how sharply the two bands diverge. A controlled simulation in which the true responses are known then establishes that the simultaneous bands' conservatism costs genuine power rather than merely trimming false rejections. The dilemma therefore shapes what kinds of economic narratives the data can credibly support.

Preliminaries. Consider a family of m scalar impulse response coefficients $\theta_1, \dots, \theta_m$, indexed across horizons, variables, and potentially multiple shocks or states. Let $\hat{\theta}_j$ denote the estimator of θ_j with standard error $\hat{\sigma}_j$. Let $\mathcal{H}_0 \subseteq \{1, \dots, m\}$ denote the subset of true null hypotheses (those j for which $\theta_j = 0$) and let $\mathcal{H}_1 = \{1, \dots, m\} \setminus \mathcal{H}_0$.

A pointwise $(1 - \alpha_{\text{pt}})$ confidence interval $\text{CI}_j^{\text{pt}}(\alpha_{\text{pt}})$ satisfies $\mathbb{P}\{\theta_j \in \text{CI}_j^{\text{pt}}(\alpha_{\text{pt}})\} \geq 1 - \alpha_{\text{pt}}$ for each fixed j , and the associated two-sided test of $H_j : \theta_j = 0$ rejects when $0 \notin \text{CI}_j^{\text{pt}}(\alpha_{\text{pt}})$. When researchers scan the full family, the probability of at least one false rejection among the true nulls,

$$\text{FWER} = \mathbb{P}\left(\sum_{j \in \mathcal{H}_0} \mathbb{I}\{0 \notin \text{CI}_j^{\text{pt}}(\alpha_{\text{pt}})\} > 0\right), \quad (1)$$

can be much larger than α_{pt} , even when each individual test is correctly sized.

A simultaneous $(1 - \alpha_{\text{FWER}})$ band $\{\text{CI}_j^{\text{sim}}(\alpha_{\text{FWER}})\}_{j=1}^m$ controls this joint error:

$$\mathbb{P}\left\{\theta_j \in \text{CI}_j^{\text{sim}}(\alpha_{\text{FWER}}) \text{ for all } j = 1, \dots, m\right\} \geq 1 - \alpha_{\text{FWER}}. \quad (2)$$

In the IRF context, a convenient implementation is the sup- t approach. Let $T_j = |\hat{\theta}_j|/\hat{\sigma}_j$ and let $T_j^{*(b)} = |\hat{\theta}_j^{*(b)} - \hat{\theta}_j|/\hat{\sigma}_j$ denote the corresponding studentized statistic computed on bootstrap draw b , where $\hat{\theta}_j^{*(b)}$ is the estimate computed on that draw and the standard error $\hat{\sigma}_j$ is held fixed across draws. The bootstrap distribution of the maximum statistic $M^{*(b)} = \max_{1 \leq j \leq m} T_j^{*(b)}$ is used to obtain a critical value $c_{1-\alpha_{\text{FWER}}}$ as the empirical $(1 - \alpha_{\text{FWER}})$ quantile of $\{M^{*(b)}\}_{b=1}^B$. The two-sided sup- t band is then

$$\text{CI}_j^{\text{sim}}(\alpha_{\text{FWER}}) = [\hat{\theta}_j - c_{1-\alpha_{\text{FWER}}} \hat{\sigma}_j, \hat{\theta}_j + c_{1-\alpha_{\text{FWER}}} \hat{\sigma}_j], \quad j = 1, \dots, m. \quad (3)$$

Because $c_{1-\alpha_{\text{FWER}}}$ is tied to an extreme-value functional, it typically increases with the effective dimension of the response family, generating power loss in large IRF grids.

Empirical Example. We illustrate the dilemma through the common empirical exercise of estimating the dynamic effects of a contractionary monetary policy shock on output, prices, and financial conditions. Our shock measure is the monetary policy shock series of [Bu et al. \(2021\)](#) (BRW), normalized to a 100bp increase in the 2-year Treasury yield on impact. The response variables are log Industrial Production (IP), log Consumer Price Index (CPI), the excess bond premium (EBP) ([Gilchrist and Zakrajšek, 2012](#)), and the cumulative BRW series, which serves as a diagnostic for the persistence of the policy stance. Unlike [Bu et al. \(2021\)](#), we do not include a Commodity Price Index. The sample is restricted to 2008m1–2021m12, a period for which [Bu et al. \(2021\)](#) show that pointwise 90% bands produce many rejections. We estimate a monthly reduced-form VAR with $p = 12$ lags on the vector $(\text{cumulated BRW}_t, \log \text{IP}_t, \log \text{CPI}_t, \text{EBP}_t)^\top$, with the monetary policy innovation identified as the first structural innovation via a Cholesky ordering.

The bootstrap distribution of the full IRF vector is generated using the bias-corrected i.i.d. residual bootstrap of Kilian (1998). The same bootstrap draws are used to form for both pointwise intervals and sup- t bands.

Figure 1 reports VAR-based responses over horizons $h = 0, \dots, 24$ months, yielding $m = 4 \times 25 = 100$ scalar IRF coefficients. The pointwise 90% confidence intervals suggest many nonzero effects: a persistent policy stance, a delayed negative effect on output around one to two years, and a persistently negative effect on prices. This narrative lines up with the conventional transmission emphasized by Bu et al. (2021). However, when we replace pointwise bands with sup- t bands controlling the FWER at 10% over the full response family, most of these narrative findings disappear. The policy stance itself remains significant over roughly the first half year, and the price response remains significant over a narrow window around nine to eleven months. The delayed output contraction is not detected at any horizon. Of the 100 coefficients in the grid, the simultaneous band declares roughly one in ten significant, against about half under pointwise bands. Once joint error control is imposed across the full grid, the scanning researcher is left with a fragment of the transmission narrative rather than the sustained output and price dynamics that the pointwise bands appear to support.

Simulation Evidence. To confirm that the pattern in Figure 1 is systematic, we examine a controlled simulation environment in which the true impulse responses are known. The DGP is a six-variable VAR(4) with block-diagonal autoregressive and impact matrices, so that all cross-block impulse responses are identically zero by construction. This provides a large and known set of true null coefficients against which false rejections can be measured exactly. The researcher estimates an unrestricted VAR(4) and computes IRFs for each variable–shock pair across a maximum of $H = 20$ horizons for three reported shocks, giving $m = 378$ coefficients per replication. Full details on the DGP parameterization, estimation, and bootstrap are provided in Appendix A.

Figure 2 reports the key performance metrics across a range of sample sizes. Pointwise 95% intervals deliver high power, but with no multiplicity control, as measured here by the FWER. The probability of at least one false rejection remains close to one across the sample-size range, illustrating the multiplicity induced by scanning the response grid.³ Simultaneous sup- t bands control the FWER by calibrating to the extreme-value

³A near-unit FWER is close to inevitable in a family of this size, and is not by itself evidence that pointwise reporting is unreliable in the sense that matters for the scanning workflow. Any procedure with non-trivial power over 378 coefficients will commit at least one false rejection with high probability. As Section 3 shows, the FDR-controlling procedure proposed in this paper exhibits the same pattern, and for the same reason. As the sample grows and the rejection set expands, the probability of at least one false

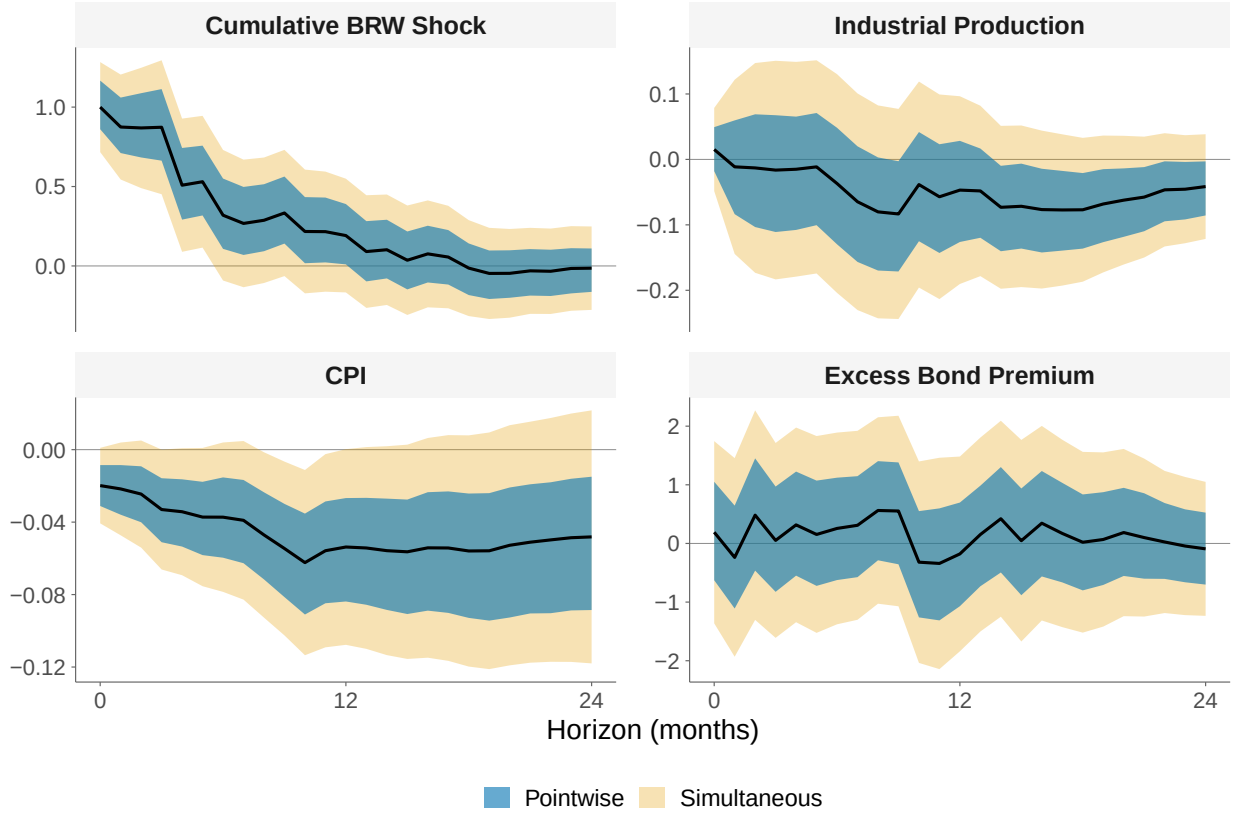


Figure 1: Responses to a contractionary monetary policy shock (VAR).

distribution of the maximum statistic, but the cost is severe — bands are roughly 1.3 to 2 times wider than their pointwise counterparts, and power drops sharply. Corresponding results using LP estimation presented in Figure A.3 yield similar patterns. The width penalty grows with the dimension of the response family, so a scanning researcher who examines more variables or longer horizons faces an increasingly uninformative simultaneous band.

Taken together, Figures 1 and 2 show that the choice between pointwise and simultaneous inference affects the substantive conclusions drawn from an IRF analysis. In the monetary policy application, the two procedures support materially different characterizations of the transmission mechanism. Pointwise bands indicate a delayed contraction in output and a persistent decline in prices, whereas simultaneous bands recover only a limited portion of these dynamics. The simulation shows that this discrepancy likely cannot be attributed solely to the removal of false pointwise rejections. Because the true IRFs are known, the results establish that simultaneous inference leads to a loss of power that becomes more pronounced as the response family expands across rejection rises while the share of false rejections stays at the target.

variables and horizons. Pointwise inference therefore leaves the error rate among selected responses uncontrolled, while simultaneous inference obtains familywise validity by imposing an increasingly severe critical-value penalty. Neither inferential target is well-matched to the empirical workflow. Researchers typically construct their substantive conclusions from a data-selected subset of the IRF grid rather than from each coefficient in isolation or from the full response family as a whole. This motivates an inferential framework that governs the reliability of the selected responses and of the intervals used to interpret them, developed in the following Section.

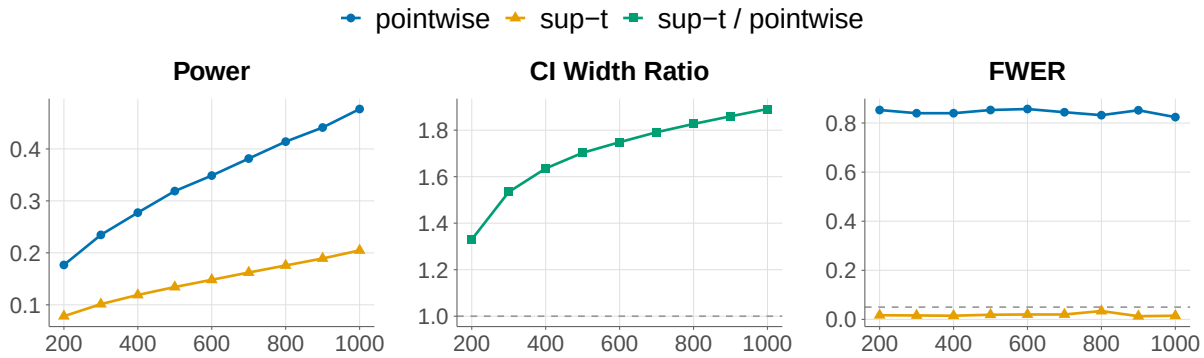


Figure 2: Monte Carlo performance (bootstrapped VAR IRFs, baseline DGP, $H = 20$): 95% pointwise bands versus sup- t bands controlling FWER at 5%.

3 False Discovery Control for Impulse Responses

The scanning workflow rests on two guarantees that its economic narratives implicitly require. First, the responses a researcher flags as significant should contain a controlled share of false rejections to be reliable. Second, the confidence intervals used to interpret them should remain valid. This section develops a two-step procedure that delivers both guarantees at a user-chosen level q . In the first step, an FDR-controlling selection rule replaces the informal practice of reading significance off pointwise bands: applied to the full vector of studentized IRF statistics, it produces a formal rejection set whose expected share of false discoveries is at most q . In the second step, marginal confidence intervals are constructed for the selected coefficients at levels recalibrated to the stringency of the selection, so that the expected share of non-covering intervals among those reported is also at most q . Because both error rates are defined on the rejection set rather than the full family, the reliability standard scales with what the researcher reports: expanding the family across variables, horizons, shocks, or states leaves both guarantees intact. The

procedure requires no new resampling. It operates on the joint bootstrap draws that standard VAR and local projection workflows already generate, and changes only the rule that converts those draws into rejections and intervals. Section 3.1 formalizes the two error rates, Section 3.2 presents the procedure, Section 3.3 revisits the monetary policy application, and Section 3.4 provides Monte Carlo evidence on finite-sample error control and power.

3.1 FDR and FCR: Error Rates Among Rejections

We adopt the notation introduced in Section 2. The family of m IRF coefficients $\theta_1, \dots, \theta_m$ is tested against the null $H_j : \theta_j = 0$ for $j = 1, \dots, m$. Let R and V denote the total number of rejections and the number of false rejections, respectively.

In the impulse response scanning workflow, the economic narrative is constructed from the rejection set: the researcher scans the estimated response grid, identifies the coefficients whose confidence intervals exclude zero, and draws substantive conclusions from the pattern of selected responses. The reliability of the resulting narrative does not depend on whether a false rejection occurs anywhere in the family, but on the proportion of false rejections among the selected responses. The appropriate error criterion is therefore defined on the rejections rather than on the full family of tested coefficients.

The false discovery proportion and false discovery rate (FDR) measure the share of false rejections among those declared significant,

$$\text{FDP} = \begin{cases} V/R, & R > 0, \\ 0, & R = 0, \end{cases} \quad \text{and} \quad \text{FDR} = \mathbb{E}[\text{FDP}]. \quad (4)$$

A procedure that controls FDR at level q guarantees that, among the responses declared significantly different from zero, the expected share of false rejections is at most q . Heuristically, a rejection set of twenty coefficients at a 10% FDR level contains at most two false discoveries on average.⁴

Analogously, to assess whether the confidence intervals attached to rejected coefficients are trustworthy, let V^{CI} count the number of rejected coefficients whose confidence interval fails to cover its true value. The false coverage proportion and false

⁴Let m_0 denote the number of true nulls in the IRF array, i.e., $\theta_j = 0$. In practice, methods implementing FDR control satisfy the asymptotic behavior $\text{FDR} \leq \frac{m_0}{m}q \leq q$, including the case where $m_0 = m$. Adaptive procedures exist to tighten these bounds and estimate m_0/m (Storey, 2002), but these lie outside the scope of this work.

coverage rate (FCR) are

$$\text{FCP} = \begin{cases} V^{\text{CI}}/R, & R > 0, \\ 0, & R = 0, \end{cases} \quad \text{and} \quad \text{FCR} = \mathbb{E}[\text{FCP}]. \quad (5)$$

FCR control at level q ensures that, among the confidence intervals attached to rejected coefficients, the expected share of non-covering intervals is at most q . Heuristically, of twenty intervals constructed at a 10% FCR level, at least eighteen cover their true parameter values on average. As with the FDR, this is an expectation over repeated samples rather than a statement conditional on the realized number of reported intervals (Benjamini and Yekutieli, 2005).

For a fixed, pre-specified IRF family that is evaluated as a whole, pointwise $(1 - \alpha_{\text{pt}})$ confidence intervals imply $\max_{j \in \{1, \dots, m\}} \mathbb{E}[\mathbf{1}\{\theta_j \notin CI_j^{\text{pt}}(\alpha_{\text{pt}})\}] \leq \alpha_{\text{pt}}$, regardless of dependence across coefficients. However, researchers select and emphasize only responses whose pointwise intervals exclude zero. Unadjusted marginal pointwise coverage does not control the expected noncoverage fraction among the intervals in that selected set, unlike FCR-controlled intervals.⁵ The definitions of FDR and FCR depend on the composition of the rejection set rather than directly on the total number of tested coefficients m . This property is what allows FDR and FCR control to scale as the response family grows. A researcher who examines six variables over fifty horizons faces the same proportional reliability standard for their rejections as one who examines two variables over twenty-one horizons. By contrast, standard FWER control becomes strictly more conservative as m increases, because it insures against any false rejection anywhere in the family.

Formally, $\text{FDR} \leq \Pr(V > 0) = \text{FWER}$, with equality under the global null when $V = R$. Consequently, a nominal FWER guarantee can uniformly imply q -FDR control only when $\alpha_{\text{FWER}} \leq q$; choosing $\alpha_{\text{FWER}} > q$ provides no such guarantee and, for an exact procedure, violates it under the global null. Any larger calibration would have to depend on the unknown number of true nulls, the dependence structure, and the signal configuration, while still retaining the maximum-statistic penalty that generally increases with effective dimension. The same target mismatch applies to intervals: simultaneous bands control the probability of any noncoverage in the full family, whereas FCR controls the expected noncoverage share among the intervals actually reported. Thus, a larger-

⁵Under the global null, any pointwise interval selected for excluding zero inherently fails to cover. Thus, the selected-set noncoverage fraction is exactly one whenever at least one interval is chosen, even though the expected noncoverage fraction over the full grid remains at most α_{pt} . Dependence can affect the dispersion of both realized fractions, but not the distinction between the two inferential targets.

α_{FWER} simultaneous band is an arbitrary relaxation of a different error criterion, not a substitute for FDR/FCR-controlled inference.⁶

The goal of the procedure developed below is to control both FDR and FCR at a user-specified level q . Initially, one determines which coefficients to declare significantly different from zero using an FDR-controlling rule. Afterward, one constructs confidence intervals for the rejected coefficients with coverage calibrated to guarantee FCR control at the same level.

3.2 Procedure

The procedure has two steps. First, an FDR-controlling decision rule is applied to the full family of m hypotheses to determine which coefficients to declare significantly different from zero, producing a rejection set $\mathcal{R}$ where $R = |\mathcal{R}|$. Second, confidence intervals are constructed for the rejected coefficients at levels calibrated to the selection rule. This subsection describes these steps in turn: (i) the FDR-controlling rejection rule and (ii) the FCR-adjusted confidence intervals. The procedure is stated independently of the IRF estimator; estimator-specific bootstrap details are discussed at the end of this subsection.

Both steps require a point estimate $\hat{\theta}_j$ and standard error $\hat{\sigma}_j$ for each of the m coefficients, together with a joint bootstrap that produces B draws of the full IRF vector, $\hat{\theta}^{*(b)} = (\hat{\theta}_1^{*(b)}, \dots, \hat{\theta}_m^{*(b)})^\top$ for $b = 1, \dots, B$. Drawing the full vector jointly in each bootstrap replication is essential: it preserves the cross-coefficient dependence structure so that the procedure can exploit the fact that many of the m tests are near-redundant rather than treating them as independent sources of multiplicity. To put all coefficients on a comparable scale, each is summarized by a two-sided studentized statistic $T_j = |\hat{\theta}_j|/\hat{\sigma}_j$. When the bootstrap delivers draws on the coefficient scale, the corresponding bootstrap statistic is $T_j^{*(b)} = |\hat{\theta}_j^{*(b)} - \hat{\theta}_j|/\hat{\sigma}_j$. A bootstrap p-value for each hypothesis is

$$\hat{p}_j = \frac{1 + \sum_{b=1}^B \mathbf{1}\{T_j^{*(b)} \geq T_j\}}{B + 1}. \quad (6)$$

Step 1: FDR-controlled selection via RSW. The first step determines which coefficients to declare significantly different from zero. We adopt the bootstrap step-down procedure of [Romano et al. \(2008\)](#) (RSW), which controls FDR under general dependence by

⁶Hendry's discussion provides an important antecedent for treating the significance level as a design choice and for questioning whether the probability of any false inclusion is informative for a selection problem ([Hendry, 1995, 2009](#)). It does not, however, make a larger FWER level equivalent to FDR or FCR control: raising α_{FWER} relaxes the tolerated probability of any error, whereas FDR and FCR govern the expected error shares among the findings and intervals actually reported.

calibrating critical values directly from the joint bootstrap distribution of the test statistic vector. Simpler alternatives exist: the [Benjamini and Hochberg \(1995\)](#) (BH) and [Benjamini and Yekutieli \(2001\)](#) (BY) step-up procedures operate only on marginal p-values and require no joint bootstrap. However, BH's FDR guarantee requires independence or positive regression dependence on the subset of true nulls (PRDS), a condition that IRF test statistics do not generically satisfy because coefficients load on common estimated objects with mixed-sign dependence across variables and horizons. BY controls FDR under general dependence but imposes a harmonic penalty that substantially reduces the number of rejections when m is large. RSW avoids both limitations by using the joint bootstrap distribution to incorporate the actual dependence structure, making it the principled default for IRF applications.⁷

The procedure begins by ordering hypotheses from least to most significant, $T_{(1)} \leq T_{(2)} \leq \dots \leq T_{(m)}$, with π denoting the permutation that maps the order-statistic index to the original hypothesis index, so that $T_{(j)} = T_{\pi(j)}$. The bootstrap statistics are reordered using the same permutation, $T_{(j)}^{*(b)} = T_{\pi(j)}^{*(b)}$, to preserve the dependence structure in the relevant ordering. For any bootstrap draw, let $T_{(r):t}^{*(b)}$ denote the r -th smallest element of $\{T_{(1)}^{*(b)}, \dots, T_{(t)}^{*(b)}\}$, and let $\hat{P}^*(\cdot)$ denote the empirical bootstrap probability operator,

$$\hat{P}^*(A) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{A \text{ holds in draw } b\}. \quad (7)$$

The procedure computes a sequence of critical values $c_1, \dots, c_m$ by recursion, working from the least significant hypothesis ($j = 1$) upward. At each step j , the critical value c_j is chosen as the smallest threshold such that the expected false discovery proportion among the j most significant hypotheses, evaluated under the bootstrap distribution, does not exceed q . Formally,

$$c_j = \inf \left\{ c \in \mathbb{R} : \sum_{r=m-j+1}^m \frac{r-m+j}{r} \hat{P}^* \left(\{T_{(j):j}^* \geq c\} \cap \bigcap_{\ell=m-r+1}^{j-1} \{T_{(\ell):j}^* \geq c_\ell\} \cap \{T_{(m-r):j}^* < c_{m-r}\} \right) \leq q \right\}. \quad (8)$$

The final set in the intersection is interpreted as automatically satisfied when its subscript is zero. The event inside the bootstrap probability corresponds to a particular configuration of how many hypotheses among the first j would be rejected by the step-

⁷BH orders marginal p-values and compares them to the linear critical line $\hat{p}_{(k)} \leq (k/m)q$; BY replaces q with q/c_m where $c_m = \sum_{\ell=1}^m 1/\ell$. Appendix B formalizes the PRDS violation for VAR and LP configurations and constructs explicit counterexamples. Section 5 compares the three rules from a practical standpoint.

down rule in the bootstrap draw. The step-down aspect is what gives the procedure its power advantage over single-step methods. Once a hypothesis is “cleared” (not rejected) at an earlier step, the procedure conditions on that outcome and recalibrates the remaining critical values, progressively sharpening the thresholds for the more significant hypotheses. The critical values are computed sequentially from $j = 1$ up to $j = m$ and are enforced to be weakly increasing to ensure a coherent step-down decision rule.

Given the ordered statistics and critical values, the rejection set is a tail set in the ordering:

$$j_0 = \min \left\{ j \in \{1, \dots, m+1\} : T_{(r)} \geq c_r \text{ for all } r = j, \dots, m \right\}, \quad (9)$$

$$\mathcal{R} = \{ \pi(r) : r = j_0, \dots, m \}. \quad (10)$$

All hypotheses indexed by $\mathcal{R}$ are rejected; the remainder are not. By construction, the procedure targets the guarantee that the expected fraction of false rejections among all rejections is at most q .

Step 2: FCR-controlled confidence intervals. The second step equips the coefficients selected in Step 1 with confidence intervals — no interval is constructed for coefficients outside $\mathcal{R}$. Reporting ordinary marginal intervals for the selected coefficients would reintroduce the selection distortion documented in Section 3.1. The coefficients in $\mathcal{R}$ were singled out precisely because their test statistics are large, and marginal intervals that ignore this fact undercover on the selected set. The remedy of [Benjamini and Yekutieli \(2005\)](#) keeps the intervals marginal but recalibrates their nominal level to reflect how stringent the selection was. In its simplest form, the recalibration multiplies the miscoverage level q by the number of selected coefficients and divides it by the size of the family scanned, so that each interval is constructed at level qR/m rather than q . The adjusted level thus interpolates between two familiar benchmarks. If only one coefficient survives ($R = 1$), selection was severe and the level q/m coincides with the Bonferroni correction; if all m coefficients survive ($R = m$), no selection took place and the unadjusted marginal level q applies. [Benjamini and Yekutieli](#) show that this recalibration controls the FCR at level q and that it is not unduly conservative: under independence, BH selection paired with these intervals yields an FCR between $q/2$ and q .

The general construction refines the level qR/m so that it remains valid for any selection rule. Write $T = (T_1, \dots, T_m)$ for the vector of studentized statistics and let $S(T) = \mathcal{R}$ denote the selection rule of Step 1. For each selected coefficient j , hold the remaining statistics T_{-j} fixed and ask how small the selected set could have been while

still containing j :

$$R_{\min,j} = \min_{t_j: j \in S(t_j, T_{-j})} |S(t_j, T_{-j})|. \quad (11)$$

The quantity $R_{\min,j}$ is the smallest selected-set size compatible with retaining coefficient j , and the FCR-adjusted level for that coefficient is $\alpha_j^* = qR_{\min,j}/m$. A selection rule is *simple* when this refinement is unnecessary: holding T_{-j} fixed, the size of the selected set takes a single value over all t_j at which j remains selected. In that case $R_{\min,j} = R$ for every selected coefficient, and the adjustment reduces to the level qR/m above. The BH and BY step-up rules are simple. The RSW step-down rule is not simple in general, because clearing a hypothesis at an earlier step recalibrates the critical values applied to the remaining ones. Its intervals therefore use the coefficient-specific levels $qR_{\min,j}/m$ directly.⁸

The interval level is thus matched to the selection rule of Step 1 and to the dependence conditions under which that rule operates:

$$\alpha_j^* = \begin{cases} qR/m, & \text{BH selection,} \\ qR/(mc_m), & \text{BY selection,} \\ qR_{\min,j}/m, & \text{RSW selection,} \end{cases} \quad (12)$$

where $c_m = \sum_{\ell=1}^m \ell^{-1}$ is the same harmonic constant that penalizes the BY testing rule, required here because the BY route places no assumption on the dependence structure. For BH and RSW selection, $1 \leq R_{\min,j} \leq R \leq m$ implies that the adjusted level lies between the Bonferroni level q/m and the unadjusted level q ; a smaller $R_{\min,j}$ signals more stringent selection for coefficient j and produces a wider interval. For BY selection, the harmonic factor reduces the level further below qR/m . A pointwise interval, by contrast, uses the fixed level α_{pt} for every coefficient, regardless of m or of the selection outcome.

Our empirical applications implement RSW selection with the coefficient-specific adjustment $qR_{\min,j}/m$. The Monte Carlo exercises instead use the simple-rule level qR/m , because computing $R_{\min,j}$ would require rerunning the step-down algorithm for every selected coefficient in every replication. This substitution is validated directly rather than covered by the general non-simple-rule guarantee: realized FCR remains at or below q

⁸The terminology follows Remark 1 of [Benjamini and Yekutieli \(2005\)](#). Under independence of the components of T , intervals constructed at level $qR_{\min,j}/m$ control the FCR at q (their Theorem 1). Under arbitrary dependence, control obtains after the level is further divided by the harmonic constant c_m (their Theorem 4). Theorems C.3 and C.4 in Appendix C.3 extend both results to intervals that are only asymptotically valid marginally, as delivered by the bootstrap.

throughout the reported simulations, and diagnostic checks find $R_{\min,j} = R$ for the vast majority of selected coefficients.

Given the adjusted levels, the intervals themselves are ordinary marginal percentile- t bootstrap intervals evaluated at level α_j^* . Let $t_j^{*(b)} = (\hat{\theta}_j^{*(b)} - \hat{\theta}_j) / \hat{\sigma}_j$ denote the signed bootstrap t -statistic, and let $t_{j,\tau}$ denote the τ quantile of $\{t_j^{*(b)}\}_{b=1}^B$. The FCR-adjusted percentile- t interval is

$$\text{CI}_j^{\text{FCR}}(\alpha_j^*) = \left[\hat{\theta}_j - \hat{\sigma}_j t_{j,1-\alpha_j^*/2}, \hat{\theta}_j - \hat{\sigma}_j t_{j,\alpha_j^*/2} \right]. \quad (13)$$

The interval for coefficient j is constructed if and only if $j \in \mathcal{R}$. Appendix C.3 states the formal construction and its FCR guarantees, adapted to the asymptotic marginal validity delivered by the bootstrap; Appendix A documents the qR/m simulation implementation.

Bootstrap requirements. The only requirement on the bootstrap is that it consistently estimates the joint distribution of the full vector of studentized IRF statistics. In Appendix C we verify this condition for four commonly used schemes: (i) the bias-corrected i.i.d. residual bootstrap for VARs (Kilian, 1998); (ii) the residual-based moving block bootstrap for VARs (Brüggemann et al., 2016); (iii) its proxy-SVAR extension (Jentsch and Lunsford, 2022); and (iv) the VAR-based wild recursive bootstrap for local projections (Montiel Olea and Plagborg-Møller, 2021). Applied researchers can pair the FDR/FCR-controlling procedure with whichever resampling method is appropriate for their identification strategy and error structure, without modifying the rejection or inference logic. The procedure is agnostic to the IRF estimator, but the bootstrap construction determines the centering of the bootstrap test statistic. Section 5 provides further practical guidance on bootstrap choice.

3.3 Empirical Example

We return to the monetary policy application from Section 2 and use it to illustrate how FDR/FCR-adjusted reporting sits between naive pointwise inference and fully simultaneous (FWER-controlled) inference. As before, the reported response family consists of the four outcomes (cumulative BRW, IP, CPI, and EBP) over horizons $h = 0, \dots, 24$, so $m = 4 \times 25 = 100$ horizon-by-outcome coefficients are implicitly scanned when interpreting the IRF plot. Figure 3 overlays three inferential objects for this same response family: conventional 90% pointwise confidence bands (blue shading), 90% simultaneous sup- t bands controlling the FWER at 0.1 over the entire grid (tan shading),

and our dependence-robust RSW procedure controlling the FDR at $q = 0.1$ with post-selection intervals that control the FCR at the same q (red intervals).

The key visual feature of the FDR/FCR-controlling approach is that it reports uncertainty only for coefficients that survive FDR-controlled selection. Concretely, the RSW step-down algorithm is applied to the full vector of studentized IRF statistics to determine a rejection set. For each selected coefficient j , we report a marginal confidence interval computed at the FCR-adjusted level $\alpha_j^* = qR_{\min,j}/m$. These FCR-adjusted intervals are plotted as red bands (error-bar style) around the corresponding IRF point estimates. Impulse responses that are not selected simply have no red band, indicating that they are not reported as nonzero after multiplicity control.

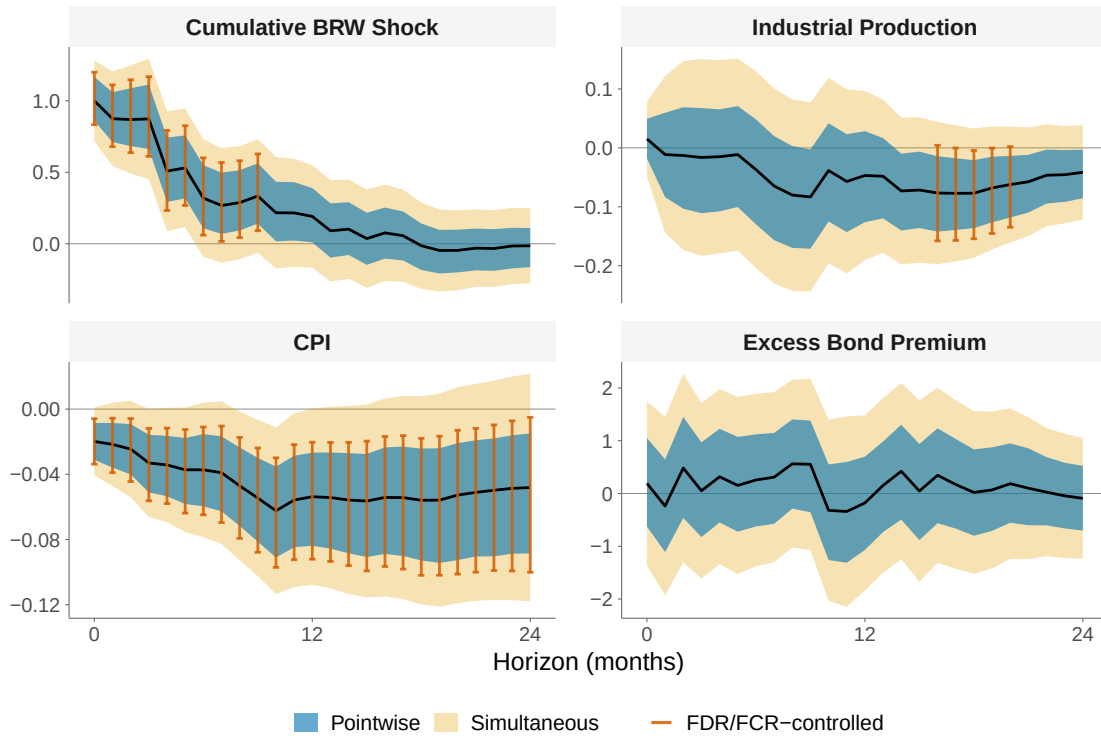


Figure 3: Responses to a contractionary monetary policy shock (VAR).

Figure 3 shows that this intermediate error criterion preserves much of the qualitative narrative suggested by pointwise bands, while avoiding the sharp loss of informativeness induced by uniform FWER control. First, the cumulative BRW response indicates that the contractionary policy stance persists for several months after impact before decaying back toward zero; the FDR/FCR selections concentrate on these early horizons, running from impact through roughly nine months, consistent with a short- to medium-lived policy tightening. Second, the CPI response displays a persistent disinflationary effect: every horizon in the reported range is selected, and the post-selection red intervals support a

sustained decline in prices relative to the baseline. The simultaneous band excludes zero for prices only over a narrow window around nine to eleven months, so the sustained disinflation is a finding of FDR control rather than of FWER control. Third, the output response shows a delayed contraction in industrial production, with selected negative coefficients between roughly sixteen and twenty months that fade again before the end of the second year. The simultaneous band detects no horizon of the output response at all, so this contraction is recovered only under FDR control. Finally, the procedure does not produce FDR-selected effects for the excess bond premium, so the method does not support a robust financial-spread response once the horizon-by-outcome multiplicity is accounted for. In total, FDR control selects about forty of the 100 coefficients, against roughly a dozen under the simultaneous band and about half under pointwise inference.

Figure 3 thus illustrates the reporting standard made available by FDR/FCR-controlled inference. Among the responses selected as significant, the expected share of false discoveries is at most $q = 0.1$. And among the corresponding confidence intervals, the expected share that fail to cover their true values is likewise at most $q = 0.1$. Relative to simultaneous inference, this alternative error criterion yields a substantially more informative selected set, retaining the delayed output contraction and sustained disinflation that the simultaneous band largely misses. It does not, however, produce any selections for the excess bond premium, for which pointwise inference is also insignificant at every horizon. The present application therefore illustrates the additional findings recovered by our proposed procedure, but not its ability to screen out findings that appear significant under pointwise inference. That screening role becomes central in Section 4, where several pointwise significance patterns do not survive FDR control. Before turning to that application, Section 3.4 evaluates the finite-sample error control and power of FDR/FCR-controlled inference in Monte Carlo designs with known true IRFs.

3.4 Monte Carlo Evidence: Error Control and Power Gains

This subsection benchmarks the proposed FDR/FCR-control procedure against pointwise inference and FWER-controlled simultaneous inference based on the sup- t approach. The comparison uses the block-diagonal simulation design introduced in Section 2. Appendix A provides full details on the design and implementation.

The baseline DGP is a six-variable VAR(4). We report responses to three shocks over horizons $h = 0, \dots, 20$. This yields $m = 6 \times 3 \times 21 = 378$ IRF coefficients in each replication. Of these coefficients, 189 are cross-block responses that are exactly zero by construction. All three methods use a nominal 5% target for their respective error

criteria. In each Monte Carlo replication, we estimate the full IRF array. For pointwise inference, we construct 95% confidence intervals and conduct the corresponding two-sided tests at $\alpha_{\text{pt}} = 0.05$. For simultaneous inference, we construct a sup- t band that controls the family-wise error rate at $\alpha_{\text{FWER}} = 0.05$ over the full IRF family. For the proposed FDR/FCR-controlling procedure, we apply the dependence-robust RSW step-down method with an FDR target of $q = 0.05$. We then construct post-selection intervals using the common nominal miscoverage level $\alpha^* = qR/m$. For computational feasibility, the Monte Carlo exercises use this simple-rule approximation rather than the coefficient-specific level $qR_{\min,j}/m$. We therefore evaluate its FCR directly rather than relying on the general result for non-simple selection rules.

We report power averaged over the true nonzero coefficients. We also report Monte Carlo estimates of FDR, FCR, and FWER. For interval width, we report the mean among reported intervals and normalize it by the pointwise benchmark. Figure 4 presents these performance measures across sample sizes for the baseline DGP estimated by VAR. Appendix A reports the complete set of results. It covers the baseline and persistent DGPs, the VAR and LP estimators, and four hypothesis-family specifications. The baseline specification uses a maximum horizon of $H = 20$ and is reported in Figures A.3–A.4. The shorter-horizon specification uses $H = 10$ and is reported in Figures A.5–A.6. The lag-misspecification results appear in Figures A.7–A.8. The results for the sparser true-null configuration appear in Figures A.9–A.10. The patterns discussed below are nearly uniform across these configurations. The only exception concerns FCR control under estimator bias, which we discuss below.

The top panels summarize the power–precision tradeoff across the three procedures. The top-left panel shows the expected power ordering. Pointwise inference has the highest power because it applies no multiplicity adjustment across the IRF family. Simultaneous sup- t inference has the lowest power because its critical value is calibrated to the maximum statistic over the full family. This calibration produces wide bands and few rejections. The FDR/FCR-control procedure lies between these two benchmarks and retains a substantial share of the power of pointwise inference. The gain relative to simultaneous inference reflects the difference between FDR and FWER control. While FWER controls the probability of any false rejection over the full family, FDR instead controls the expected proportion of false rejections among all rejections. This less stringent criterion yields a meaningful power gain in IRF families of empirically relevant size. The top-right panel shows how the different multiplicity adjustments translate into interval width. The RSW post-selection intervals are only modestly wider than pointwise intervals and remain substantially narrower than simultaneous sup- t bands. The width

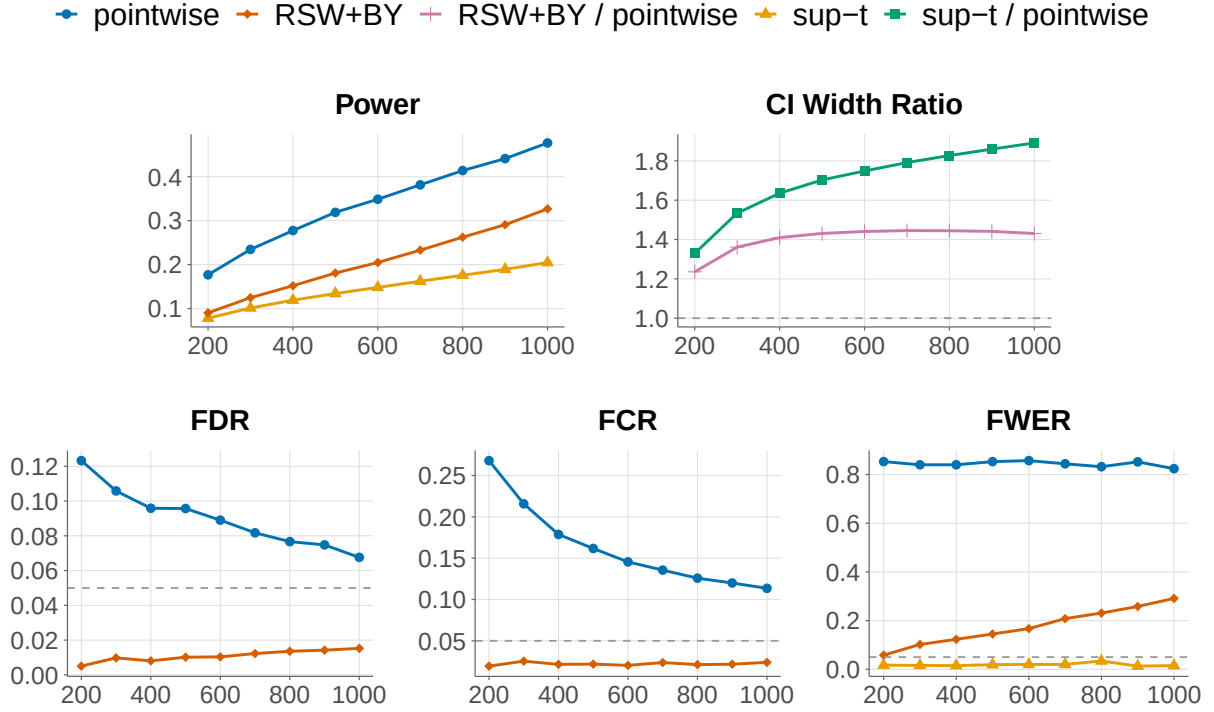


Figure 4: Monte Carlo comparison of IRF inference methods (baseline DGP, $H = 20$): 95% pointwise bands, sup- t bands controlling FWER at 5%, and RSW selection controlling FDR at $q = 0.05$ with post-selection intervals using the simple qR/m FCR adjustment. Panels report power, average interval width (relative to pointwise), and realized FDR, FCR, and FWER.

penalty from sup- t inference increases with the size of the hypothesis family. At $H = 10$ and $m = 198$, the sup- t -to-pointwise width ratio ranges from roughly 1.3 to 1.6. At $H = 20$ and $m = 378$, the ratio rises to approximately 1.6 to 2.0. This increase reflects the larger critical value required to control the maximum statistic over a larger family. By contrast, the RSW-to-pointwise width ratio remains between 1.0 and 1.3 across both horizon specifications. Figures A.5–A.3 report the corresponding comparisons.

The bottom panels compare the error-control performance of the three procedures. In the baseline design, the RSW procedure keeps both FDR and FCR close to the 5% target. Pointwise tests and intervals produce uniformly elevated error rates because their coefficientwise calibration does not account for multiplicity or selection. Across the broader set of designs, FDR and FCR respond differently to estimator misspecification.

FDR remains at or very near the target in every design we examine.⁹ This pattern

⁹The baseline procedure uses RSW for discovery selection. BH yields comparable power and empirical

extends to the persistent DGP, the LP estimator, and the lag-misspecification design in which a VAR(2) is fitted to data generated by a VAR(4). Even under the wrong lag order, the cross-block null responses remain centered near zero. The RSW selection step therefore continues to control FDR.

FCR places an additional requirement on the coverage of the reported intervals. Its performance therefore depends on the centering of the underlying IRF estimator.¹⁰ When a VAR(2) is fitted to data generated by a VAR(4), omitted dynamics bias the estimated nonzero responses. For the VAR estimator, the estimated FCR exceeds the target in the baseline DGP. It rises with sample size as the intervals contract around biased estimates. FCR also remains above the target throughout the persistent DGP. Under a correctly specified lag order, smaller exceedances appear in the persistent DGP. These deviations diminish as sample size increases and are consistent with finite-sample estimation bias. The comparison with pointwise inference supports this interpretation. Pointwise intervals centered on the same estimates have substantially higher noncoverage rates among selected coefficients. The LP estimator is less sensitive to lag-order misspecification, and its FCR declines toward the target as the sample grows. Figures A.7–A.8 report the corresponding lag-misspecification results. Taken together, the simulations distinguish the performance of the selection step from the coverage properties of the underlying estimator. RSW continues to control FDR under lag misspecification, while FCR reflects the centering of the estimator and the resulting interval coverage. Section 5 discusses the implications for lag-length choice.

Notice that the proposed procedure does not control the FWER, and the bottom-right panel shows that the probability of at least one false rejection increases with sample size. As the sample grows, the procedure gains power to detect true effects, and the rejection set expands. A larger rejection set means the FDR budget can absorb a small number of false rejections while keeping their share well below the target q . To see the arithmetic concretely, suppose that in a given replication the procedure rejects 40 of the 378 coefficients and two of those rejections are false. The replication registers a familywise error (a false rejection occurred), but its realized false discovery proportion is $2/40 = 0.05$, exactly at the target level. FWER and FDR characterize different dimensions of multiplicity. FWER is the probability of at least one false rejection anywhere in the IRF

FDR near the target across all configurations examined. BH relies on stronger dependence conditions than RSW. BY is more conservative throughout. Appendix A reports the full comparison in Figures A.11–A.12 and the corresponding figures for the remaining specifications. For applications that prioritize a simpler implementation, BH combined with the same FCR adjustment is a practical alternative.

¹⁰The general FCR result requires asymptotically valid marginal intervals under Assumption C.6. In the lag-misspecified VAR designs, nonvanishing bias in the nonzero IRF estimates means that this condition does not hold.

grid. FDR is the expected proportion of false rejections among the responses declared significant. For a researcher who builds an economic interpretation from the selected responses, FDR is more closely aligned with the reliability of the reported conclusions. It evaluates the composition of the selected set rather than the occurrence of any false rejection anywhere in the full family. This distinction is central to the paper and has direct practical consequences. The Monte Carlo results show that the proposed procedure keeps the false discovery share low and stable even as the probability of at least one false rejection approaches one.

4 Empirical Application: Oil Supply Shocks

We now apply the FDR/FCR-adjusted reporting rule to a prominent empirical IRF design. [Känzig \(2021\)](#) identifies oil supply news shocks in an external-instrument VAR, using high-frequency oil futures surprises around OPEC production announcements as the instrument. A single identified shock traced across six variables and 51 horizons generates a 306-dimensional response surface, so the implicit hypothesis family is large even though the underlying design is standard. As in [Section 3.3](#), we report three inferential summaries of the same response family. These are pointwise bands, simultaneous sup- t bands that control the FWER, and dependence-robust RSW selections with post-selection intervals that control the FDR and FCR.

We replicate [Känzig’s \(2021\)](#) baseline monthly six-variable proxy-SVAR, using the original replication data and the same reduced-form choices for ordering, lag length, and deterministic terms. We use the original replication data and proxy construction so that the empirical design remains fixed across the three inferential procedures.¹¹ Let y_t denote the 6×1 vector collecting the real oil price, world oil production, world oil inventories, world industrial production, U.S. industrial production, and the U.S. CPI, all in log levels. The VAR has $p = 12$ lags and a constant. The proxy-SVAR restriction requires the monthly oil-supply-surprise instrument z_t to be correlated with the oil supply news shock and uncorrelated with the remaining structural shocks. The key identifying feature is that a news shock raises oil prices and causes inventories to accumulate in anticipation of future scarcity, whereas a physical disruption depletes them immediately. We normalize the shock to a 10% impact increase in the real oil price, matching [Känzig \(2021\)](#).

¹¹[Kilian \(2024\)](#) raises concerns about the liquidity of WTI futures before April 1989 and about the construction of the monthly instrument from daily surprises. Removing the early observations also weakens the first stage. Holding these data and measurement choices fixed isolates the effect of multiplicity adjustment on the reported findings.

The response family is $\mathcal{F} = \{\theta_k(h) : k = 1, \dots, 6; h = 0, \dots, 50\}$ with $m = |\mathcal{F}| = 306$. Each procedure uses a 5% target for its respective error criterion. All three inferential summaries are constructed from the same draws of the [Jentsch and Lunsford \(2022\)](#) moving block bootstrap for proxy SVARs.¹² Figure 5 reports the resulting estimates over horizons $h = 0, \dots, 50$ months, with pointwise and sup- t bands drawn as shaded regions. FDR selections appear as whiskers. Each whisker is a post-selection interval, drawn only at coefficients the RSW procedure selects and constructed at the coefficient-specific FCR-adjusted level $\alpha_j^* = qR_{\min,j}/m$.

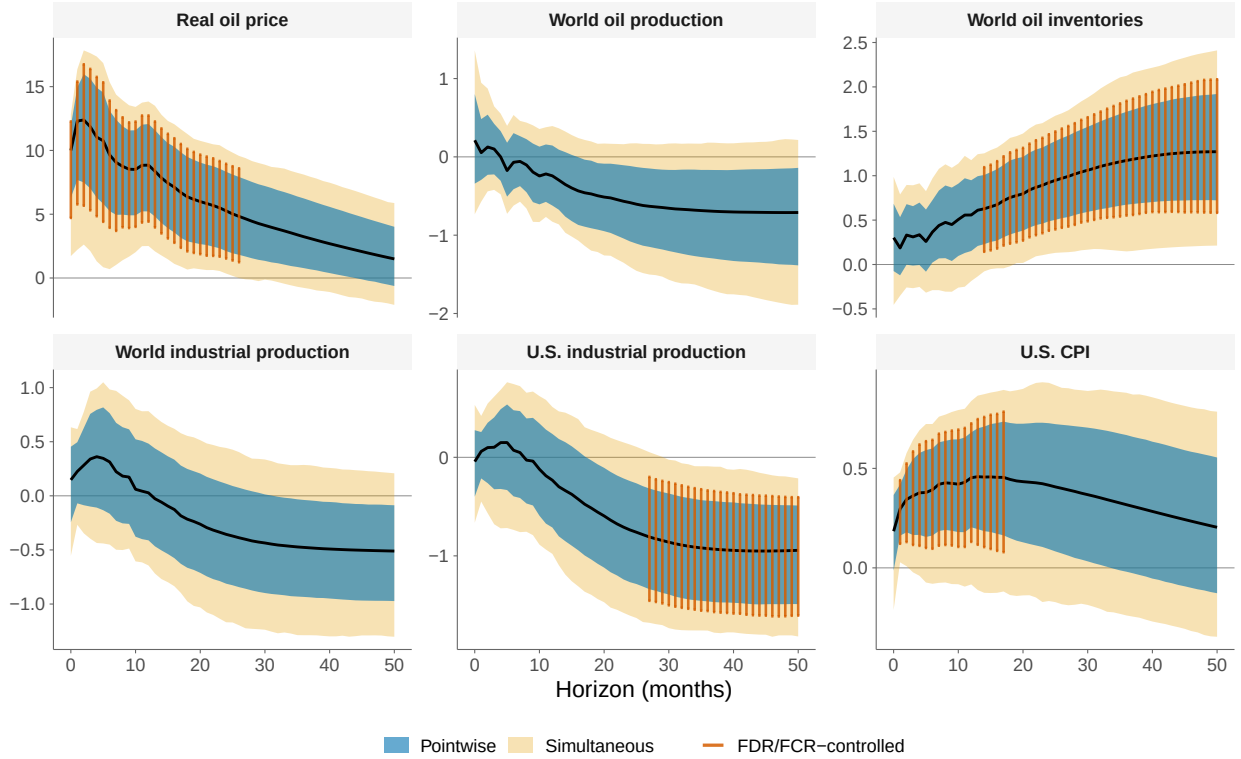


Figure 5: Impulse responses to an oil supply news shock. Pointwise bands are 95% bootstrap confidence intervals ($\alpha_{\text{pt}} = 0.05$). Simultaneous sup- t bands control the FWER at $\alpha_{\text{FWER}} = 0.05$ over the full response family. RSW whiskers mark coefficients selected under FDR control at $q = 0.05$ and display post-selection intervals constructed at the coefficient-specific FCR-adjusted level $\alpha_j^* = qR_{\min,j}/m$.

The point estimates in Figure 5 replicate the qualitative dynamics emphasized by [Känzig \(2021\)](#). The three inferential layers, however, support different subsets of those dynamics. Pointwise bands declare roughly 200 of the 306 coefficients significant, FDR

¹²The moving block bootstrap jointly resamples blocks of reduced-form residuals and the external instrument, preserving serial dependence and the temporal alignment required for valid proxy-SVAR inference.

control selects about 100, and the simultaneous band about 75. What matters for the economic narrative is not the count but which variables each layer supports, and here the three diverge sharply. The real oil price response is statistically distinguishable from zero under all three procedures at impact and over roughly the first two years. The impact result partly reflects the normalization. RSW selections extend through about horizon 25. The simultaneous band excludes zero through about horizon 23. Pointwise bands extend the finding to about horizon 40. The close agreement between RSW and simultaneous inference shows that the early oil price response is among the most robust features of the response surface.

The inventory response is where the methods begin to diverge. Pointwise bands show a broad, persistent buildup, significant from about horizon 6 onward. The simultaneous sup- t band controls the probability of at least one false rejection over the full 306-coefficient family. It excludes zero for the inventory response only from about horizon 20 onward. FDR selection recovers an earlier portion of the buildup as well. RSW selects the inventory response from roughly horizon 14 onward. It therefore retains an earlier portion of the inventory buildup than simultaneous inference. The inventory response remains selected after accounting for the full response family. This strengthens the empirical basis for the news-shock interpretation.

World oil production is the first variable where FDR selection departs from the pointwise narrative. Pointwise bands show a persistent decline in production over the medium to long run. The RSW procedure selects no horizon of the world oil production response. The persistent decline indicated by the pointwise bands is therefore not retained after multiplicity adjustment. The near-zero impact response reflects the 30-day implementation lag of OPEC production decisions and is consistent across all three inferential layers.

The multiplicity-adjusted evidence differs across the two activity measures. Pointwise bands suggest persistent declines in both world and U.S. industrial production. For world industrial production, [Känzig \(2021\)](#) emphasizes that global activity changes little during the first year and then declines persistently. The paper interprets this delayed decline as evidence that adverse general-equilibrium effects on oil-exporting countries eventually dominate their short-run gains from higher prices. The RSW procedure selects no horizon of this response. The simultaneous band also excludes zero at no horizon. The pointwise evidence for the decline is concentrated after about horizon 30, but none of these later coefficients is retained after multiplicity adjustment. Among the principal responses, the delayed decline in global activity is therefore the most sensitive to multiplicity adjustment.

The U.S. responses differ from the global activity response. U.S. industrial production is selected from roughly horizon 27 through the end of the response window. U.S. consumer prices are selected from near impact through about horizon 17. These selections retain the buildup of inflationary pass-through, although not its later persistence. Together, the two responses support a U.S. stagflationary pattern. Simultaneous inference retains the U.S. output contraction from about horizon 30 onward. The U.S. price response, however, is limited to one or two horizons immediately after impact. The joint stagflationary pattern is therefore retained under FDR control but not under FWER control.

The FDR analysis complements the specification-based robustness checks in [Känzig \(2021\)](#). It isolates how multiplicity adjustment changes the set of findings supported by the original empirical design. The oil price and inventory signatures and the U.S. stagflationary pattern remain selected. The persistent declines in world oil production and world industrial production do not. FWER-controlled simultaneous inference produces a more conservative empirical summary. It retains the oil price and inventory responses and the U.S. output contraction. It largely removes the U.S. price response along with the two global production findings. The resulting interpretation therefore differs from both the pointwise and FDR-controlled summaries. [Kilian \(2024\)](#) reaches a related conclusion through a reassessment of the futures data and proxy construction. His analysis also finds that the world oil production and world industrial production responses are not robust. Multiplicity adjustment and measurement-based reassessment therefore identify the same components of the original interpretation as fragile.

5 Practical Guidance

This section summarizes the main implementation decisions that arise when applying FDR/FCR-controlled inference to IRFs. They concern the test family, the target level q , the selection rule, the bootstrap, and the visual presentation of results.

Specifying the test family. The definition of the response family $\mathcal{F}$ is the most consequential design choice because the error-rate guarantees apply only to hypotheses included in that family. The family should be specified before the results are inspected. It should include every coefficient that may be used to support an inferential conclusion. A scan of six response variables over horizons $h = 0, \dots, 50$ implies $m = 6 \times 51 = 306$ tests. If conclusions are drawn jointly across two regimes, the pooled family contains $m = 612$ tests. Separate families are appropriate only when they correspond to distinct inferential

exercises specified in advance. Any coefficient examined for statistical significance but omitted from $\mathcal{F}$ falls outside the multiplicity adjustment.

Choosing the target level q . The target q sets the FDR level for the chosen selection rule. Through the corresponding FCR adjustment, it also determines the target expected share of non-covering intervals among those reported. Lower values of q generally produce fewer selections and wider post-selection intervals. The appropriate value therefore reflects the cost of false discoveries in the application. At $q = 0.05$, the procedure targets an expected false-discovery proportion no greater than 5% under the stated conditions. A value of $q = 0.10$ may be suitable for exploratory work that will be followed by independent confirmation. Values of $q = 0.05$ or $q = 0.01$ place greater weight on limiting false discoveries.

For RSW selection, the FCR adjustment depends on the coefficient-specific quantity $R_{\min,j}$. Smaller values of q or $R_{\min,j}$ produce wider intervals. The exact nominal level should match the dependence conditions used for the FCR guarantee. We recommend choosing a primary value of q before inspecting the selection results. Additional values may be reported as sensitivity analyses.

Choosing the selection rule. Section 3 describes the RSW, BH, and BY selection rules. Table 1 summarizes their statistical and computational tradeoffs. For empirical work, our baseline selection rule is RSW. RSW calibrates its critical values from the joint bootstrap distribution of the full impulse response coefficient family. Its formal FDR result requires the bootstrap-validity and null-exchangeability conditions stated in Appendices B and C. When these conditions are plausible and joint bootstrap draws are available, RSW provides a natural baseline for IRF applications. The RSW selection step can reuse the same joint bootstrap draws used for a sup- t band. Computing the coefficient-specific values $R_{\min,j}$ requires an additional post-selection calculation for each reported coefficient.

BH requires only marginal p -values and performs comparably to RSW in our simulations. Its standard FDR guarantee requires independence or PRDS. BY also requires only marginal p -values and controls FDR under general dependence. It produces fewer selections in our simulations. The dependence conditions for the selection rule and the post-selection interval adjustment should be reported separately.

Implementing the bootstrap. The procedure can use the same joint bootstrap draws employed to construct pointwise intervals and sup- t bands. At the level of the general

Table 1: Selection rule comparison

	RSW	BH	BY
FDR guarantee	Under general dependence	Under independence or PRDS	Under general dependence
Inputs required	Joint bootstrap draws of full test statistic vector	Marginal p -values only	Marginal p -values only
Power	High	High, comparable to RSW	Lower than RSW and BH
Computation	Requires joint bootstrap	Minimal	Minimal

theory, the central requirement is that the bootstrap consistently estimate the joint distribution of the studentized IRF vector. Assumption C.4 in Appendix C states this requirement formally. Table 2 summarizes suitable bootstrap schemes for common empirical settings.

Three implementation details are especially important. First, the number of bootstrap draws B must be large enough to approximate the relevant tails accurately. Both the RSW step-down recursion and the FCR-adjusted interval endpoints depend on tail quantiles. We use $B = 10,000$ throughout for families of a few hundred tests. For each selected coefficient j , the quantity $B\alpha_j^*/2$ gives the expected number of bootstrap draws in each tail. Small values indicate limited numerical resolution for the interval endpoints. For larger families or smaller adjusted levels, B should be increased until both the selected set and the reported intervals are stable across independent bootstrap runs.

Second, bias correction of the bootstrap DGP can improve the centering of the bootstrap distribution in small samples. Bias correction changes the bootstrap DGP but does not change the multiplicity formulas. We therefore recommend it in small-sample applications.

Third, interval coverage depends on the centering of the underlying IRF estimator. In our lag-misspecification designs, FDR remains near its target while FCR for the VAR estimator rises when omitted dynamics bias the estimated responses. Standard information criteria provide a natural starting point for selecting the lag order. Researchers should also examine nearby specifications with additional lags and report whether the selected set is stable. When lag-order uncertainty is substantial, LP estimates provide a useful robustness comparison because their coverage deteriorates less under misspecification in our experiments.

Table 2: Bootstrap schemes compatible with FDR/FCR-adjusted inference

Setting	Bootstrap	Key assumption
Recursively identified VAR, i.i.d. innovations	Bias-corrected residual bootstrap (Kilian, 1998)	Innovations are i.i.d. with finite fourth moments
VAR or SVAR with conditionally heteroskedastic or serially dependent (MDS) innovations	Moving block bootstrap (Brüggemann et al., 2016)	Innovations are MDS and the block length grows with T
Proxy SVAR (external instrument)	Moving block bootstrap for proxy SVARs (Jentsch and Lunsford, 2022)	As above, plus a relevant and exogenous instrument
Local projections	VAR-based wild recursive bootstrap (Montiel Olea and Plagborg-Møller, 2021)	DGP is well approximated by a finite-order VAR

Visualizing FDR/FCR-controlled IRFs. An FDR/FCR-controlled figure reports intervals only for selected coefficients. The point estimates should remain visible at all horizons and for all response variables. Plotting them as connected lines preserves the full dynamic profile of each response. Overlay the FDR-selected coefficients with error bars at the corresponding FCR-controlled levels. The absence of a whisker indicates that the coefficient was not selected after FDR adjustment. It does not provide evidence that the true response is zero. A selected coefficient should retain its whisker even when the associated FCR-controlled interval includes zero. Selection and post-selection interval coverage are separate components of the procedure. Pointwise and sup- t bands may be displayed in the background as visual benchmarks, as in Figures 3 and 5. At minimum, an empirical application should report the definition of $\mathcal{F}$, the family size m , the target q , the selection rule, the bootstrap scheme, the number of bootstrap draws, the number of selected coefficients, and the adjustment used for the reported intervals.

6 Conclusion

Empirical impulse response analysis often proceeds by scanning estimated impulse responses over a large grid of response variables and horizons and interpreting the coefficients that appear significant. This practice creates a multiple-testing problem. The pointwise intervals most frequently seen in the literature control only the marginal coverage for each coefficient in isolation. The less commonly used FWER-controlled simultaneous inference instead controls the probability of at least one false rejection

over the full response family. While the simultaneous approach remains appropriate when the empirical claim concerns an entire response path, neither criterion is tailored to the common empirical workflow of forming narratives based on the features of a data-selected subset of the estimated IRF grid. The FDR/FCR-control framework developed in this paper targets the reliability of that set of coefficients precisely, by controlling the expected proportion of false rejections among selected responses and the expected proportion of noncovering intervals among those reported.

This change in inferential target matters in practice. In Monte Carlo simulations, the procedure delivers substantial power gains relative to simultaneous bands while maintaining control of false discoveries and false coverage across a range of empirically relevant designs. In the monetary policy application, it confirms the qualitative transmission narrative and places it on firmer statistical footing, recovering the delayed output contraction and the full disinflationary path, neither of which the simultaneous band detects. In the oil supply news application, it retains the core oil price, inventory, and U.S. stagflationary responses. The simultaneous band supports the oil price and inventory findings and the U.S. output contraction, but not the U.S. price response, so it cannot deliver the stagflationary reading. The pointwise evidence of persistent declines in world oil production and world industrial production does not survive the adjustment. The same procedure thus confirms one pointwise narrative and revises another, in each case at a controlled error rate.

The framework leaves the underlying VAR or LP estimator unchanged. Its RSW implementation can reuse the joint bootstrap draws employed for simultaneous inference. The main change in applied practice is to define the response family explicitly. Researchers then select coefficients under FDR control and report FCR-adjusted intervals for the selected responses. The full IRF path remains visible throughout. FDR and FCR therefore align the inferential guarantee with the selected findings used to support the economic interpretation.

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Appendices

A Monte Carlo Simulations

This appendix documents the simulation design and presents the full set of Monte Carlo results summarized in Section 2. Notation and the replication workflow follow Section 2. Indexing conventions are: sample size $N \in \mathcal{N}$; replication $r \in \{1, \dots, N_{\text{rep}}\}$; bootstrap draw $b \in \{1, \dots, B\}$; horizon $h \in \{0, \dots, H\}$; variable $k \in \{1, \dots, K\}$; shock $s \in \mathcal{S}$; coefficient $j \in \{1, \dots, m\}$. Simulations were run using the Wake Forest University High Performance Computing Facility ([Information Systems and Wake Forest University, 2021](#)).

A.1 Design

Design parameters. Table A.1 collects the scalar design choices.

Table A.1: Simulation design parameters

Symbol	Value	Description
K	6	Number of endogenous variables
p	4 (DGP); $p_{\text{est}} = 2$ in misspec	VAR lag order (varies only in lag-misspecification estimation)
H	20 (baseline); also 10	Maximum IRF horizon (varies across family specifications)
S	3	Number of shocks reported/tested
$\mathcal{S}$	$\{1, \dots, S\}$	Shock index set
m	378 (baseline); also 198	Total number of scalar IRF coefficients (varies with family)
N_{rep}	1000	Monte Carlo replications
B	10000	Bootstrap draws
T_{burn}	1000	Burn-in length
α_{pt}	0.05	Pointwise test/interval level
α_{FWER}	0.05	FWER target for the sup- t band
q	0.05	FDR and FCR target for RSW reporting
$\mathcal{T}$	$\{200, 300, \dots, 1000\}$	Sample-size grid

DGP specification. Data generation follows a $\text{VAR}(p)$ recursion with burn-in. The DGP is

$$y_t = \sum_{\ell=1}^4 A_{\ell} y_{t-\ell} + u_t, \quad u_t = C \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, I_6). \quad (\text{A.1})$$

The simulation uses two DGPs, both with block-diagonal lag and impact matrices:

$$A_\ell = \begin{bmatrix} A_{\ell,1} & 0 \\ 0 & A_{\ell,2} \end{bmatrix}, \quad \ell = 1, \dots, 4, \quad C = \begin{bmatrix} C_1 & 0 \\ 0 & C_2 \end{bmatrix}, \quad (\text{A.2})$$

with two 3×3 blocks. True IRFs θ^0 are computed from the standard MA recursion $\Psi_h = \Phi_h C$ and vectorized with index map $j(k, s, h) = ((k - 1)S + (s - 1))(H + 1) + (h + 1)$, ordering k major, s middle, h fastest. The baseline DGP chooses $(A_1, \dots, A_4)$ to be stable and non-oscillatory, with companion-form spectral radius strictly below one. The persistent DGP keeps the same block structure and impact matrix C , and rescales the lag matrices so that the companion spectral radius is approximately 0.98. Cross-block responses are exactly zero by construction, providing clean ground truth for FDR evaluation. Figures A.1 and A.2 display the true IRF grids for the two DGPs.

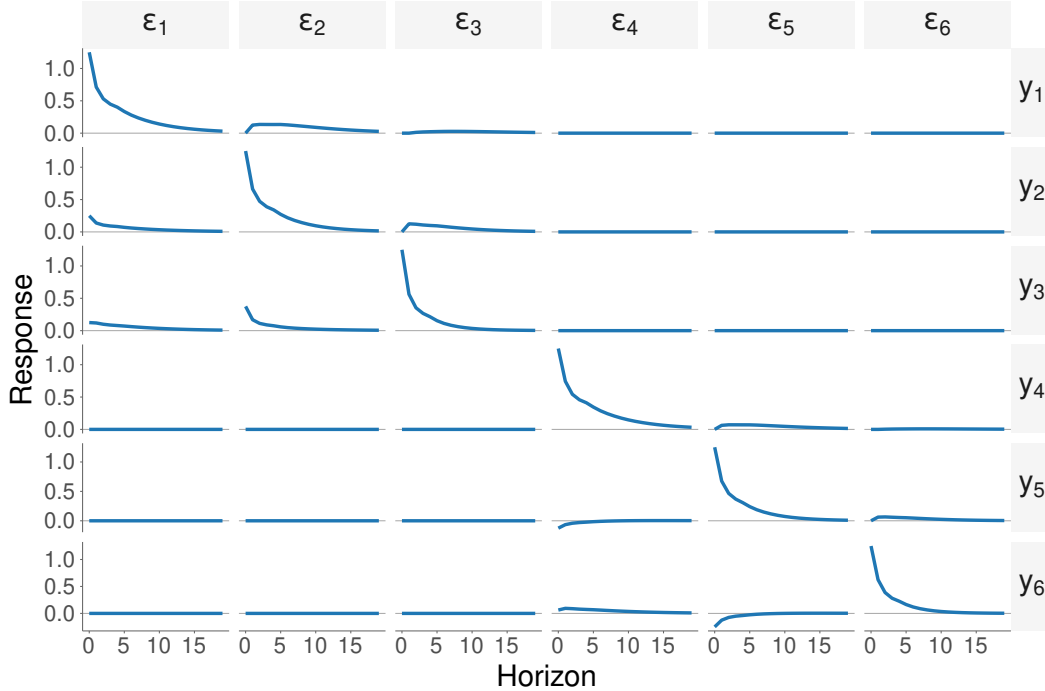


Figure A.1: True IRF grid for the baseline DGP (dgp_baseline).

Hypothesis families. For each DGP, five hypothesis-family specifications are run. In each case the coefficient index set is $j \in \{1, \dots, m\}$ with $m = KS(H + 1)$ unless a subset is specified.

- **Baseline family (H20):** horizons $h \in \{0, \dots, H\}$ with $H = 20$ (21 horizons); full coefficient set; $m = 378$. Estimator lag order $p = 4$.

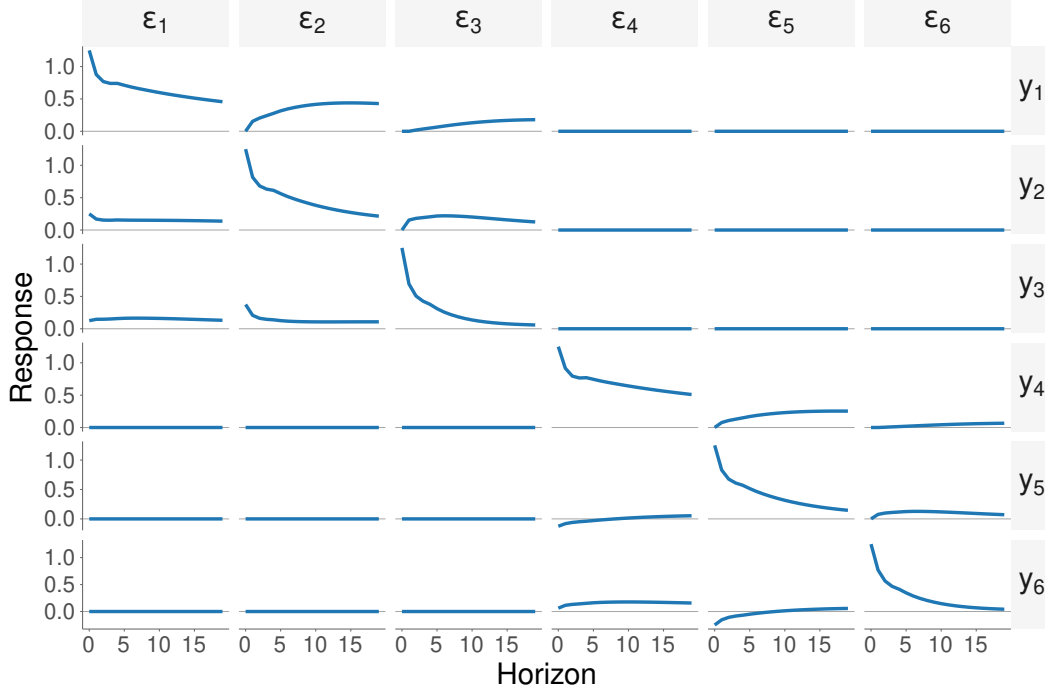


Figure A.2: True IRF grid for the persistent DGP (dgp_persistent).

- **Fewer horizons (H10):** $H = 10$ (11 horizons); full set; $m = 198$; $p = 4$.
- **Lag misspecification (misspec):** $H = 20$, full set; $m = 378$. The estimator uses $p_{\text{est}} = 2$ (VAR(2) and LP with 2 lags) while the DGP remains VAR(4).
- **True-null sparsity (63nulls):** $H = 20$; hypothesis family restricted to variables $k \in \{1, \dots, 4\}$, shocks $s \in \{1, 2, 3\}$, horizons $h \in \{0, \dots, 20\}$; $m = 4 \times 3 \times 20 = 240$. By the block-diagonal structure, 60 of these coefficients are true nulls (cross-block responses).

Estimation. VAR and LP estimation follow Section 3 (“VAR and LP implementation”). For LP, the usable sample at horizon h is $t \in \{p + 2, \dots, T - h\}$, giving $T - h - (p + 1)$ observations. For each DGP and each hypothesis family, results are produced for both the **VAR** estimator (bias-corrected residual bootstrap) and the **LP** estimator (VAR-based wild recursive bootstrap). Within each figure, the VAR panel appears on top and the LP panel on the bottom.

Bootstrap and inference details. All procedures target $H_j : \theta_j^0 = 0$ for $j = 1, \dots, m$. The general bootstrap framework follows Section 3; this subsection records estimator-specific implementation choices.

VAR bootstrap. The VAR bootstrap distribution is generated using the [Kilian \(1998\)](#) bias-corrected i.i.d. residual bootstrap. Residuals $\hat{u}_t$ ($t = p + 1, \dots, T$) are resampled i.i.d. with replacement, and bootstrap pseudo-data are generated recursively from bias-corrected coefficients $(\tilde{A}_\ell, \tilde{c})$ with initial conditions $y_{1:p}^{*(b)} = y_{1:p}$:

$$y_t^{*(b)} = \tilde{c} + \sum_{\ell=1}^p \tilde{A}_\ell y_{t-\ell}^{*(b)} + \hat{u}_{t-p}^{*(b)}, \quad t = p + 1, \dots, T. \quad (\text{A.3})$$

Bias correction uses an internal residual bootstrap of size $B_{\text{bias}} = B = 10000$:

$$A_\ell^{\text{bc}} = 2\hat{A}_\ell - \frac{1}{B_{\text{bias}}} \sum_{b=1}^{B_{\text{bias}}} \hat{A}_\ell^{*(b)}, \quad c^{\text{bc}} = 2\hat{c} - \frac{1}{B_{\text{bias}}} \sum_{b=1}^{B_{\text{bias}}} \hat{c}^{*(b)}. \quad (\text{A.4})$$

Stability is enforced by requiring $\rho(\text{companion}(A^{\text{bc}})) < 0.999$; if this fails, the correction is halved repeatedly (up to 25 steps), falling back to uncorrected OLS if necessary.

The bootstrap standard error $\hat{\sigma}_j = \text{sd}_b(\theta_j^{*(b)})$ is floored at 10^{-12} , and bootstrap- t statistics are

$$T_j^{\text{obs}} = \hat{\theta}_j / \hat{\sigma}_j, \quad T_j^{*(b)} = (\theta_j^{*(b)} - \hat{\theta}_j) / \hat{\sigma}_j. \quad (\text{A.5})$$

LP bootstrap. LP inference uses the VAR-based wild recursive bootstrap of [Montiel Olea and Plagborg-Møller \(2021\)](#). A VAR(p) is fit to the data by OLS, and the VAR coefficients are bias-corrected using the [Pope \(1990\)](#) formula, following the same procedure described in the VAR bootstrap paragraph above. Let $(\tilde{A}_\ell, \tilde{c})$ denote the bias-corrected coefficients. For each bootstrap replication b , wild bootstrap residuals are constructed as $\hat{u}_t^{*(b)} = U_t^{(b)} \hat{u}_t$, where $U_t^{(b)} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$ independently of the data. A block of p initial observations $(y_1^{*(b)}, \dots, y_p^{*(b)})$ is drawn uniformly at random from the $T - p + 1$ overlapping blocks of p consecutive observations in the original sample. Bootstrap pseudo-data are then generated recursively:

$$y_t^{*(b)} = \tilde{c} + \sum_{\ell=1}^p \tilde{A}_\ell y_{t-\ell}^{*(b)} + \hat{u}_{t-p}^{*(b)}, \quad t = p + 1, \dots, T, \quad (\text{A.6})$$

with structural shocks $\varepsilon_t^{*(b)} = C^{-1} \hat{u}_{t-p}^{*(b)}$. LP regressions are then re-estimated on each bootstrap sample. Following [Montiel Olea and Plagborg-Møller \(2021\)](#), bootstrap- t

statistics are centered at the VAR-implied IRF rather than the LP point estimate:

$$T_j^{*(b)} = \frac{\hat{\theta}_j^{*(b)} - \hat{\theta}_j^{\text{VAR}}}{\hat{\sigma}_j}, \quad (\text{A.7})$$

where $\hat{\sigma}_j = B^{-1} \sum_{b=1}^B \hat{\sigma}_j^{*(b)}$ averages the replication-level standard errors across draws, and the recentered draws $\theta_j^{*(b)} = \hat{\theta}_j + (\hat{\theta}_j^{*(b)} - \hat{\theta}_j^{\text{VAR}})$ yield the bias-corrected LP estimate $\hat{\theta}_j^{\text{bc}} = 2\hat{\theta}_j - B^{-1} \sum_b \theta_j^{*(b)}$.

sup- t band. For each draw, $M_j^{*(b)} = \max_j |T_j^{*(b)}|$. Let $c_{1-\alpha_{\text{FWER}}}$ denote the empirical $(1 - \alpha_{\text{FWER}})$ quantile of $\{M_j^{*(b)}\}_{b=1}^B$. The simultaneous band is $\text{Band}_j^{\text{sup-}t} = [\hat{\theta}_j - c_{1-\alpha_{\text{FWER}}} \hat{\sigma}_j, \hat{\theta}_j + c_{1-\alpha_{\text{FWER}}} \hat{\sigma}_j]$ with $\hat{\sigma}_j$ corresponding to the respective VAR or LP estimate.

RSW step-down and post-selection intervals. Bootstrap standard errors $\hat{\sigma}_j^{\text{RSW}}$ (floored at 10^{-12}) yield absolute t -statistics $T_j^{\text{obs}} = |\hat{\theta}_j|/\hat{\sigma}_j^{\text{RSW}}$ and bootstrap counterparts $T_j^{*(b)} = |\theta_j^{*(b)} - \hat{\theta}_j|/\hat{\sigma}_j^{\text{RSW}}$, ordered as $T_{(1)}^{\text{obs}} \leq \dots \leq T_{(m)}^{\text{obs}}$.

Prefix-maximum step-down construction. For each $j = 1, \dots, m$, define

$$M_j^{(b)} = \max_{1 \leq t \leq j} T_{(t)}^{*(b)}. \quad (\text{A.8})$$

For $j \geq 2$, set

$$t_{\text{fail}}^{(b)}(j) = \max \left\{ t \in \{1, \dots, j-1\} : T_{(t)}^{*(b)} < c_t \right\}, \quad (\text{A.9})$$

with convention $t_{\text{fail}}^{(b)}(j) = 0$ if the set is empty, then

$$k^{(b)}(j) = j - t_{\text{fail}}^{(b)}(j), \quad w^{(b)}(j) = \frac{k^{(b)}(j)}{m - j + k^{(b)}(j)}. \quad (\text{A.10})$$

The objective used to select each critical value is

$$g_j(c) = \frac{1}{B} \sum_{b=1}^B w^{(b)}(j) \mathbb{I}\{M_j^{(b)} \geq c\}. \quad (\text{A.11})$$

The rejection index is $j_0 = \min\{j : T_{(i)}^{\text{obs}} \geq c_i \text{ for all } i = j, \dots, m\}$, with no rejections if the set is empty.

FCR-adjusted intervals. Let $S_{\text{RSW}}(T)$ denote the RSW rejection set in a Monte Carlo

replication, and let $R = |S_{\text{RSW}}(T)|$. All RSW-based simulations use the simple-selection-rule adjustment

$$\alpha^* = qR/m, \quad (\text{A.12})$$

rather than recomputing the coefficient-specific $qR_{\min,j}/m$ level required by the general construction for a non-simple selection rule. This substitution makes the Monte Carlo feasible, because computing $R_{\min,j}$ would require repeatedly rerunning the RSW rule for every selected coefficient in every replication. Diagnostic calculations show that $R_{\min,j} = R$ for the vast majority of simulated selected coefficients, so the two levels usually coincide; the reported results also verify directly that FCR remains at or below q in the correctly specified baseline-DGP designs (Section A.2 documents the persistent-DGP and lag-misspecification cases, in which estimator bias raises noncoverage under every interval construction). Reported intervals for selected coefficients are percentile- t intervals,

$$\text{CI}_j^{\text{FCR}}(\alpha^*) = \left[\hat{\theta}_j - \hat{\sigma}_j Q_{1-\alpha^*/2}(t_j^{*(b)}), \hat{\theta}_j + \hat{\sigma}_j Q_{\alpha^*/2}(t_j^{*(b)}) \right], \quad \text{if } \hat{r}_j = 1, \quad (\text{A.13})$$

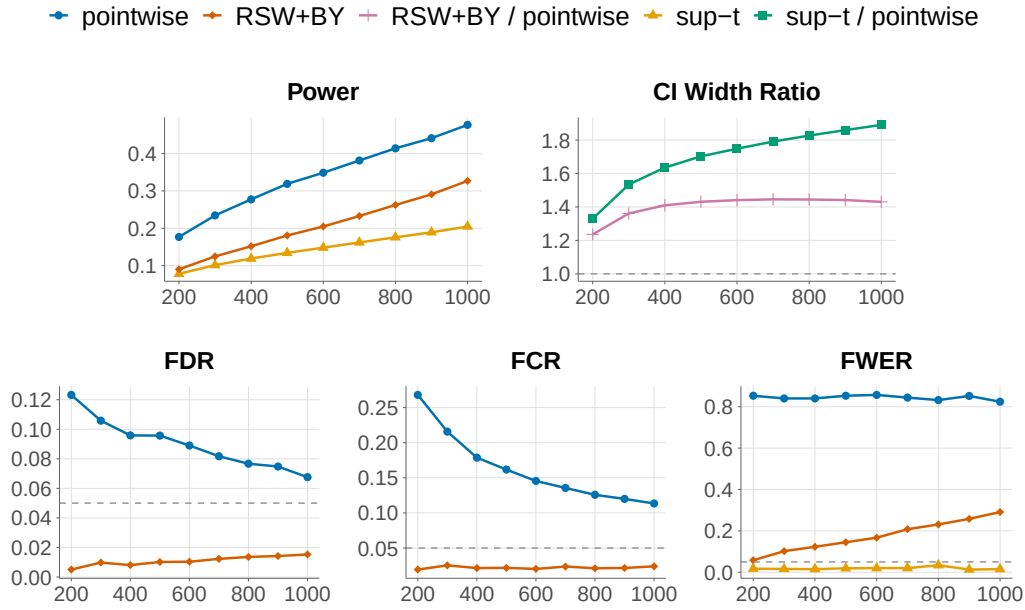
where $t_j^{*(b)} = (\theta_j^{*(b)} - \hat{\theta}_j)/\hat{\sigma}_j$ is the signed bootstrap t -statistic and $Q_\tau(\cdot)$ denotes the empirical τ quantile across draws b .

Performance criteria. Metrics are computed at the replication level and averaged across N_{rep} replications (with Monte Carlo standard errors). A coefficient is classified as a true null if $|\theta_j^0| \leq 10^{-12}$; write $\mathcal{H}_0$ and $\mathcal{H}_1$ for the null and non-null index sets. For a method with rejection indicators $\hat{r}_j$ and intervals $\mathcal{I}_j = [L_j, U_j]$, define: discoveries $R = \sum_j \hat{r}_j$; false discoveries $V = \sum_{j \in \mathcal{H}_0} \hat{r}_j$; true positives $\text{TP} = \sum_{j \in \mathcal{H}_1} \hat{r}_j$; and interval non-coverage count $V_{\text{CI}} = \sum_j \hat{r}_j \mathbb{I}\{\theta_j^0 \notin \mathcal{I}_j\}$. The reported metrics are:

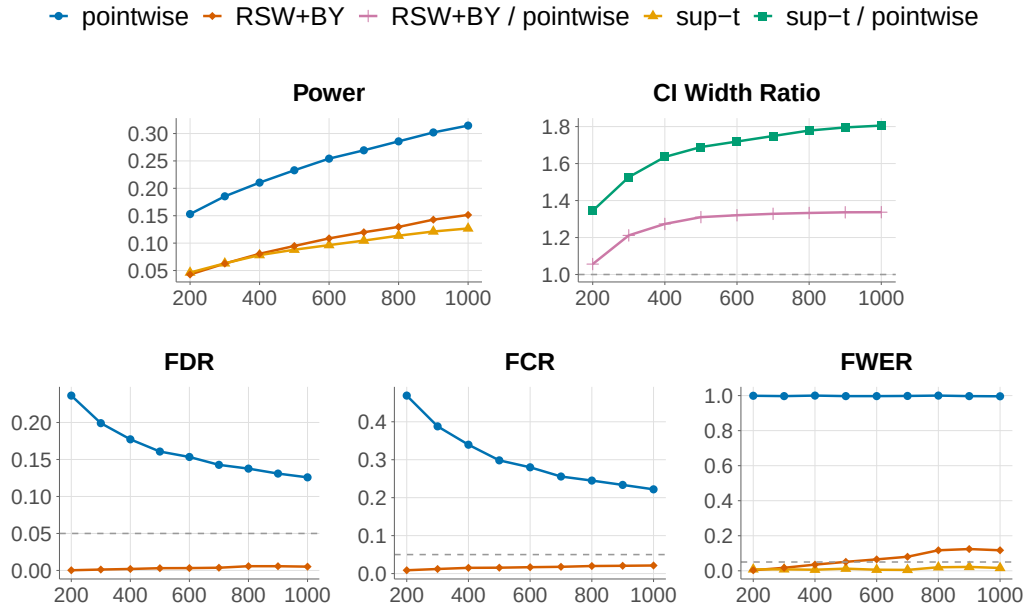
$$\begin{aligned} \text{FDP} &= V/R, & \text{FWER} &= \mathbb{I}\{V > 0\}, & \text{Power} &= \text{TP}/|\mathcal{H}_1|, \\ \text{FCP} &= V_{\text{CI}}/R, & \text{MeanWidth} &= \frac{1}{R} \sum_{j: \hat{r}_j=1} (U_j - L_j), & \text{SimulCov} &= \mathbb{I}\{\theta_j^0 \in \text{Band}_j^{\text{sup-}t} \forall j\}, \end{aligned} \quad (\text{A.14})$$

with the convention that ratios involving R are zero when $R = 0$.

A.2 Results: Pointwise vs sup- t vs RSW



(a) VAR



(b) LP

Figure A.3: Pointwise vs. sup- t vs. RSW, baseline DGP, baseline family ($H = 20$).

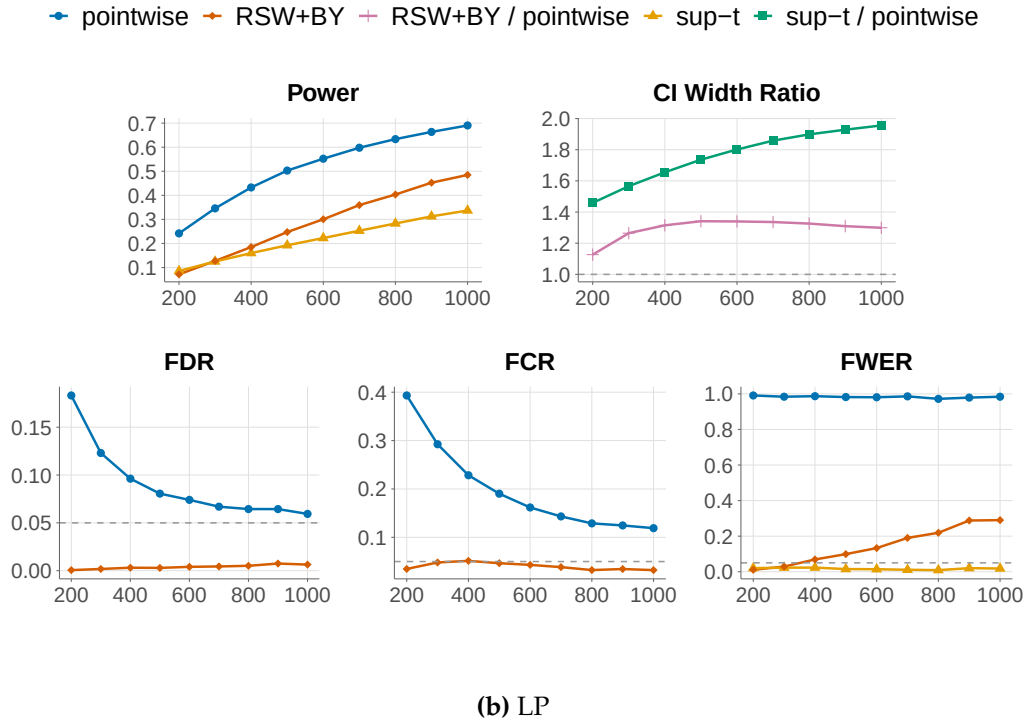
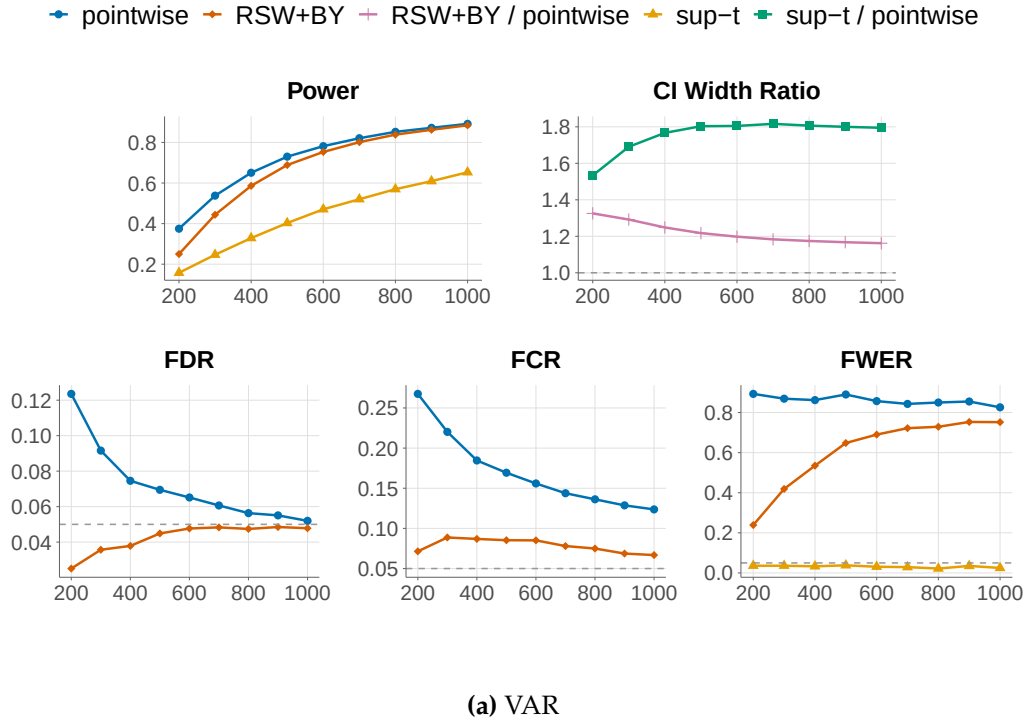


Figure A.4: Pointwise vs. sup- t vs. RSW, persistent DGP, baseline family ($H = 20$).

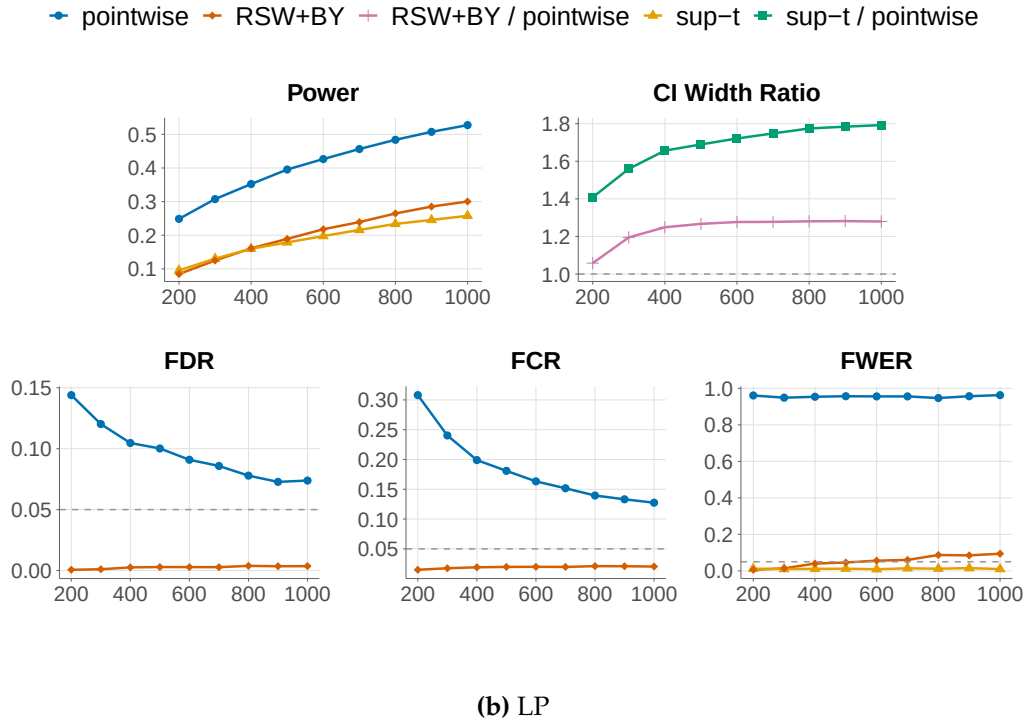
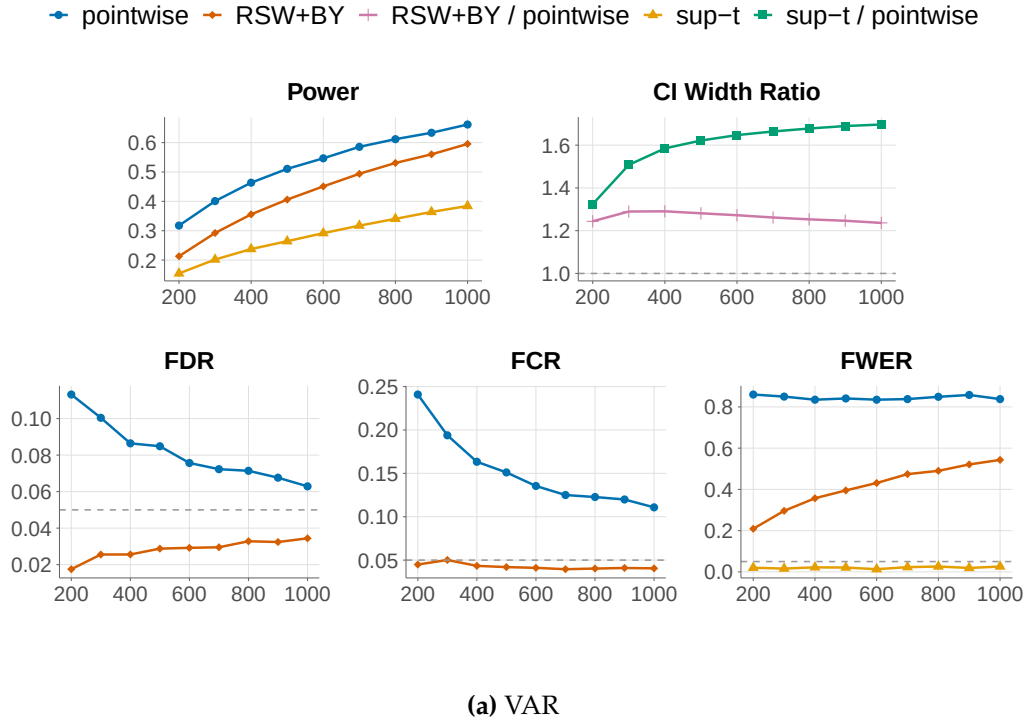


Figure A.5: Pointwise vs. sup- t vs. RSW, baseline DGP, fewer horizons ($H = 10$).

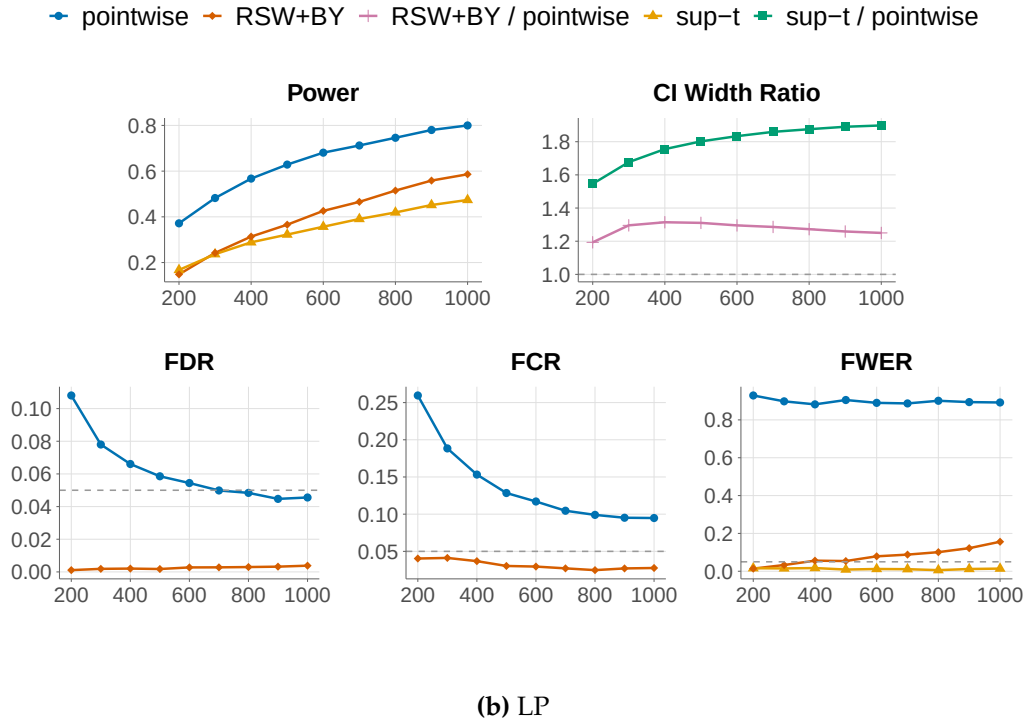
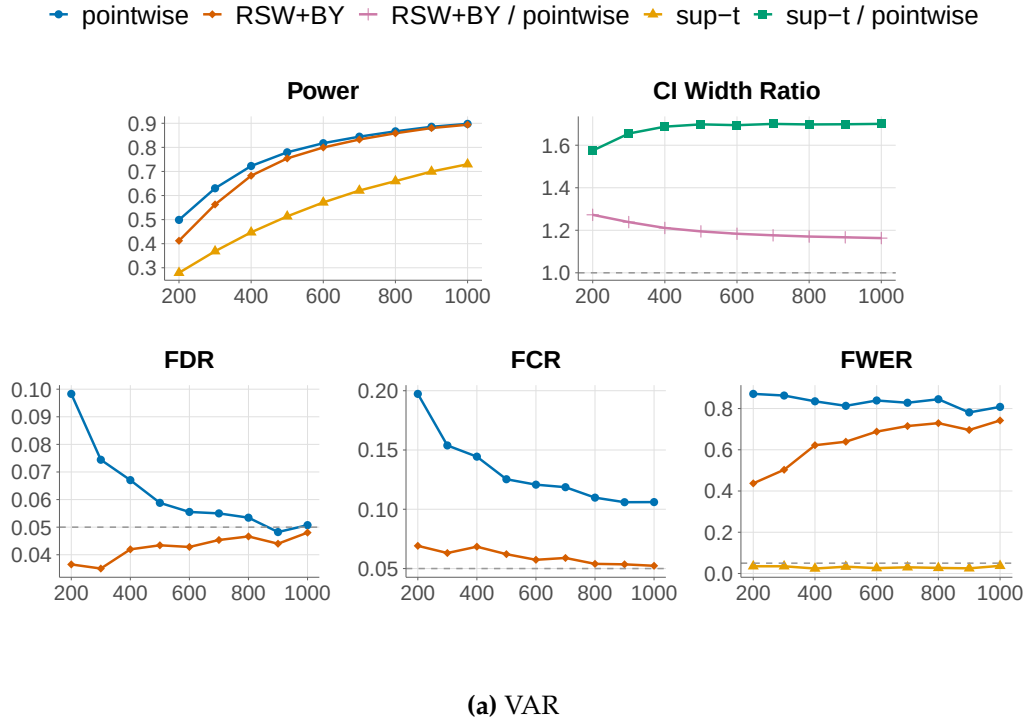


Figure A.6: Pointwise vs. sup- t vs. RSW, persistent DGP, fewer horizons ($H = 10$).

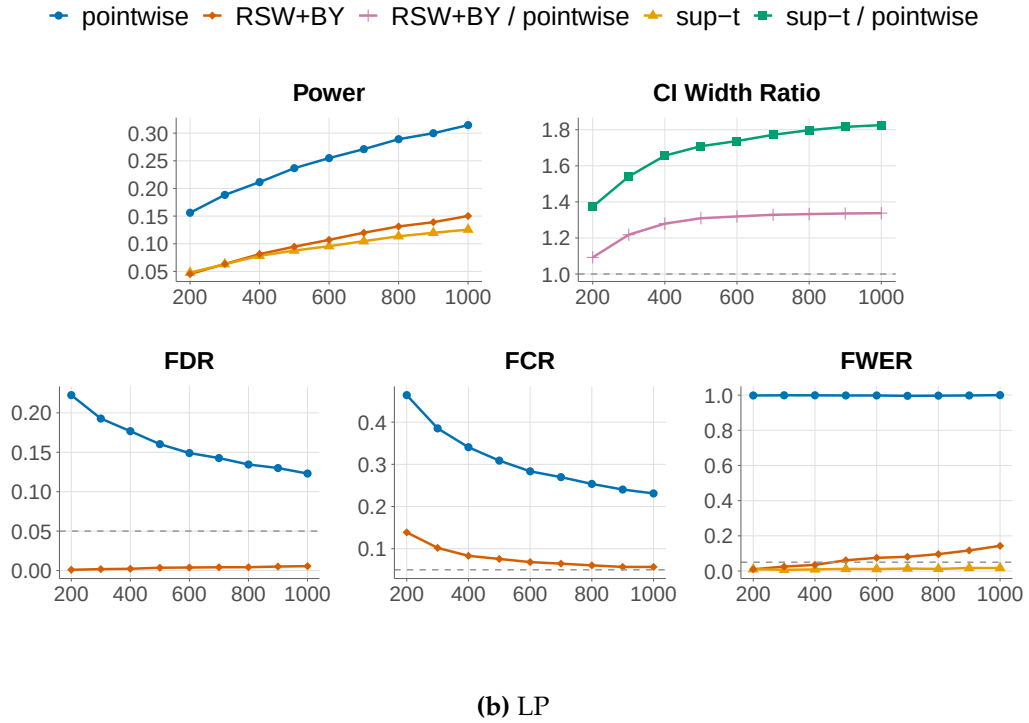
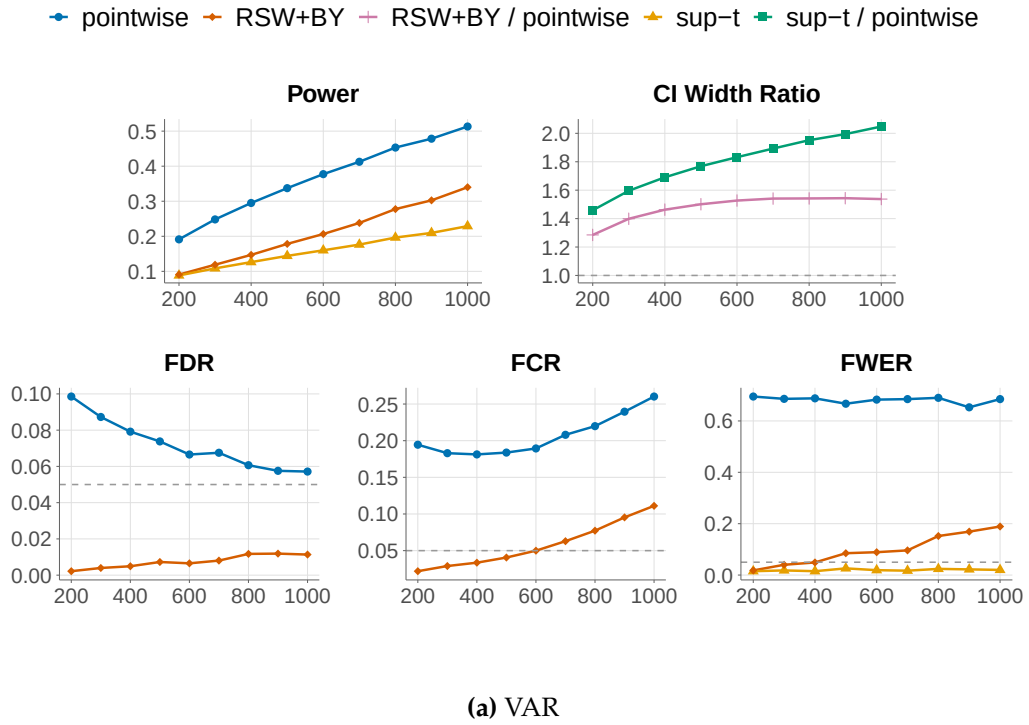


Figure A.7: Pointwise vs. sup- t vs. RSW, baseline DGP, lag misspecification.

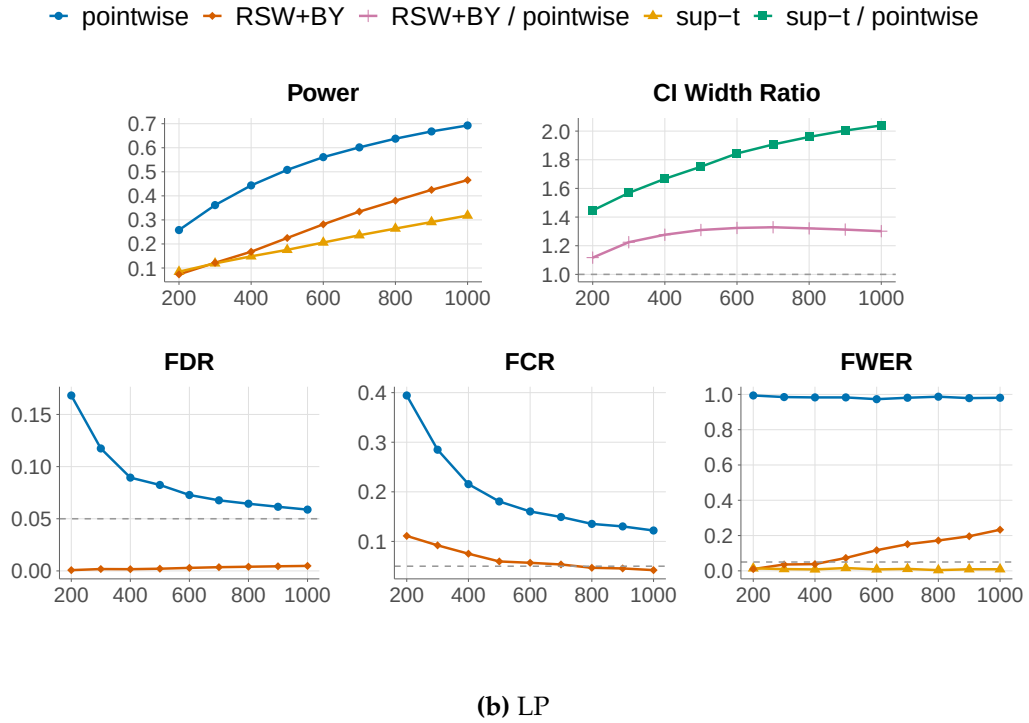
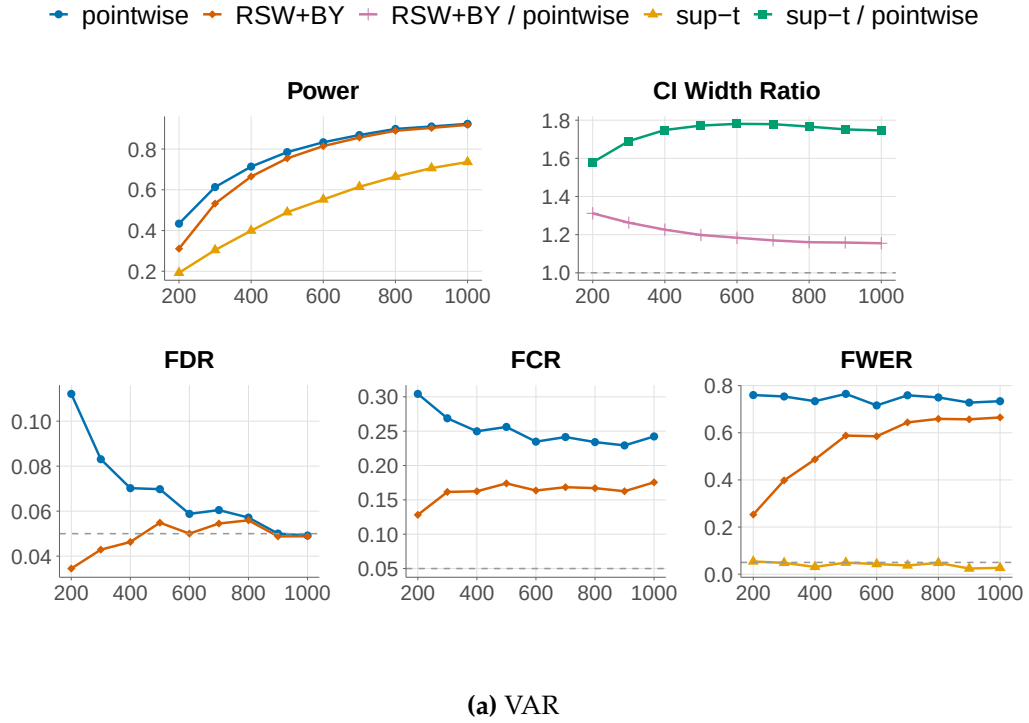


Figure A.8: Pointwise vs. sup- t vs. RSW, persistent DGP, lag misspecification.

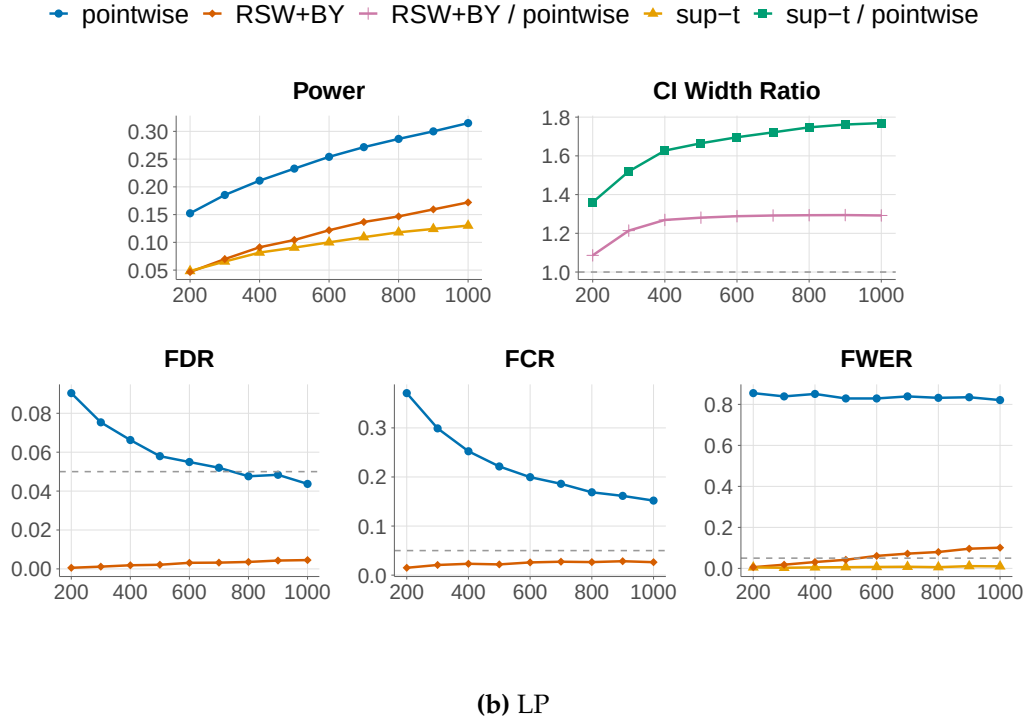
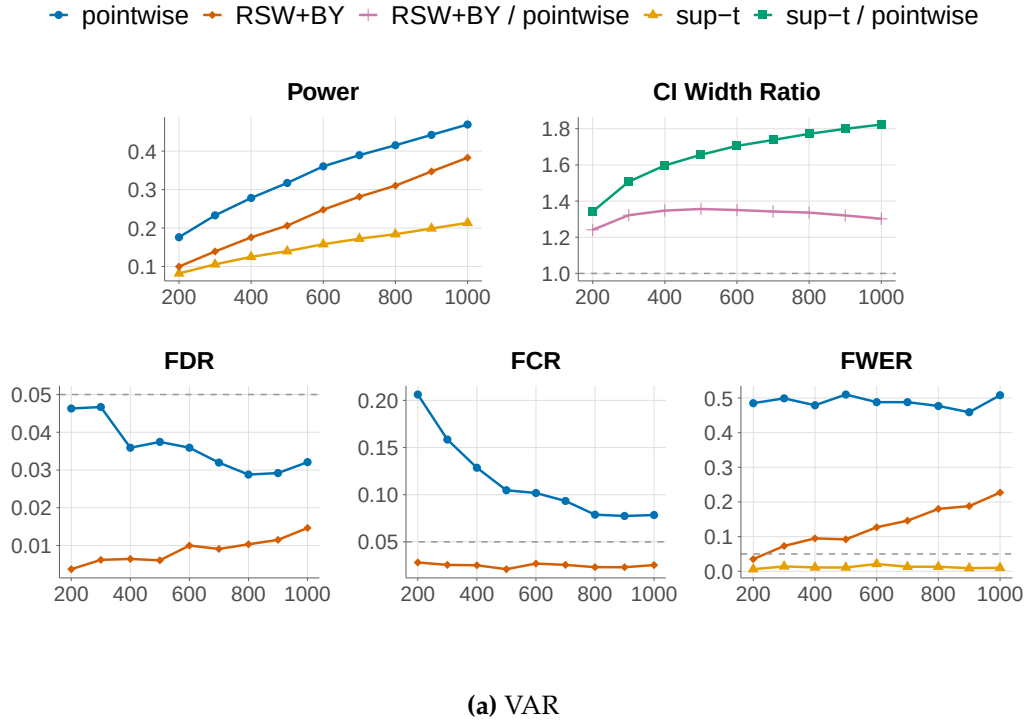


Figure A.9: Pointwise vs. sup- t vs. RSW, baseline DGP, true-null sparsity (63nulls).

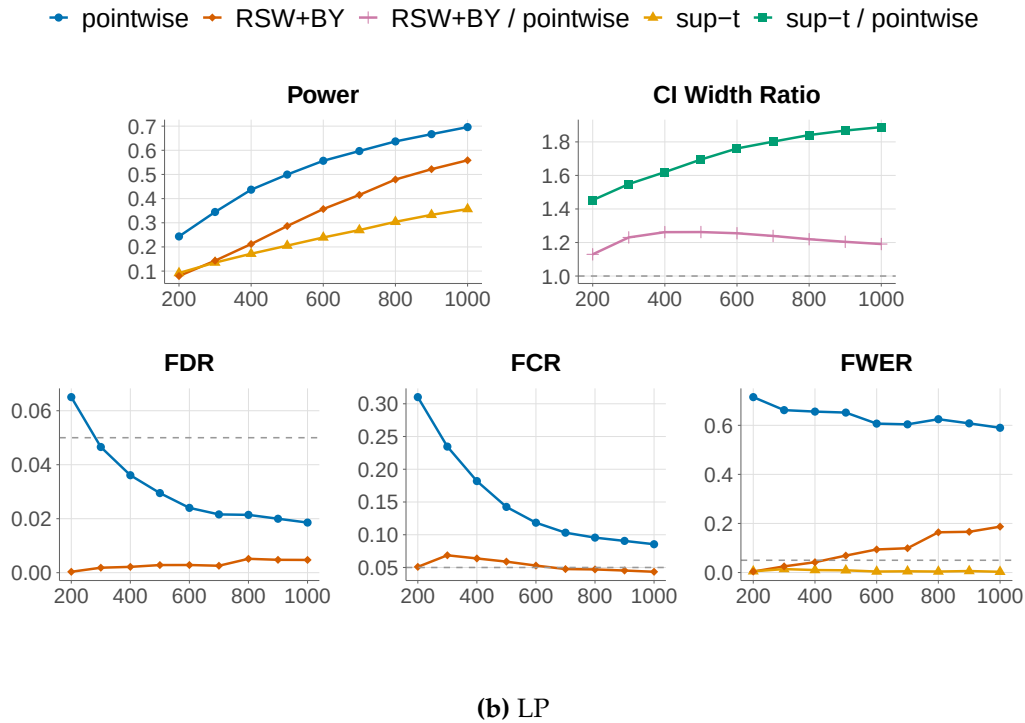
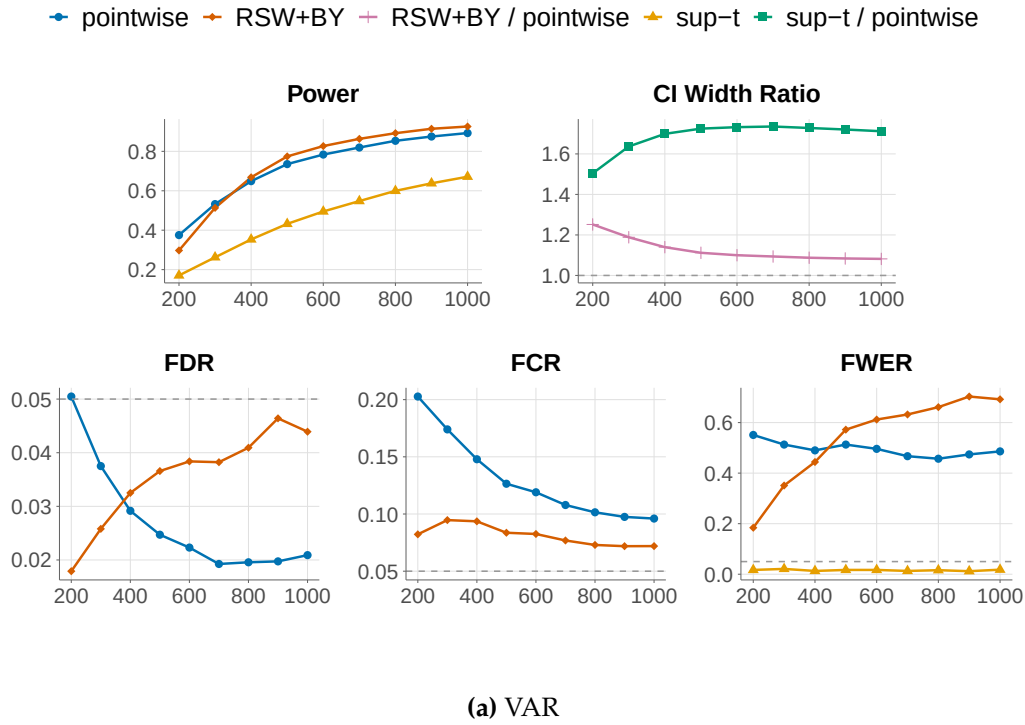


Figure A.10: Pointwise vs. sup- t vs. RSW, persistent DGP, true-null sparsity (63nulls).

A.3 Results: RSW vs BH vs BY

For these comparisons, RSW- and BH-selected intervals use qR/m , while BY-selected intervals use $qR/(mc_m)$.

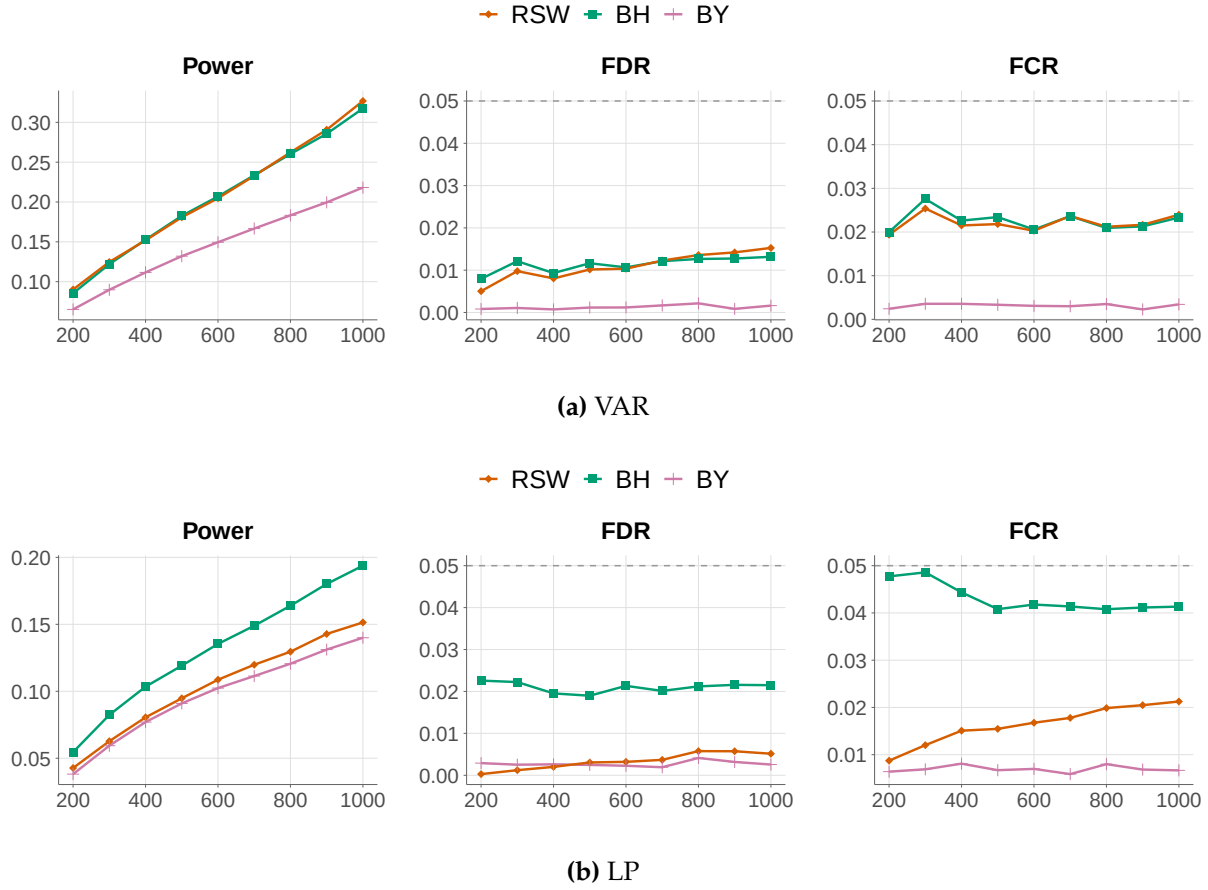
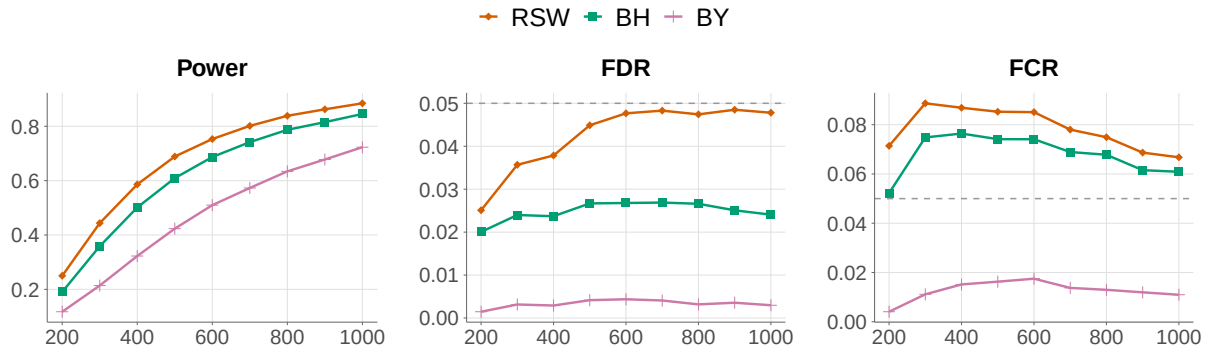
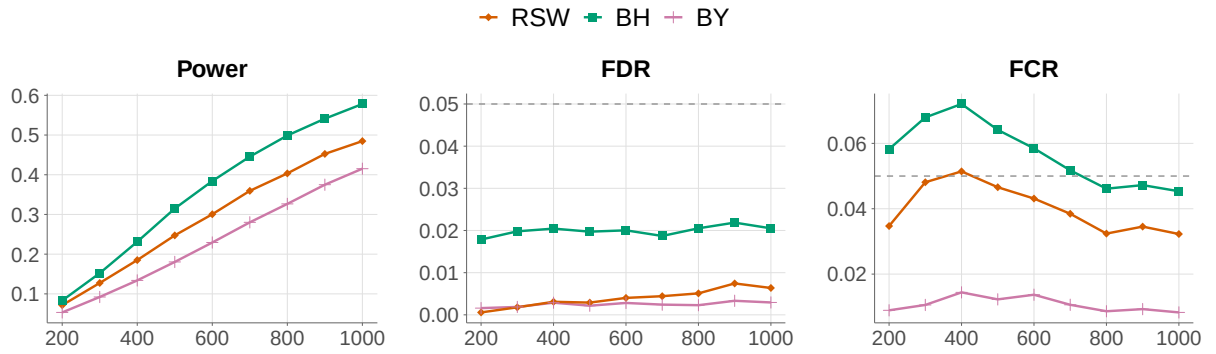


Figure A.11: RSW vs. BH vs. BY, baseline DGP, baseline family ($H = 20$).

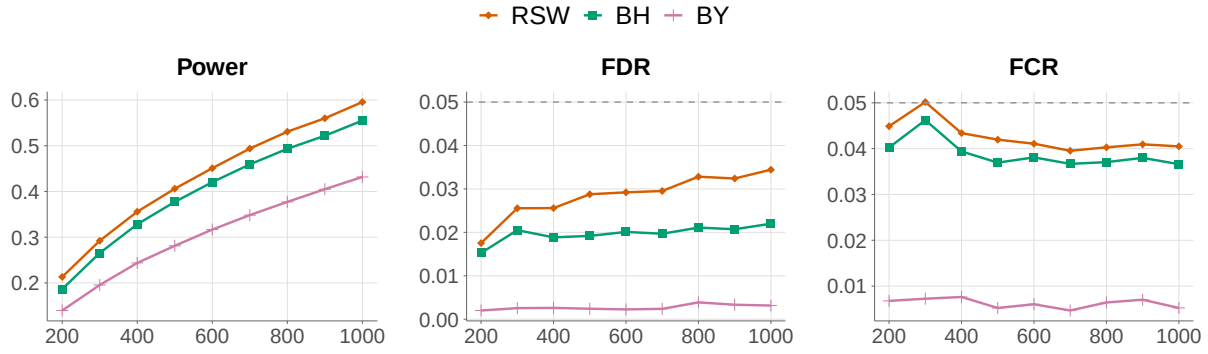


(a) VAR

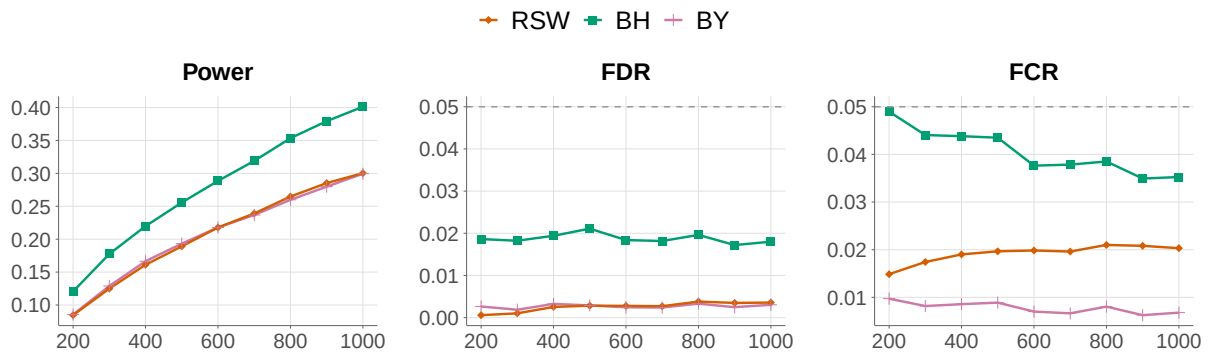


(b) LP

Figure A.12: RSW vs. BH vs. BY, persistent DGP, baseline family ($H = 20$).

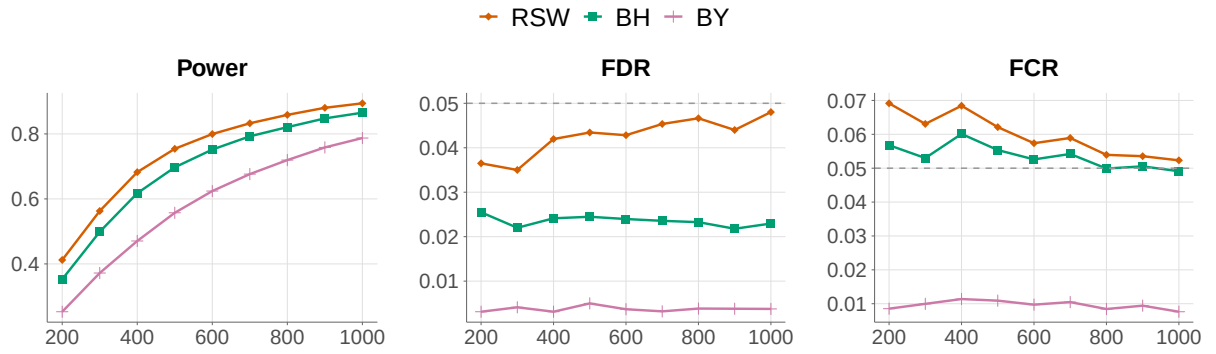


(a) VAR

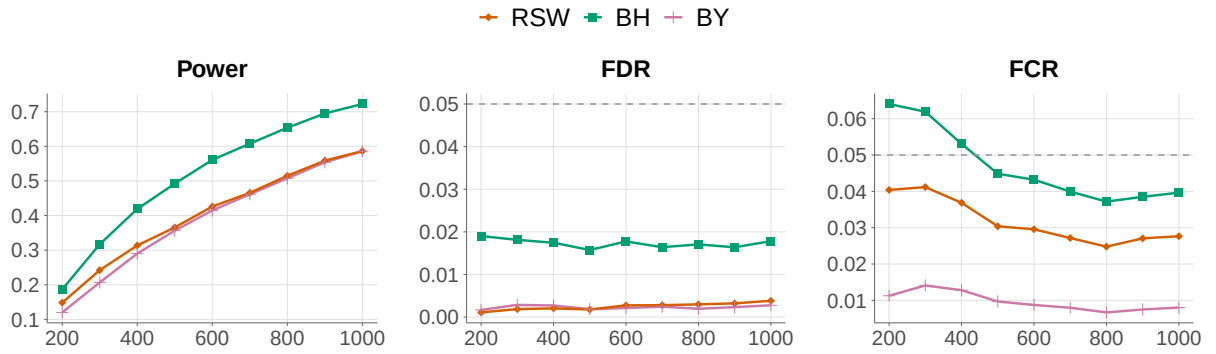


(b) LP

Figure A.13: RSW vs. BH vs. BY, baseline DGP, fewer horizons ($H = 10$).

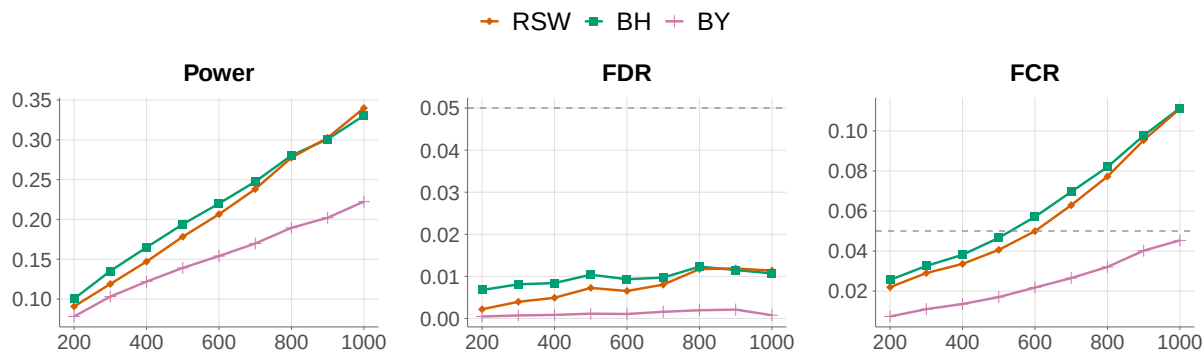


(a) VAR

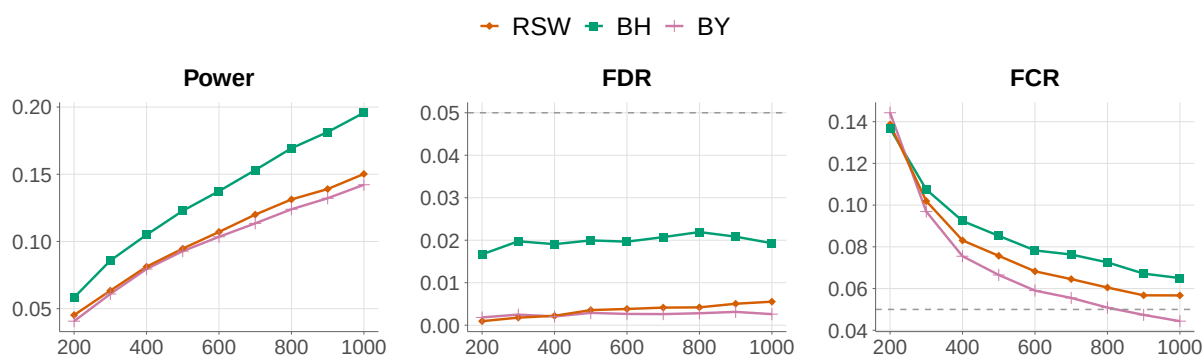


(b) LP

Figure A.14: RSW vs. BH vs. BY, persistent DGP, fewer horizons ($H = 10$).

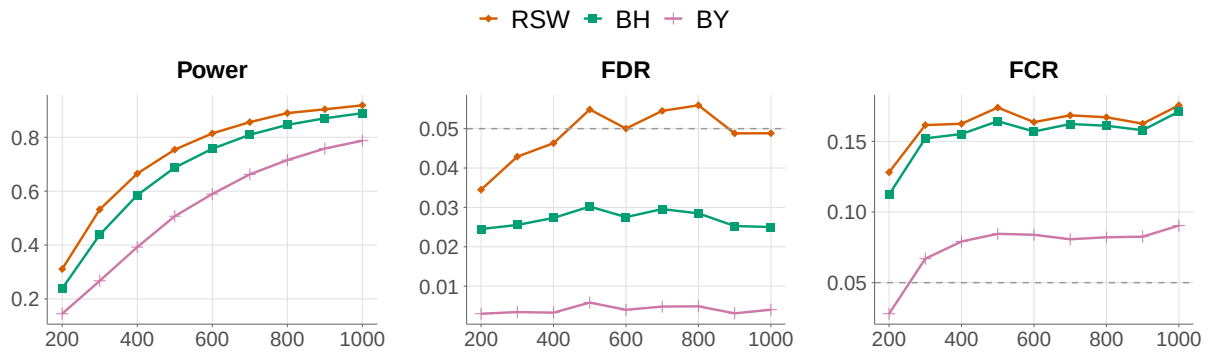


(a) VAR

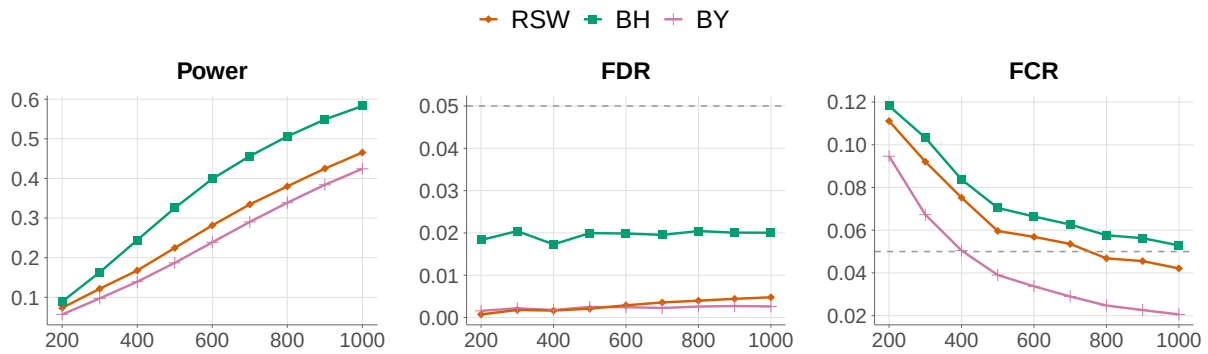


(b) LP

Figure A.15: RSW vs. BH vs. BY, baseline DGP, lag misspecification.

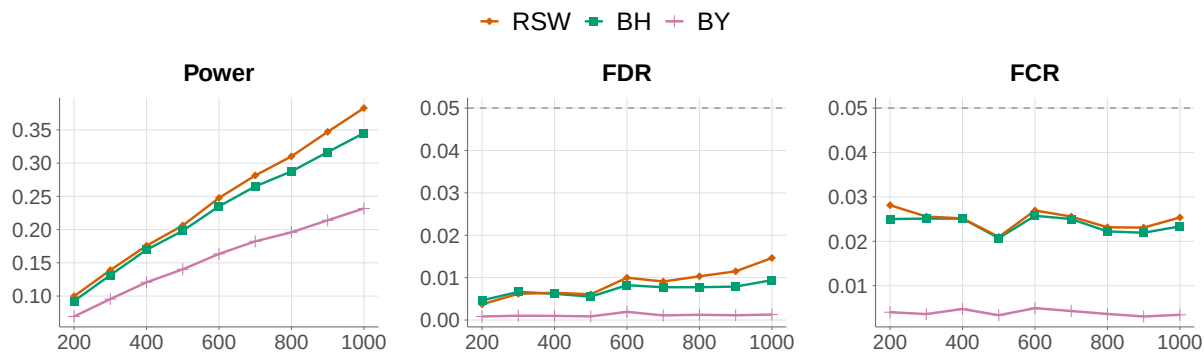


(a) VAR

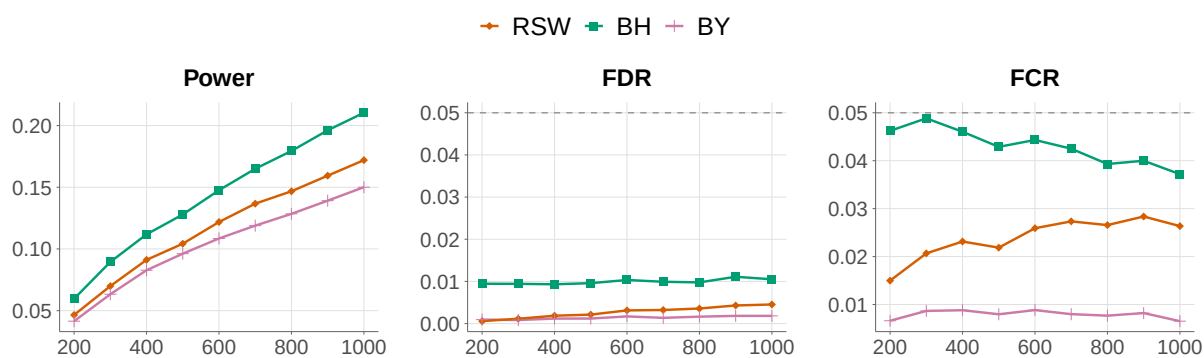


(b) LP

Figure A.16: RSW vs. BH vs. BY, persistent DGP, lag misspecification.

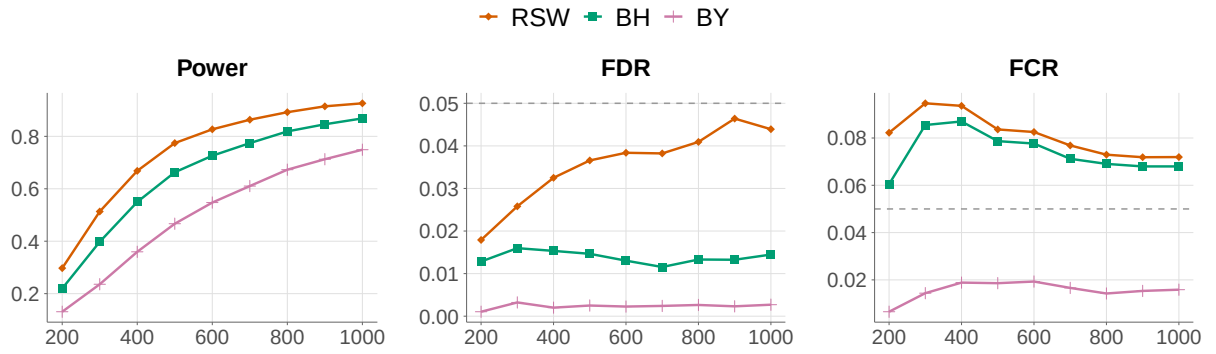


(a) VAR

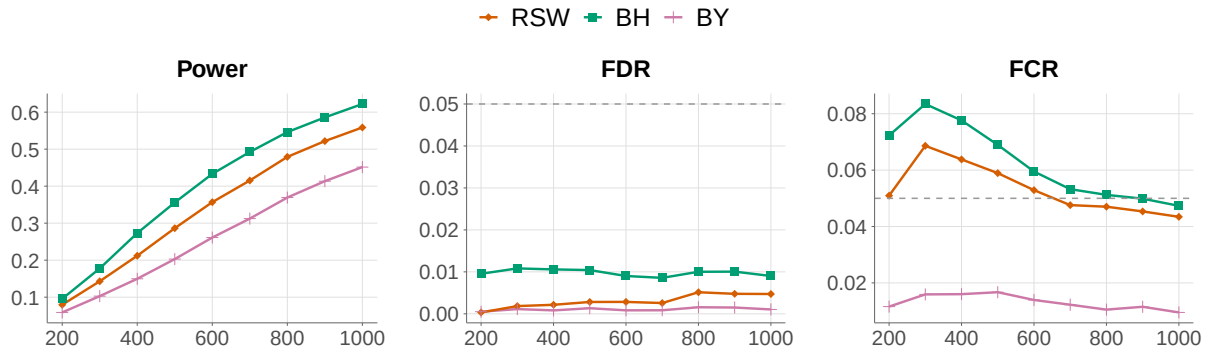


(b) LP

Figure A.17: RSW vs. BH vs. BY, baseline DGP, true-null sparsity (63nulls).



(a) VAR



(b) LP

Figure A.18: RSW vs. BH vs. BY, persistent DGP, true-null sparsity (63nulls).

B BH vs RSW: Dependence in IRF t-statistics

Impulse response function (IRF) t-statistics are typically dependent across horizons, response variables, shocks, and states. Because Benjamini–Hochberg (BH) FDR control relies on independence or on a specific positive dependence condition (PRDS) that is not generically implied by IRF dependence, BH may not be theoretically justified for IRF discovery selection. The Romano–Shaikh–Wolf (RSW) step-down procedure is designed for dependent studentized statistics, because it calibrates critical values using the empirically estimated joint distribution of IRF t-statistics via bootstrap.

B.1 Setup and notation

We observe a time series sample of length T . All probability statements are under the true data-generating process. Throughout, “large” test statistics mean “more evidence against the null.”

Let $\theta_0 \in \mathbb{R}^d$ denote the population parameter vector that is sufficient to define the IRFs of interest. Let $\hat{\theta}$ denote an estimator of θ_0 computed from the T observations. We interpret $\hat{\theta}$ broadly as a stacked estimator containing every estimated object needed to construct the reported IRF panel.

Let $\mathcal{I}$ denote the index set for response variables. Let $\mathcal{H}$ denote the index set for horizons. Let $\mathcal{J}$ denote the index set for structural shocks. Let $\mathcal{K}$ denote the index set for states or regimes. Let $a = (i, h, j, k)$ denote a generic index quadruple with $i \in \mathcal{I}$, $h \in \mathcal{H}$, $j \in \mathcal{J}$, and $k \in \mathcal{K}$. Let $\mathcal{A} = \mathcal{I} \times \mathcal{H} \times \mathcal{J} \times \mathcal{K}$ denote the full index set of reported IRF objects. Let $m = |\mathcal{A}|$ denote the number of simultaneous hypotheses.

For each $a \in \mathcal{A}$, let $\psi_a(\theta_0) \in \mathbb{R}$ denote the corresponding population IRF coefficient. In a state-invariant analysis, $|\mathcal{K}| = 1$ and the index k is redundant. Define the vector-valued IRF mapping $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^m$ by $\Psi(\theta) = (\psi_a(\theta))_{a \in \mathcal{A}}$.

Let $\hat{\psi}_a$ denote the estimator of $\psi_a(\theta_0)$ that is reported in practice. In a VAR, $\hat{\psi}_a$ is a plug-in estimator $\psi_a(\hat{\theta})$. In a local projection (LP), $\hat{\psi}_a$ is an estimated regression coefficient. For uniformity, define the stacked estimator $\hat{\Psi} = (\hat{\psi}_a)_{a \in \mathcal{A}} \in \mathbb{R}^m$. When a plug-in representation is available, $\hat{\Psi} = \Psi(\hat{\theta})$ by construction.

For each $a \in \mathcal{A}$, let $\hat{\sigma}_a$ denote a standard error estimator for $\hat{\psi}_a$. Assume $\hat{\sigma}_a > 0$ with probability tending to one. Define the studentized statistic and its absolute value by

$$T_a = \frac{\hat{\psi}_a - \psi_a(\theta_0)}{\hat{\sigma}_a}, \quad S_a = |T_a|. \quad (\text{B.1})$$

When the null is $\psi_a(\theta_0) = 0$, the statistic simplifies to $T_a = \hat{\psi}_a / \hat{\sigma}_a$. Define the associated two-sided normal-reference p-value by

$$\hat{p}_a = 2\left(1 - \Phi(S_a)\right), \quad (\text{B.2})$$

where $\Phi(\cdot)$ denotes the standard normal distribution function.

For each $a \in \mathcal{A}$, define the null hypothesis and alternative hypothesis by

$$H_{0,a} : \psi_a(\theta_0) = 0, \quad H_{1,a} : \psi_a(\theta_0) \neq 0. \quad (\text{B.3})$$

A multiple testing procedure outputs a random empirical rejection set $\mathcal{R} \subseteq \mathcal{A}$. Let $\mathcal{A}_0 \subseteq \mathcal{A}$ denote the (unknown) set of indices corresponding to true null hypotheses. Define the number of rejections and false rejections by

$$R = |\mathcal{R}|, \quad F = |\mathcal{R} \cap \mathcal{A}_0|. \quad (\text{B.4})$$

Define the false discovery proportion (FDP) and false discovery rate (FDR) by

$$\text{FDP} = \frac{F}{\max\{R, 1\}}, \quad \text{FDR} = \mathbb{E}[\text{FDP}]. \quad (\text{B.5})$$

A procedure controls the FDR at level $\alpha \in (0, 1)$ if $\text{FDR} \leq \alpha$.

B.2 Joint asymptotics and generic dependence of IRF estimators

Assume $\hat{\theta}$ admits an asymptotic linear representation with a nondegenerate Gaussian limit. In particular, assume

$$\sqrt{T}(\hat{\theta} - \theta_0) \Rightarrow \mathcal{N}(0, V_\theta), \quad (\text{B.6})$$

where $V_\theta \in \mathbb{R}^{d \times d}$ is the asymptotic covariance matrix of $\sqrt{T}(\hat{\theta} - \theta_0)$.

Assume that $\Psi(\theta)$ is differentiable in a neighborhood of θ_0 . Define the Jacobian matrix $G \in \mathbb{R}^{m \times d}$ by

$$G = \left. \frac{\partial \Psi(\theta)}{\partial \theta^\top} \right|_{\theta=\theta_0}. \quad (\text{B.7})$$

Let $g_a^\top \in \mathbb{R}^{1 \times d}$ denote the a th row of G .

Under the multivariate delta method,

$$\sqrt{T} (\Psi(\hat{\theta}) - \Psi(\theta_0)) \Rightarrow \mathcal{N}(0, \Sigma), \quad (\text{B.8})$$

where

$$\Sigma = G V_{\theta} G^{\top}. \quad (\text{B.9})$$

Let Σ_{ab} denote the (a, b) entry of Σ for $a, b \in \mathcal{A}$.

For any two indices $a, b \in \mathcal{A}$, the leading covariance approximation implied by the delta method is

$$\text{Cov}(\psi_a(\hat{\theta}), \psi_b(\hat{\theta})) \approx \frac{1}{T} g_a^{\top} V_{\theta} g_b. \quad (\text{B.10})$$

This covariance is generically nonzero because (i) V_{θ} is generically non-diagonal and (ii) distinct gradients g_a and g_b load on common components of $\hat{\theta}$. The sign of $g_a^{\top} V_{\theta} g_b$ is not restricted in general. Therefore, the induced dependence across IRF elements is not generically “positive” in any monotone sense.

To translate this to studentized statistics, assume the marginal standard error estimators are consistent in the sense that

$$\hat{\sigma}_a^2 \xrightarrow{p} \frac{\Sigma_{aa}}{T}, \quad a \in \mathcal{A}. \quad (\text{B.11})$$

Then the large-sample correlation between the marginal t-statistics satisfies

$$\text{Corr}(T_a, T_b) \approx \frac{\Sigma_{ab}}{\sqrt{\Sigma_{aa}\Sigma_{bb}}} = \frac{g_a^{\top} V_{\theta} g_b}{\sqrt{(g_a^{\top} V_{\theta} g_a)(g_b^{\top} V_{\theta} g_b)}}. \quad (\text{B.12})$$

In particular, $(T_a)_{a \in \mathcal{A}}$ is typically a strongly dependent vector.

B.3 The source of IRF dependence in common implementations

This section links the generic covariance formula $\Sigma = G V_{\theta} G^{\top}$ to standard VAR and LP constructions. The goal is not to catalog special cases where dependence disappears. The goal is to emphasize that dependence is the default outcome across horizons, variables, shocks, and states.

B.3.1 VAR IRFs

Consider a stable reduced-form VAR(p) in n_y variables. Let $y_t \in \mathbb{R}^{n_y}$ satisfy

$$y_t = A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \quad \mathbb{E}[u_t] = 0, \quad \mathbb{E}[u_t u_t^\top] = \Sigma_u. \quad (\text{B.13})$$

Let u_t be linked to structural shocks $\varepsilon_t \in \mathbb{R}^{n_\varepsilon}$ by

$$u_t = B \varepsilon_t, \quad \mathbb{E}[\varepsilon_t] = 0, \quad \mathbb{E}[\varepsilon_t \varepsilon_t^\top] = I_{n_\varepsilon}. \quad (\text{B.14})$$

Let Φ_h denote the reduced-form moving-average coefficient at horizon h . Define $\Phi_0 = I_{n_y}$ and define Φ_h recursively from the VAR coefficients for $h \geq 1$.

The structural IRF of variable i at horizon h to shock j is

$$\psi_{i,h}^{(j)} = e_i^\top \Phi_h B e_j, \quad (\text{B.15})$$

where e_i and e_j denote standard basis vectors of appropriate dimensions.

All horizons share the same estimated VAR coefficients $(\hat{A}_1, \dots, \hat{A}_p)$ through $\hat{\Phi}_h$. All response variables share the same estimated system objects. All shocks share the same estimated impact matrix $\hat{B}$ and any common identification objects used to construct it. Therefore, the full IRF panel is a nonlinear transformation of a common estimated parameter vector. This is exactly the setting in which $\Sigma = G V_\theta G^\top$ is generically dense, rather than diagonal.

B.3.2 Local projections across horizons

Local projections estimate separate regressions for each horizon, but separate regressions do not imply independent estimators. Fix a scalar shock regressor x_t and controls $w_t \in \mathbb{R}^q$. For each horizon $h \in \mathcal{H}$ and response variable $i \in \mathcal{I}$, consider

$$y_{i,t+h} = \alpha_{i,h} + \beta_{i,h} x_t + \gamma_{i,h}^\top w_t + \varepsilon_{i,h,t}, \quad (\text{B.16})$$

where $\varepsilon_{i,h,t}$ is the regression disturbance. Define the LP IRF coefficient as $\psi_{i,h}^{(j)} = \beta_{i,h}$ when x_t is the empirical analog of shock j .

Even if each $\hat{\beta}_{i,h}$ is computed by OLS in a horizon-specific regression, the random variables $\{y_{i,t+h}\}_t$ and $\{y_{i,t+h'}\}_t$ are overlapping leads of the same underlying time series. This overlap generally induces correlation between $\varepsilon_{i,h,t}$ and $\varepsilon_{i,h',t}$. This overlap also induces serial correlation in the score contributions used to estimate $(\beta_{i,h})_{h \in \mathcal{H}}$.

A compact representation stacks the horizon-specific score contributions. Let $g_{i,h,t} = x_t \varepsilon_{i,h,t}$ denote the score contribution for $\beta_{i,h}$ at time t . Let $g_{i,t} = (g_{i,h,t})_{h \in \mathcal{H}}$ denote the stacked score vector across horizons for fixed i . Under standard conditions, the joint asymptotic covariance of $(\hat{\beta}_{i,h})_{h \in \mathcal{H}}$ depends on the long-run covariance matrix of $g_{i,t}$, which is typically non-diagonal. Define the long-run covariance matrix $\Omega_i \in \mathbb{R}^{|\mathcal{H}| \times |\mathcal{H}|}$ by

$$\Omega_i = \sum_{\ell=-\infty}^{\infty} \text{Cov}(g_{i,t}, g_{i,t-\ell}). \quad (\text{B.17})$$

The off-diagonal elements of Ω_i encode cross-horizon dependence through terms such as the following.

$$\sum_{\ell=-\infty}^{\infty} \text{Cov}(x_t \varepsilon_{i,h,t}, x_{t-\ell} \varepsilon_{i,h',t-\ell}). \quad (\text{B.18})$$

B.3.3 Dependence across response variables and shocks

Across response variables, dependence arises mechanically because the same shock regressor x_t and the same sample are reused across all response equations. Across response variables, dependence is amplified when the innovation processes of different $y_{i,t}$ are contemporaneously correlated. In a VAR, this enters through Σ_u and through the joint estimation error of the system parameters. In an LP system, this enters through cross-equation covariance in $(\varepsilon_{i,h,t})_{i \in \mathcal{I}}$.

Across shocks, dependence arises whenever the identification of multiple shocks uses common estimated objects. In a VAR, the estimated columns of $\hat{B}$ typically co-move because they are functions of the same reduced-form covariance estimator and any shared identifying restrictions. In an LP setting with external instruments, multiple shock series may share estimated first-stage objects. Therefore, even when shocks are orthonormal in the population, the estimated shock-specific IRFs are not generally independent.

B.3.4 State dependence

Let $s_t \in \{0, 1\}$ denote a state indicator. Consider a state-dependent LP specification with state-specific coefficients:

$$y_{i,t+h} = \alpha_{i,h} + \beta_{i,h,0} x_t(1 - s_t) + \beta_{i,h,1} x_t s_t + \gamma_{i,h}^\top w_t + \varepsilon_{i,h,t}. \quad (\text{B.19})$$

The regressors $x_t(1 - s_t)$ and $x_t s_t$ are mechanically linked because they partition the same x_t . The two state-specific estimators are computed on the same realized sample path and therefore inherit common randomness. The overlapping-outcome dependence across horizons remains present within each state and across states.

If the state is itself estimated, then $\hat{s}_t$ enters multiple horizons, variables, and shocks simultaneously. This introduces an additional shared estimated object and therefore an additional dependence channel.

B.4 The limitations of BH for FDR-controlled IRF discovery selection

The BH step-up procedure uses only the ordered marginal p-values. Let $\hat{p}_{(1)} \leq \dots \leq \hat{p}_{(m)}$ denote the p-values ordered from smallest to largest. Let $(1), \dots, (m)$ denote the corresponding ordering of indices in $\mathcal{A}$. Define the BH cutoff rank by

$$\hat{r}_{\text{BH}} = \max \left\{ 0 \leq r \leq m : \hat{p}_{(r)} \leq \frac{r}{m} \alpha \right\}, \quad (\text{B.20})$$

with the convention that $\hat{p}_{(0)} = 0$. The BH rejection set is

$$\mathcal{R}_{\text{BH}} = \{(1), \dots, (\hat{r}_{\text{BH}})\}. \quad (\text{B.21})$$

BH controls the FDR at level α under independence of the null p-values. BH also controls the FDR under certain positive dependence conditions, including PRDS on the subset of true null hypotheses ([Benjamini and Hochberg, 1995](#); [Benjamini and Yekutieli, 2001](#)).

In the present IRF setting, the issue is not that BH “fails” mechanically. The issue is that BH requires a dependence restriction that is not implied by the econometric dependence structure of IRF t-statistics.

To make the dependence condition explicit, define the vector of evidence statistics as follows.

$$S = (S_a)_{a \in \mathcal{A}} \in \mathbb{R}^m. \quad (\text{B.22})$$

A set $D \subseteq \mathbb{R}^m$ is called increasing if $x \in D$ and $y \geq x$ componentwise imply $y \in D$. The vector S is said to be *positive regression dependent on a subset* (PRDS) on $\mathcal{A}_0$ if, for every increasing set D and every $a \in \mathcal{A}_0$, the map

$$s \mapsto \mathbb{P}(S \in D \mid S_a = s) \quad (\text{B.23})$$

is nondecreasing in s . Because $\hat{p}_a$ is a strictly decreasing function of S_a , one can equivalently formulate PRDS directly in terms of p-values by reversing the partial order.

The delta-method representation $\Sigma = GV_\theta G^\top$ implies that the dependence of $(T_a)_{a \in \mathcal{A}}$ can be strong and can involve both positive and negative correlations. Because PRDS is a global monotonicity restriction on conditional distributions, it is not generically implied by nonzero covariances. Therefore, even if a specific IRF application happened to satisfy PRDS, it is not a transparent or verifiable implication of standard VAR or LP implementations. In other words, BH's sufficient conditions for FDR control do not map cleanly to IRF practice.

A dependence-robust alternative is the Benjamini–Yekutieli (BY) modification, which replaces α by α/c_m with $c_m = \sum_{\ell=1}^m 1/\ell$. The BY procedure controls the FDR under general dependence (Benjamini and Yekutieli, 2001). The tradeoff is that BY can be very conservative when m is large.

B.4.1 A concrete PRDS violation in VAR and LP IRF panels

PRDS is a joint distributional restriction on the full evidence vector $S = (S_a)_{a \in \mathcal{A}}$ under the true nulls. In particular, PRDS fails as soon as it fails for any finite subvector corresponding to indices in $\mathcal{A}_0$. This subsection gives an explicit three-hypothesis construction in which PRDS is violated even under an (asymptotically) Gaussian approximation.

A three-statistic Gaussian counterexample. Let U and V be independent $\mathcal{N}(0, 1)$ random variables. Define three (null) studentized statistics by

$$T_1 = U, \quad T_2 = \frac{U + V}{\sqrt{2}}, \quad T_3 = \frac{-U + V}{\sqrt{2}}, \quad (\text{B.24})$$

and define evidence statistics $S_\ell = |T_\ell|$ for $\ell \in \{1, 2, 3\}$. Consider the increasing set

$$D_c = \{(s_1, s_2, s_3) \in \mathbb{R}_+^3 : s_2 > c, s_3 > c\}, \quad c > 0. \quad (\text{B.25})$$

PRDS on the (true) null corresponding to S_1 would require the map

$$s \mapsto \mathbb{P}(S \in D_c \mid S_1 = s) \quad (\text{B.26})$$

to be nondecreasing in s .

To see that this monotonicity can fail, fix $c > 0$ and write $a = c\sqrt{2}$. Because conditioning on $S_1 = s$ means $U = \pm s$ with equal probability, and because the event

in D_c is invariant to the sign of U , we have

$$\mathbb{P}(S \in D_c \mid S_1 = s) = \mathbb{P}\left(\left|\frac{s+V}{\sqrt{2}}\right| > c, \left|\frac{-s+V}{\sqrt{2}}\right| > c\right) \quad (\text{B.27})$$

$$= \mathbb{P}(|V+s| > a, |V-s| > a) \quad (\text{B.28})$$

$$= \mathbb{P}(V \notin [s-a, s+a] \cup [-s-a, -s+a]). \quad (\text{B.29})$$

At $s = 0$, the two intervals coincide and

$$\mathbb{P}(S \in D_c \mid S_1 = 0) = \mathbb{P}(|V| > a) = 2(1 - \Phi(a)). \quad (\text{B.30})$$

At $s = a$, the two intervals exactly meet at 0 and their union is $[-2a, 2a]$, so

$$\mathbb{P}(S \in D_c \mid S_1 = a) = \mathbb{P}(|V| > 2a) = 2(1 - \Phi(2a)) < 2(1 - \Phi(a)) = \mathbb{P}(S \in D_c \mid S_1 = 0). \quad (\text{B.31})$$

On the other hand, as $s \rightarrow \infty$ the union $[s-a, s+a] \cup [-s-a, -s+a]$ moves out to $\pm\infty$ and leaves the central region uncovered, implying

$$\lim_{s \rightarrow \infty} \mathbb{P}(S \in D_c \mid S_1 = s) = 1. \quad (\text{B.32})$$

Therefore, the function $s \mapsto \mathbb{P}(S \in D_c \mid S_1 = s)$ is not nondecreasing (it falls from $s = 0$ to $s = a$ but eventually rises toward 1), which violates PRDS. This shows that PRDS can fail in a simple Gaussian setting driven purely by linear combinations of common estimated objects.

How this arises in a VAR IRF. Consider a stable VAR(1) in two variables with coefficient matrix

$$A = \rho \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad 0 < \rho < \frac{1}{\sqrt{2}}, \quad (\text{B.33})$$

so that the reduced-form MA coefficient at horizon 1 is $\Phi_1 = A$. Let the structural impact vector for shock $j = 1$ be $b = Be_1 = (b_1, b_2)^\top$. Then three IRF entries in the reported panel satisfy

$$\psi_{1,0}^{(1)} = b_1, \quad (\text{B.34})$$

$$\psi_{1,1}^{(1)} = e_1^\top \Phi_1 b = \rho(b_1 + b_2), \quad (\text{B.35})$$

$$\psi_{2,1}^{(1)} = e_2^\top \Phi_1 b = \rho(-b_1 + b_2). \quad (\text{B.36})$$

Suppose these three hypotheses are true nulls (e.g. $b_1 = b_2 = 0$), and suppose $(\hat{b}_1, \hat{b}_2)$ is asymptotically normal with a nonsingular covariance matrix (as is standard when b is an estimated function of reduced-form moments and identifying objects). After studentization, the associated t-statistics are asymptotically a standardized linear transformation of $(\hat{b}_1 - b_1, \hat{b}_2 - b_2)$. In particular, in the special case where the studentized errors of $\hat{b}_1$ and $\hat{b}_2$ are asymptotically independent standard normals, the three t-statistics above converge to the (T_1, T_2, T_3) construction. Hence PRDS fails for this three-element IRF subvector, and therefore PRDS is not a generic implication of VAR IRF practice.

How this arises in an LP IRF. Consider three LP regressions at a fixed horizon h with a common shock regressor x_t and no loss of generality normalize $\mathbb{E}[x_t^2] = 1$:

$$y_{\ell,t+h} = \beta_{\ell,h} x_t + \varepsilon_{\ell,t}, \quad \ell \in \{1, 2, 3\}. \quad (\text{B.37})$$

Suppose the three null hypotheses $\beta_{\ell,h} = 0$ are true, and suppose the (time- t) disturbance vector has a simple two-factor representation

$$\varepsilon_{1,t} = \eta_t, \quad \varepsilon_{2,t} = \eta_t + \tilde{\zeta}_t, \quad \varepsilon_{3,t} = -\eta_t + \tilde{\zeta}_t, \quad (\text{B.38})$$

where $(\eta_t, \tilde{\zeta}_t, x_t)$ are i.i.d. over t , mean-zero, mutually independent, and (for concreteness) Gaussian with unit variances. Then the OLS score contributions are $x_t \varepsilon_{\ell,t}$, and the joint CLT implies that the studentized LP t-statistics converge to the same (T_1, T_2, T_3) construction above (with U and V arising from the partial sums of $x_t \eta_t$ and $x_t \tilde{\zeta}_t$). Consequently, the associated evidence vector $(|T_1|, |T_2|, |T_3|)$ violates PRDS. This factor structure is economically interpretable: two responses (or two state-specific equations, or two shock series built from a shared first stage) can load with opposite signs on a common disturbance component, while also sharing additional common variation. Thus, even in a textbook LP setup with Gaussian primitives and valid (marginal) studentization, PRDS need not hold for the IRF evidence vector.

B.5 Why the RSW step-down procedure is well-suited for IRFs

RSW construct step-down multiple testing procedures that achieve asymptotic FDR control under dependence by calibrating critical values from a *joint* resampling distribution of the full vector of studentized statistics (Romano et al., 2008). This is a direct match to IRF panels: the empirically relevant dependence across horizons, response variables, shocks, and states is induced by common estimated objects (e.g. shared reduced-form parameters, shared first-stage shocks, shared state classification)

and by reuse of the same time-series sample. Unlike BH, which reduces the problem to marginal p-values and therefore requires additional dependence restrictions for validity, RSW explicitly uses the joint distribution of the entire evidence vector in its calibration step.

B.5.1 Step-down structure

Let $S_{(1)} \geq \dots \geq S_{(m)}$ denote the evidence statistics ordered from largest to smallest, with associated ordered hypotheses $H_{0,(1)}, \dots, H_{0,(m)}$. Given critical values $c_1 \geq \dots \geq c_m$, a step-down rule rejects the most significant hypotheses until the first comparison fails. Define

$$j_0 = \max \left\{ 0 \leq j \leq m : S_{(1)} \geq c_1, \dots, S_{(j)} \geq c_j \right\}, \quad (\text{B.39})$$

and the rejection set

$$\mathcal{R} = \{(1), \dots, (j_0)\}. \quad (\text{B.40})$$

The substantive problem is therefore the construction of $(c_j)_{j=1}^m$ in a way that controls the FDR when $(S_a)_{a \in \mathcal{A}}$ is dependent.

RSW choose $(\hat{c}_j)$ via a recursion that is defined in terms of the *joint* distribution of order statistics of a resampled evidence vector. Operationally, the recursion uses resampling probabilities of events involving the ordered resampled statistics evaluated on index sets tied to the observed ordering of S (see, e.g., their bootstrap critical values in Eq. (17) and surrounding discussion) (Romano et al., 2008). Because these probabilities depend on the full joint law of the resampled vector, the procedure incorporates cross-coordinate dependence by construction.

B.5.2 Resampling, centered studentization, and dependence preservation

Let $X_{1:T}$ denote the observed sample, and let $\hat{P}_T$ denote an estimated distribution used to generate a resample $X_{1:T}^*$ (for time series data, $\hat{P}_T$ should be induced by a bootstrap that preserves serial dependence). Given $X_{1:T}^*$, recompute the full IRF panel using the same estimation pipeline as in the original data:

$$\hat{\Psi}^* = (\hat{\psi}_a^*)_{a \in \mathcal{A}}. \quad (\text{B.41})$$

To approximate the joint distribution of the studentized estimation error, use centered resampled studentized statistics,

$$T_a^* = \frac{\hat{\psi}_a^* - \hat{\psi}_a}{\hat{\sigma}_a}, \quad S_a^* = |T_a^*|, \quad S^* = (S_a^*)_{a \in \mathcal{A}}. \quad (\text{B.42})$$

The studentization uses the same $\hat{\sigma}_a$ as the observed statistics, held fixed across resamples. A single resample produces the entire vector S^* . Because S^* is computed jointly from one resampled dataset and through the same IRF construction map used in the original sample, it inherits (by design) the same cross-horizon, cross-variable, cross-shock, and cross-state dependence present in the empirical IRF pipeline. This is exactly the dependence information needed to calibrate critical values for a step-down rule in a setting where marginal evidence statistics are not independent and where PRDS is not generically implied.

B.5.3 Where exchangeability enters, and what it means for IRFs

The RSW bootstrap step-down relies on an exchangeability condition for the limiting joint distribution of the studentized statistics corresponding to *true* nulls (Romano et al., 2008). In this context, exchangeability means that (after studentization) the joint limiting law of $(T_a)_{a \in \mathcal{A}_0}$ is invariant to permutations of the indices in $\mathcal{A}_0$. Intuitively, this symmetry allows the procedure to use the *least significant* coordinates (those with small observed S_a) as an asymptotic proxy for the unknown null subset in the recursion that determines $(\hat{c}_j)$. Absent such symmetry, the recursion can implicitly overweight or underweight certain null coordinates because “small S_a ” would no longer be an approximately uninformative label for “likely null” uniformly across a .

For IRF panels, exchangeability is most plausible when the analyst deliberately enforces comparability of null distributions across coordinates by (i) using a single, uniform studentization recipe across all (i, h, j, k) (same variance estimator class and tuning rules, same lag-selection conventions, same normalization of shocks), and (ii) avoiding mixtures of fundamentally different testing objects within the same FDR family (e.g. combining statistics produced by different estimators or different identification regimes in one pooled multiple-testing problem).

B.5.4 What this delivers in practice

In an IRF discovery-selection problem, the object of interest is a rejection set that is robust to the empirically unavoidable dependence induced by the IRF construction.

RSW is well-suited to this task because it (a) uses a joint resampling distribution of the entire evidence vector and therefore carries dependence into the calibration step, and (b) produces a step-down rejection set with critical values chosen to target FDR control under dependence ([Romano et al., 2008](#)). Consequently, when the bootstrap is valid for the joint studentized IRF vector and exchangeability is a reasonable approximation for true-null coordinates, RSW provides a principled alternative to BH in settings where BH's dependence conditions are not transparently satisfied.

C Asymptotic Validity

This appendix establishes the theoretical guarantees for the bootstrap step-down procedure in Section 3 applied to VAR- and LP-based IRF statistics. We operate under the standard asymptotic regime where the number of parameters in the declared IRF family is fixed ($m < \infty$) and the sample size grows ($n \rightarrow \infty$).

C.1 Preliminaries

Let $\theta = (\theta_1, \dots, \theta_m)^\top$ denote the stacked IRF vector defined in Section 3, and $\hat{\theta} = (\hat{\theta}_1, \dots, \hat{\theta}_m)^\top$ its estimator computed from a sample of length n . We test the m null hypotheses $H_{0,j} : \theta_j = 0, j \in \{1, \dots, m\}$, against their two-sided alternatives $H_{1,j} : \theta_j \neq 0$. Let $\mathcal{H}_0 = \{j : \theta_j = 0\} \subseteq \{1, \dots, m\}$ denote the (unknown) index set of true nulls. Let $T_j = \hat{\theta}_j / \hat{\sigma}_j$ be the signed studentized test statistics, where $\hat{\sigma}_j$ is the estimated standard error of the respective estimator and $|T_j|$ is used for two-sided testing of each hypothesis.

Let $\mathcal{R} \subseteq \{1, \dots, m\}$ denote the set of rejections produced by the RSW step-down algorithm at target level q (Section 3). Write $R = |\mathcal{R}|$ as the total number of rejections and $V = |\mathcal{R} \cap \mathcal{H}_0|$ as the number of false rejections. By definition, the false discovery rate under the true probability measure P is $\text{FDR} = \mathbb{E}[V / \max\{R, 1\}]$.

Let $\hat{P}_n$ denote the law used to generate the joint bootstrap IRF draws. Let $\hat{\theta}^*$ denote the IRF estimator computed on a bootstrap sample drawn from $\hat{P}_n$. Bootstrap statistics are studentized by the same fixed standard errors $\hat{\sigma}_j$ as the sample statistics.

Directly following Romano et al. (2008), the bootstrap step-down theory is stated in terms of $|T_j^*|$, where T_j^* is given in Equation (C.1).

$$T_j^* = \frac{\hat{\theta}_j^* - \theta_j(\hat{P}_n)}{\hat{\sigma}_j} \quad (\text{C.1})$$

It is possible to center T_j^* at $\hat{\theta}_j$, rather than $\theta_j(\hat{P}_n)$, and obtain identical asymptotic results if $\hat{\theta}_j - \theta_j(\hat{P}_n) = o_p(n^{-1/2})$. For the standard residual bootstrap, $\theta_j(\hat{P}_n) = \hat{\theta}_j$ by definition. For further comments on this choice, see Romano et al. (2008), Remark 5.

For any $K \subseteq \{1, \dots, m\}$ and probability measure P governing the data, let $J_{n,K}(P)$ denote the joint distribution under P of the (signed) studentized estimation errors $(\hat{\theta}_j - \theta_j(P)) / \hat{\sigma}_j$ for all $j \in K$. We rely on the following assumptions for theoretical results (Romano et al., 2008).

Assumption C.1. For every $j \in \{1, \dots, m\}$, the estimated standard errors satisfy $\sqrt{n}\hat{\sigma}_j \xrightarrow{P} \sigma_j$, where $\sigma_j \in (0, \infty)$.

Assumption C.2. As $n \rightarrow \infty$, $J_{n,\{1,\dots,m\}}(P)$ weakly converges to a limiting distribution $J_{\{1,\dots,m\}}(P)$, which implies $J_{n,\mathcal{H}_0}(P) \Rightarrow J_{\mathcal{H}_0}(P)$ by the continuous mapping theorem. Further, $J_{\mathcal{H}_0}(P)$ has continuous marginal distributions and connected support.

Assumption C.3. The limiting joint distribution $J_{\mathcal{H}_0}(P)$ is exchangeable (i.e., invariant under permutations of indices in $\mathcal{H}_0$).

Let d denote any metric characterizing weak convergence on $\mathbb{R}^m$.

Assumption C.4. The bootstrap probability measure $\hat{P}_n$ consistently estimates P in the sense that $d\left(J_{n,\{1,\dots,m\}}(P), J_{n,\{1,\dots,m\}}(\hat{P}_n)\right) \xrightarrow{P} 0$.

Theorem C.1 establishes pointwise asymptotic FDR control for a fixed probability measure P . It is a direct result of Romano et al. (2008); thus, we provide the statement without proof.

Theorem C.1. Suppose Assumptions C.1 – C.4 are satisfied. Then, the RSW bootstrap step-down procedure applied to $T = (T_1, \dots, T_m)$ at target level $q \in (0, 1)$ satisfies $\limsup_{n \rightarrow \infty} \text{FDR} \leq q$.

Theorem C.2 establishes the limiting power of the test tending to one as $n \rightarrow \infty$. This is a standard property of $\sqrt{n}$ -consistent estimators.

Theorem C.2. Suppose Assumptions C.1 and C.2 are satisfied. Then, as $n \rightarrow \infty$ for any fixed $\theta_j \neq 0$, $|T_j| \xrightarrow{P} \infty$.

Proof of Theorem C.2. As a consequence of Assumption C.2, $(\hat{\theta}_j - \theta_j)/\hat{\sigma}_j$ converges in distribution, hence is $O_p(1)$. From Assumption C.1, $\hat{\sigma}_j = n^{-1/2}(\sigma_j + o_p(1))$. Multiplying yields, $\hat{\theta}_j - \theta_j = \hat{\sigma}_j \cdot O_p(1) = O_p(n^{-1/2})$, establishing $\sqrt{n}$ -consistency. Then,

$$|T_j| = \left| \frac{\hat{\theta}_j}{\hat{\sigma}_j} \right| = \left| \frac{\theta_j + O_p(n^{-1/2})}{n^{-1/2}(\sigma_j + o_p(1))} \right| = \sqrt{n} \left| \frac{\theta_j + O_p(n^{-1/2})}{\sigma_j + o_p(1)} \right|.$$

By Slutsky's theorem,

$$|T_j| = \sqrt{n} \left(\left| \frac{\theta_j}{\sigma_j} \right| + o_p(1) \right) = \sqrt{n} \left| \frac{\theta_j}{\sigma_j} \right| + o_p(\sqrt{n}).$$

Since $\theta_j \neq 0$ and $\sigma_j \in (0, \infty)$, $\sqrt{n}|\theta_j/\sigma_j|$ diverges to infinity as $n \rightarrow \infty$. □

Section C.2 verifies the conditions required by Theorems C.1 and C.2 when applying the RSW step-down procedure to VAR and LP estimators across different bootstrap specifications. Assumption C.5 places restrictions on the data generating process

for compatibility with the bootstrap methods highlighted in this section. For LP estimation, it is possible to relax this strict parametric condition on the DGP and achieve identical asymptotic results. We highlight these specific cases to illustrate the step-down procedure's application to popular resampling schemes in the applied macroeconomics literature.

Assumption C.5. The observed vector $\{y_t\}_{t=1}^n$ is generated by a stable VAR(p) process of the form $y_t = \Phi_0 + \sum_{k=1}^p \Phi_k y_{t-k} + u_t$. The error process $\{u_t\}$ satisfies the following conditions: $\{u_t\}$ is strictly stationary, geometric α -mixing, $\mathbb{E}[u_t] = 0$, $\mathbb{E}[\|u_t\|^4] < \infty$, and $\Sigma_u = \mathbb{E}[u_t u_t']$ positive definite.

Remark C.1. Assumption C.5 guarantees that a VAR(p)-based bootstrap approximates the true probability measure P . This specifically encompasses standard residual bootstraps for i.i.d. $\{u_t\}$ (Kilian, 1998), residual-based moving block bootstraps for martingale difference sequence (MDS) errors in VAR and SVAR-IV models (Brüggemann et al., 2016; Jentsch and Lunsford, 2022), and VAR-based bootstraps for LP estimators with MDS errors (Montiel Olea and Plagborg-Møller, 2021). While LPs are frequently utilized precisely because their estimation is robust to general DGPs and does not require Assumption C.5, applying a VAR(p)-based bootstrap to LPs strictly relies on this parametric restriction. Although we state our main results in terms of this DGP, several other bootstrap methods (e.g., moving or circular block bootstraps (Kunsch, 1989; Politis and Romano, 1992)) exist that satisfy the requirements of Theorems C.1 and C.2 under more general conditions.

C.2 Verification of assumptions for IRF inference

To verify the assumptions for structural IRF inference, we formalize estimation of the m -dimensional stacked structural impulse response vector θ across a fixed horizon grid $h \in \{0, \dots, H\}$. Under Assumption C.5, θ can be consistently estimated using either a VAR or LP framework.

In the VAR approach, let $\hat{\theta} = f(\hat{\eta})$ denote the plug-in estimator, where η collects the reduced-form parameters and any auxiliary moments used to identify the invertible structural impact matrix $A = g(\Sigma_u)$. Provided $f(\cdot)$ and $g(\cdot)$ are continuously differentiable, asymptotic normality follows from the delta method (precluding non-smooth set-identification schemes).

Alternatively, in the LP approach, the parameters are estimated directly from OLS via (lag-augmented) horizon-specific regressions. Reduced-form responses are mapped to the structural responses via the invertible structural impact matrix $A = g(\Sigma_u)$. Standard

asymptotic theory under weak dependence ensures that the LP estimator admits an asymptotically linear representation (Montiel Olea and Plagborg-Møller, 2021).

Regardless of the chosen framework, the generic estimator $\hat{\theta}$ can be written as $\sqrt{n}(\hat{\theta} - \theta) = n^{-1/2} \sum_{t=1}^n \psi_t + o_p(1)$, where the influence process $\{\psi_t\}$ is strictly stationary with finite second moments. Geometric α -mixing of $\{u_t\}$ in Assumption C.5 extends to $\{y_t\}$, and since ψ_t is a measurable function of $\{y_t\}$ and its lags, to $\{\psi_t\}$. Thus, this admits a multivariate central limit theorem (Davidson, 1994), $\sqrt{n}(\hat{\theta} - \theta) \Rightarrow \mathcal{N}_m(0, \Sigma_\theta)$ where $[\Sigma_\theta]_{jj} = \sigma_j^2$.

Proposition C.1 verifies Assumption C.1 for both VAR and LP estimation schemes.

Proposition C.1. *Suppose Assumption C.5 holds, the structural impact matrix A is invertible, and the specified impulse responses do not lie in the null space of the Jacobian $\nabla f(\eta)$. Let $\hat{\sigma}_j$ denote the standard error estimator derived via the delta method for the VAR framework, or the OLS standard error for the lag-augmented LP framework. Then, $\sqrt{n}\hat{\sigma}_j \xrightarrow{p} \sigma_j$ with $\sigma_j \in (0, \infty)$, satisfying Assumption C.1.*

Proof of Proposition C.1. We verify the conditions for the VAR and LP frameworks separately.

In the VAR framework, let $\hat{\Sigma}_\eta$ be a standard consistent estimator for the asymptotic variance of the reduced-form parameters, such that $\hat{\Sigma}_\eta \xrightarrow{p} \Sigma_\eta$. Because the mapping $f(\cdot)$ is continuously differentiable in a neighborhood of the truth, $\nabla f(\eta)$ is continuous. By the continuous mapping theorem, $\hat{\Sigma}_\theta = \nabla f(\hat{\eta})\hat{\Sigma}_\eta\nabla f(\hat{\eta})^\top \xrightarrow{p} \nabla f(\eta)\Sigma_\eta\nabla f(\eta)^\top = \Sigma_\theta$. Because $\sqrt{n}\hat{\sigma}_j$ corresponds to the square root of the j -th diagonal element of $\hat{\Sigma}_\theta$, consistency follows immediately, $\sqrt{n}\hat{\sigma}_j \xrightarrow{p} \sigma_j$. Non-degeneracy, $\sigma_j \in (0, \infty)$, follows from invertibility of A and the assumption that the selected impulse responses do not lie in the null space of $\nabla f(\eta)$, which together ensure Σ_θ has strictly positive diagonal entries.

In the lag-augmented LP framework, Montiel Olea and Plagborg-Møller (2021) establish that lag augmentation renders serial correlation correction unnecessary, so standard OLS standard errors consistently estimate the asymptotic variance of the reduced-form LP estimator. Mapping through the continuously differentiable transformation $A^{-1} = g(\Sigma_u)^{-1}$ via the delta method then yields $\sqrt{n}\hat{\sigma}_j \xrightarrow{p} \sigma_j$ for the structural estimator. Non-degeneracy, $\sigma_j \in (0, \infty)$, follows from invertibility of A , positive-definiteness of Σ_u under Assumption C.5, and non-degeneracy of the LP regressor matrix under Assumption C.5, as the stable VAR implies the regressors have a non-singular second-moment matrix. $\square$

Proposition C.2 verifies Assumption C.2 for both VAR and LP estimation schemes.

Proposition C.2. *Suppose the conditions of Proposition C.1 hold. Then,*

$$\left(\frac{\hat{\theta}_1 - \theta_1}{\hat{\sigma}_1}, \dots, \frac{\hat{\theta}_m - \theta_m}{\hat{\sigma}_m} \right)^\top \Rightarrow \mathcal{N}_m(0, R_\theta)$$

where R_θ has ones on the diagonal, satisfying Assumption C.2.

Proof of Proposition C.2. As established above, the stacked structural estimators satisfy $\sqrt{n}(\hat{\theta} - \theta) \Rightarrow \mathcal{N}_m(0, \Sigma_\theta)$. Note that the studentized test statistic for the j -th impulse response can be written as

$$\frac{\hat{\theta}_j - \theta_j}{\hat{\sigma}_j} = \frac{\sqrt{n}(\hat{\theta}_j - \theta_j)}{\sqrt{n}\hat{\sigma}_j}.$$

By Proposition C.1, $\sqrt{n}\hat{\sigma}_j \xrightarrow{p} \sigma_j > 0$ for all $j \in \{1, \dots, m\}$. Applying the continuous mapping theorem to the jointly converging sequence $(\sqrt{n}(\hat{\theta} - \theta), \sqrt{n}\hat{\sigma})$ with the continuous map $(x, d) \mapsto D^{-1}x$, where $D = \text{diag}(\sqrt{n}\hat{\sigma}_1, \dots, \sqrt{n}\hat{\sigma}_m)$, yields joint weak convergence of the studentized statistics to $\mathcal{N}_m(0, R_\theta)$. The limiting covariance matrix R_θ is the correlation matrix associated with Σ_θ , which by definition has ones on the main diagonal. Finally, the multivariate normal limiting distribution $\mathcal{N}_m(0, R_\theta)$ inherently possesses continuous marginal distributions and connected support, satisfying all regularity requirements of Assumption C.2. $\square$

Proposition C.3 verifies Assumption C.4 for both VAR and LP estimation schemes for a list of popular resampling schemes in applied macroeconomics literature. We specifically prove Proposition C.3 for four settings: VAR estimation with i.i.d. innovations using a residual bootstrap (Kilian, 1998); VAR and SVAR-IV estimation with MDS innovations using a moving block bootstrap (Brüggemann et al., 2016; Jentsch and Lunsford, 2022); and LP estimation with MDS innovations using a VAR(p)-based residual bootstrap (Montiel Olea and Plagborg-Møller, 2021). Similar proofs naturally extend to other bootstraps in the literature.

Proposition C.3. *Let $\hat{P}_n$ denote the law of the bootstrap and suppose the conditions of Proposition C.1 hold. Then, $d\left(J_{n,\{1,\dots,m\}}(P), J_{n,\{1,\dots,m\}}(\hat{P}_n)\right) \xrightarrow{p} 0$, satisfying Assumption C.4.*

Proof of Proposition C.3. To prove Proposition C.3, it is enough to show that the joint law of $T^* = (T_1^*, \dots, T_m^*)^\top$ converges to the continuous limiting distribution established in Proposition C.2, $\mathcal{N}_m(0, R_\theta)$. If both $J_{n,\{1,\dots,m\}}(P)$ and $J_{n,\{1,\dots,m\}}(\hat{P}_n)$ converge weakly

to the same continuous limit distribution, $d\left(J_{n,\{1,\dots,m\}}(P), J_{n,\{1,\dots,m\}}(\hat{P}_n)\right) \xrightarrow{p} 0$ follows immediately for any metric characterizing weak convergence on $\mathbb{R}^m$.

For the VAR frameworks, Kilian (1998) establish $\sqrt{n}(\hat{\eta}^* - \hat{\eta}) \Rightarrow \mathcal{N}(0, \Sigma_\eta)$ conditionally for i.i.d. residual bootstraps, Brüggemann et al. (2016) for moving block bootstraps, and Jentsch and Lunsford (2022) for SVAR-IV moving block bootstraps. Because $\theta = f(\eta)$ is continuously differentiable, the bootstrap delta method yields $\sqrt{n}(\hat{\theta}^* - \hat{\theta}) \Rightarrow \mathcal{N}_m(0, \Sigma_\theta)$ conditionally. Studentizing by the fixed standard errors $\hat{\sigma}_j$, which satisfy $\sqrt{n}\hat{\sigma}_j \xrightarrow{p} \sigma_j$ by Proposition C.1, and applying the continuous mapping theorem then yields $T^* \Rightarrow \mathcal{N}_m(0, R_\theta)$.

For the LP framework using the VAR(p)-based wild recursive bootstrap, Montiel Olea and Plagborg-Møller (2021) propose this resampling strategy but leave its formal pointwise validity to future work as a straightforward extension of their main results. We formalize this step here. The DGP under $\hat{P}_n$ is, by construction, a finite-order VAR process driven by bias-corrected sample estimates, with bootstrap innovations $\hat{u}_t^* = U_t \hat{u}_t$, where $U_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$ independently of the data. Because the multipliers possess finite moments of all orders and the empirical moments of the sample residuals converge to the finite population moments, the bootstrap innovations inherit the required finite conditional moments in probability. The conditional covariance matrix of $\hat{u}_t^*$ equals $\text{diag}(\hat{u}_t \hat{u}_t^\top)$, and the sample average $n^{-1} \sum_t \hat{u}_t \hat{u}_t^\top$ converges to Σ_u , which is positive definite by Assumption C.5. Consequently, this bootstrap DGP satisfies the regularity conditions required by Montiel Olea and Plagborg-Møller (2021). Applying their result to the probability measure $\hat{P}_n$, along with the consistency of the sample VAR estimates, yields that the unscaled bootstrap LP estimators conditionally mimic the asymptotic covariance of the reduced-form estimator. Because $A^{-1} = g(\Sigma_u)^{-1}$ is continuously differentiable and consistently estimated under $\hat{P}_n$, the bootstrap delta method extends this to Σ_θ . Studentizing by the same fixed standard errors again yields $T^* \Rightarrow \mathcal{N}_m(0, R_\theta)$, establishing pointwise bootstrap validity. $\square$

We state Corollaries C.1 and C.2 as analogs to the results in Theorems C.1 and C.2.

Corollary C.1. *Let $\hat{P}_n$ denote the law of the bootstrap and suppose the conditions of Proposition C.1 hold, along with Assumption C.3. Then, the RSW bootstrap step-down procedure applied to $T = (T_1, \dots, T_m)$ at target level $q \in (0, 1)$ satisfies $\limsup_{n \rightarrow \infty} \text{FDR} \leq q$.*

Corollary C.2. *Let $\hat{P}_n$ denote the law of the bootstrap and suppose the conditions of Proposition C.1 hold. Then, for any fixed $\theta_j \neq 0$, $|T_j| \xrightarrow{p} \infty$.*

Rather than exclusively limiting the discussion to the FDR, we comment on the tail

behavior of the false discovery proportion, $\text{FDP} = V / \max\{R, 1\}$, where $\text{FDR} = \mathbb{E}[\text{FDP}]$. We establish firm bounds in Proposition C.4.

Proposition C.4. *Let $\hat{P}_n$ denote the law of the bootstrap and suppose the conditions of Proposition C.1 hold, along with Assumption C.3. Then, the RSW bootstrap step-down procedure applied to $T = (T_1, \dots, T_m)$ at target level $q \in (0, 1)$ satisfies $\limsup_{n \rightarrow \infty} \mathbb{P}(\text{FDP} \geq \gamma) \leq q/\gamma$.*

Proof of Proposition C.4. The proof is a direct application of Markov's inequality and Corollary C.1: $\limsup_{n \rightarrow \infty} \mathbb{P}(\text{FDP} \geq \gamma) \leq \limsup_{n \rightarrow \infty} \mathbb{E}[\text{FDP}]/\gamma = \limsup_{n \rightarrow \infty} \text{FDR}/\gamma \leq q/\gamma$. $\square$

C.2.1 Verification of Assumption C.3

Assumption C.3 is an additional symmetry restriction on the dependence structure of the true-null studentized statistics. For a Gaussian vector, exchangeability is equivalent to invariance of its correlation matrix R_0 under permutations of the indices in $\mathcal{H}_0$. A sufficient condition is compound symmetry, $R_0 = (1 - \rho)I + \rho \mathbf{1}\mathbf{1}'$ for some $\rho \in (-1/(|\mathcal{H}_0| - 1), 1)$.

In the context of structural impulse responses, this assumption may be violated, especially for contiguous horizons. Even when the true impulse response parameters are strictly zero, estimation errors are correlated across adjacent horizons. In the VAR framework, this dependence arises from the iterative calculation of the impulse response functions. In the LP framework, the multi-step projections mechanically generate overlapping moving average residuals. Exchangeability becomes much more plausible for non-contiguous or distant horizons.

The RSW bootstrap step-down procedure, however, remains practically justified. Correlation between adjacent impulse response horizons is, in general, positive. While this alone does not imply all conditions required for formal FDR control, positive dependence structures are often associated with conservative behavior in step-up procedures (Benjamini and Yekutieli, 2001). Step-down procedures do not admit a general dominance relationship for FDR control, but are typically more conservative in their rejection thresholds. This provides heuristic support that the realized FDR is unlikely to exceed the nominal level q , even in the absence of formal guarantees.

For strict γ -tail control of the FDP without exchangeability (an analog to Proposition C.4 for a given q , i.e., $\mathbb{P}(\text{FDP} \geq \gamma) \leq q$), we cite the step-down procedure of Romano and Wolf (2007), Algorithm 4.1. Romano et al. (2008) dynamically scales the target level while continuously evaluating the distribution of the maximum test statistic,

whereas [Romano and Wolf \(2007\)](#) maintains a fixed level but shifts the chosen order statistic.

C.3 FCR-adjusted post-selection confidence intervals

This subsection justifies the post-selection confidence intervals reported in the second step of the paper's two-step procedure. The relevant error criterion is the false coverage rate (FCR) of [Benjamini and Yekutieli \(2005\)](#).

For each $j \in \mathcal{R}$, let $\text{CI}_j(\alpha)$ report the marginal confidence interval for θ_j with nominal coverage level $1 - \alpha$. Let U be the number of false coverage statements, $U = \sum_{j \in \mathcal{R}} \mathbf{1}\{\theta_j \notin \text{CI}_j(\alpha)\}$. Define the false coverage proportion (FCP) as $\text{FCP} = U / \max\{R, 1\}$ and the FCR under measure P as $\text{FCR} = \mathbb{E}[\text{FCP}]$.

To control the FCR at target level q , we construct FCR-adjusted confidence intervals by dynamically adapting the nominal coverage level based on the proportion of rejected hypotheses. The FCR adjustment presumes access to a confidence interval construction that is (at least asymptotically) valid marginally for each coefficient at any desired level. Throughout this paper, such intervals are obtained from the bootstrap distribution of the studentized estimator (percentile- t).

Assumption C.6. As $n \rightarrow \infty$, $\sup_{j \in \{1, \dots, m\}, \alpha \in (0, 1)} |\mathbb{P}\{\theta_j \notin \text{CI}_j(\alpha)\} - \alpha| \rightarrow 0$. Further, if $\alpha \geq \alpha'$, then $\text{CI}_j(\alpha) \subseteq \text{CI}_j(\alpha')$ almost surely.

Let $T = (T_1, \dots, T_m)$ denote the random vector on which the selection rule is based, and write $\mathcal{R} = S(T)$ for a measurable selection mapping S . For each j , partition T into (T_j, T_{-j}) , where $T_{-j} = T \setminus T_j$. For each selected index $j \in S(T)$, define $R_{\min, j} = \min_{t_j \in \mathbb{R}} \{|S(t_j, T_{-j})| : j \in S(t_j, T_{-j})\}$. The level- q FCR-adjusted selective confidence interval for θ_j is then $\text{CI}_j^*(q) = \text{CI}_j(\alpha_j^*)$, where $\alpha_j^* = qR_{\min, j}/m$ for $j \in \mathcal{R}$. When $j \notin \mathcal{R}$, no interval is reported. The adjusted level α_j^* increases with the number of reported intervals, and is equivalent to the Bonferroni level when $R_{\min, j} = 1$.

Remark C.2. If the selection rule is simple (e.g., the step-up procedure of [Benjamini and Hochberg \(1995\)](#), see Remark 1 of [Benjamini and Yekutieli \(2005\)](#)), or $|S(T_j, T_{-j})|$ is constant over all T_j , then $R_{\min, j} = R$ and $\text{CI}_j^*(q) = \text{CI}_j(qR/m)$. For complex selection rules (e.g., the RSW bootstrap step-down procedure of Section 3 and [Romano et al. \(2008\)](#)), the equality does not necessarily hold.

Theorems C.3 and C.4 summarize the FCR guarantees established by [Benjamini and Yekutieli \(2005\)](#) under independence and general dependence of the test statistics, respectively.

Theorem C.3. Suppose the components of T are independent and Assumption C.6 holds. Then, the FCR-adjusted confidence intervals $\text{CI}_j^*(q)$ satisfy $\limsup_{n \rightarrow \infty} \text{FCR} \leq q$ for $q \in (0, 1)$. If Assumption C.6 holds with exact marginal coverage, $\mathbb{P}\{\theta_j \notin \text{CI}_j(\alpha)\} \leq \alpha$ for every $j \in \{1, \dots, m\}$ and $\alpha \in (0, 1)$, then $\text{FCR} \leq q$ for $q \in (0, 1)$.

Proof. The finite-sample statement is Theorem 1 of Benjamini and Yekutieli (2005). Let $\epsilon_n = \sup_{j \in \{1, \dots, m\}, \alpha \in (0, 1)} |\mathbb{P}\{\theta_j \notin \text{CI}_j(\alpha)\} - \alpha|$, so that $\epsilon_n \rightarrow 0$ by Assumption C.6.

The FCR decomposes as

$$\text{FCR} = \sum_{j=1}^m \mathbb{E} \left[\frac{\mathbf{1}\{j \in \mathcal{R}, \theta_j \notin \text{CI}_j(\alpha_j^*)\}}{\max\{R, 1\}} \right].$$

For each term, condition on T_{-j} and use the fact that $\alpha_j^* = qR_{\min,j}/m$ depends only on T_{-j} .

$$\mathbb{E} \left[\frac{\mathbf{1}\{j \in \mathcal{R}, \theta_j \notin \text{CI}_j(\alpha_j^*)\}}{\max\{R, 1\}} \right] \leq \mathbb{E} \left[\frac{\mathbf{1}\{j \in \mathcal{R}\}}{\max\{R, 1\}} \cdot \mathbb{P}\{\theta_j \notin \text{CI}_j(\alpha_j^*) \mid T_{-j}\} \right]$$

Under the independence of the components of T , conditioning on T_{-j} renders T_j independent of α_j^* , so that $\mathbb{P}\{\theta_j \notin \text{CI}_j(\alpha_j^*) \mid T_{-j}\} \leq \alpha_j^* + \epsilon_n = qR_{\min,j}/m + \epsilon_n$. Substituting and summing over j ,

$$\text{FCR} \leq \sum_{j=1}^m \mathbb{E} \left[\frac{\mathbf{1}\{j \in \mathcal{R}\}}{\max\{R, 1\}} \cdot \left(\frac{qR_{\min,j}}{m} + \epsilon_n \right) \right].$$

The first term recovers the exact finite-sample bound of Benjamini and Yekutieli (2005), yielding q . For the second term, since $\mathbf{1}\{j \in \mathcal{R}\} / \max\{R, 1\} \leq 1$ deterministically and $m < \infty$ is fixed, $\text{FCR} \leq q + m \cdot \epsilon_n$. Taking the limit supremum and applying $\epsilon_n \rightarrow 0$ yields $\limsup_{n \rightarrow \infty} \text{FCR} \leq q$ for $q \in (0, 1)$. $\square$

Theorem C.4. Suppose Assumption C.6 holds and define $c_m = \sum_{\ell=1}^m \ell^{-1}$. Then, the FCR-adjusted confidence intervals $\text{CI}_j^*(q/c_m)$ satisfy $\limsup_{n \rightarrow \infty} \text{FCR} \leq q$ for $q \in (0, 1)$. If Assumption C.6 holds with exact marginal coverage, $\mathbb{P}\{\theta_j \notin \text{CI}_j(\alpha)\} \leq \alpha$ for every $j \in \{1, \dots, m\}$ and $\alpha \in (0, 1)$, then $\text{FCR} \leq q$ for $q \in (0, 1)$.

Proof. The finite-sample statement is Theorem 4 of Benjamini and Yekutieli (2005), which holds under general dependence via a harmonic series argument yielding the c_m correction. The asymptotic extension follows by substituting the asymptotic coverage bound $\alpha_j^* + \epsilon_n$ for the exact bound α_j^* , exactly as in the proof of Theorem C.3, yielding $\limsup_{n \rightarrow \infty} \text{FCR} \leq q$ for $q \in (0, 1)$. $\square$

C.3.1 Verification of Assumption C.6

Proposition C.5 verifies Assumption C.6 for both VAR and LP estimation schemes.

Proposition C.5. *Let $\hat{P}_n$ denote the law of the bootstrap and suppose the conditions of Proposition C.1 hold. Further, let $z_j^*(\alpha)$ denote the α quantile of T_j^* . Then, the bootstrap percentile-t confidence interval $\text{CI}_j(\alpha) = [\hat{\theta}_j - \hat{\sigma}_j \cdot z_j^*(1 - \alpha/2), \hat{\theta}_j - \hat{\sigma}_j \cdot z_j^*(\alpha/2)]$ satisfies Assumption C.6.*

Proof of Proposition C.5. By Proposition C.3, the conditional bootstrap distribution of T_j^* converges weakly to the standard normal Φ in probability. Because Φ is continuous, Pólya's theorem gives $\sup_{x \in \mathbb{R}} |F_{T_j^*}(x) - \Phi(x)| \xrightarrow{P} 0$. Uniform convergence of CDFs to a strictly increasing continuous limit implies uniform convergence of quantiles, so $\sup_{\alpha \in (0,1)} |z_j^*(\alpha) - z_j(\alpha)| \xrightarrow{P} 0$.

Since $(\hat{\theta}_j - \theta_j)/\hat{\sigma}_j \Rightarrow \mathcal{N}(0,1)$ by Proposition C.2, and $\sup_{\alpha \in (0,1)} |z_j^*(\alpha) - z_j(\alpha)| \xrightarrow{P} 0$, it follows by Slutsky's theorem that $\mathbb{P}\{\theta_j \in \text{CI}_j(\alpha)\} = \mathbb{P}\{z_j^*(\alpha/2) \leq (\hat{\theta}_j - \theta_j)/\hat{\sigma}_j \leq z_j^*(1 - \alpha/2)\} \rightarrow \mathbb{P}\{z_j(\alpha/2) \leq Z \leq z_j(1 - \alpha/2)\} = 1 - \alpha$, where $Z \sim \mathcal{N}(0,1)$. Because the convergence of both $(\hat{\theta}_j - \theta_j)/\hat{\sigma}_j$ to $\mathcal{N}(0,1)$ and $z_j^*(\alpha)$ to $z_j(\alpha)$ is uniform over $\alpha \in (0,1)$, the coverage convergence is uniform over $\alpha \in (0,1)$. Since $m < \infty$ is fixed, uniformity also holds across all j , satisfying the first condition of Assumption C.6.

The monotonicity condition is mechanically satisfied. Because $\alpha \mapsto z_j^*(\alpha)$ is strictly non-decreasing, selecting $\alpha' \leq \alpha$ implies $z_j^*(\alpha'/2) \leq z_j^*(\alpha/2)$ and $z_j^*(1 - \alpha'/2) \geq z_j^*(1 - \alpha/2)$. Thus, the interval bounds expand (or remain equal), so $\text{CI}_j(\alpha) \subseteq \text{CI}_j(\alpha')$ almost surely, satisfying the second condition of Assumption C.6. $\square$

Consequently, we state Corollary C.3.

Corollary C.3. *Let $\hat{P}_n$ denote the law of the bootstrap and suppose the conditions of Proposition C.1 hold. Further, let $z_j^*(\alpha)$ denote the α quantile of T_j^* . Then, the FCR-adjusted confidence intervals $\text{CI}_j^*(q/c_m)$ satisfy $\limsup_{n \rightarrow \infty} \text{FCR} \leq q$ for $q \in (0,1)$.*